Lazy Realism FX

Getting Started

Installation

1. Use Blender 5.2 or newer In **Edit → Preferences → Get Extensions**, choose **Install from Disk** and select the supplied add-on ZIP. If the package is distributed as a legacy add-on, Blender also accepts its ZIP from **Preferences → Add-ons → Install from Disk**. Enable **Lazy Realism FX** after installation. See the Blender 5.2 extension installation guide for Blender's installation controls.
2. Keep the packaged `.blend` source libraries alongside the add-on's `__init__.py`; they contain the generator objects, Geometry Nodes, materials and collections used by the tools. Installing the Python file alone will leave effects unavailable. In the add-on preferences, check **Bundled source libraries** and run **Validate Source** if an effect reports a missing file.
3. Open a **3D Viewport**, press **N** to show the Sidebar, and select the **Lazy Realism FX** tab. The **Effects Browser** is where you add effects; **Effect Settings** shows the selected object's effects and their controls.

Interface and viewport overlays

The **Effects Browser** offers search, Grid and List views, category filters such as **Generators** and **Shaders**, plus **Favorites** and **Recent**. Open an effect's card and click its action button; some cards offer separate modes, such as **Scatter** and **Draw**, or a pair of add and remove actions. The star marks a favorite. You can also use **Shift+A → Quick Realism FX** in the 3D Viewport. Its menu always has **Open Sidebar Panel**; choose other shortcuts in the add-on's **Quick Access** preferences.

Several placement and drawing tools display a temporary instruction overlay in the viewport and a hint in the header. Follow the prompt for that particular tool: one may ask you to click objects, another to draw a stroke and press **Enter**, and another to turn the mouse wheel before finishing. **Esc** cancels drawing tools that show that option; check the on-screen hint before assuming a control applies to every tool. Once an effect has been created, select its object and open **Effect Settings**. You can also find many controls in the selected object's modifier or material settings.

Finding and organizing effects. Open **Edit → Preferences → Add-ons** (or the corresponding installed-extension preferences) to configure where Lazy Realism FX appears and which generators are duplicated under **Quick Access**. Save your preferences after changing the shortcuts. The Sidebar browser can be resized, and its **Grid View** makes effect thumbnails easier to recognize. Search also matches predefined tags, not only exact asset names. Use **Favorites** for frequently used effects, **Recent** for recently accessed assets, and **Manage Favorites** to organize them without browsing the entire catalog.

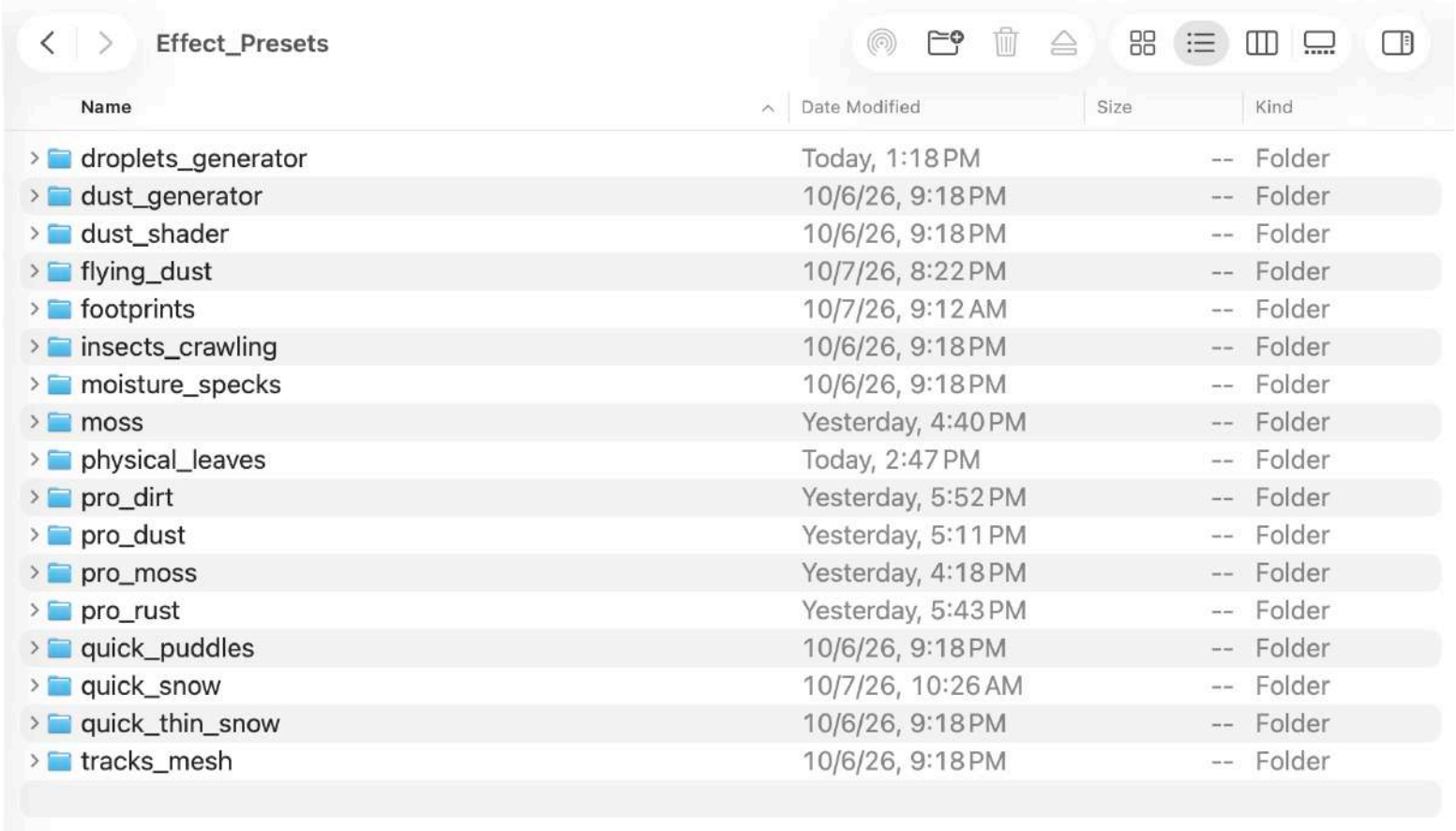
Effect hierarchy and settings. The **Effect Settings** panel lists Lazy Realism FX node groups, materials, and Geometry Nodes effects found on the selected object. A single object may hold several separate effects; expand its entry to edit one or use the **X** button beside an effect to remove it. The toolbar can refresh, expand, or collapse the hierarchy. Standalone generator objects, such as the Airplane domain, may present their settings without a detailed per-material hierarchy. For most Geometry Nodes effects, the same settings remain accessible in Blender's **Modifiers** tab.

Basic workflow

1. **Prepare the scene.** Add the objects or ground that the effect should use. Select the target mesh for an object effect or shader. For domain and emitter effects, ensure the scene has the geometry needed for placement and scale.
2. **Add an asset.** Find it in the Effects Browser or the configured Quick Realism FX menu, choose the appropriate action, then complete any on-screen placement, selection or drawing prompt. The add-on appends a prepared object or material, or copies a Geometry Nodes modifier onto the chosen object.

3. **Adjust the effect.** Select the resulting generator or original target, then change its named settings in **Effect Settings**. Shape controls change the area or surface; density and count controls change how many elements appear; seed changes their repeatable distribution; speed, wind and timing controls affect animation. Shader effects appear through the object's material and may depend on lighting and the render engine.
4. **Preview motion and refine.** Move through the timeline to see animated effects, then adjust the domain, direction, density, trails and materials to fit the shot. Some effects use simulation or stored history; when their result depends on prior frames, evaluate them from the start of the relevant frame range before judging a later frame.

Presets and sharing. Each effect offers a presets area. Adjust the effect, use the save icon, enter a preset name, and save to capture that combination of values. Select a saved preset to restore its parameters. The folder icon opens the location containing the preset **TXT files** in File Explorer (Windows) or Finder (macOS); these files can be backed up, copied to another computer, shared, or deleted. If you are following the video tutorial, open the separately supplied **Tutorial Scene** to reproduce its demonstrations; the scene is not required for normal use.



Applying effects. Where **Apply Effect** is available, apply a finished procedural effect to regular geometry to avoid continued live evaluation. This trades editability for a potentially lighter scene. Keep an unapplied copy or save a separate version before applying. For animated effects built on Simulation Nodes, review the relevant caching controls before expecting a stored animation to work out of sequence.

Working with material effects

Many Lazy Realism FX shaders can be applied to **one or several selected materials** on the same object. When the material-selection dialog appears, choose exactly the materials that should receive the effect.

Unified shader controls in the add-on. If multiple materials on the object contain the effect's shader node group, **Effect Settings** in the Lazy Realism FX Sidebar controls them together. Changing a setting in the add-on panel applies that value to the corresponding shader groups in **all affected materials at once**. This is convenient for keeping a consistent look without adjusting each material separately.

Editing one material independently. If you want a particular material to use different values, open the **Shader Editor**, select that material, find its Lazy Realism FX shader node group, and edit **the inputs on that material's group node** manually. These material-specific adjustments should be made in the Shader Editor rather than through the add-on panel, which applies unified changes. If you later adjust the same setting in the add-on panel, it may overwrite that material's individual value. Avoid editing the *internal definition* of a shared node group unless you intend to change all materials using that group.

You can also open the **Shader Editor** to connect a custom shader wherever the effect provides a material input, or to edit the base material underneath an effect. The original material, shader node groups, and any required auxiliary Geometry Nodes modifiers are separate parts of the setup; do not remove companion modifiers without checking whether the mask relies on their data.

A useful workflow for layered materials is to begin with one effect at a time: establish its overall strength, inspect its placement mask, adjust texture scale and breakup, and only then tune bump, roughness, and color. When several effects overlap, temporarily disable the others to see which mask is responsible for the result. Shader effects that use geometry-based attributes (such as **Surface Flatness**, **Triplanar Projection**, or **Edges Data**) need their supporting modifier or node data on the receiving object.

How the assets work

Think of each effect as a way to make a scene element automatically. Place a box, select a surface, or draw a path; the add-on fills it with the effect. Change the controls whenever you want a different result.

Type	What you provide	What the add-on does
Domain generator	A box or closed mesh defining a region	Populates the region with elements such as planes, birds, ash or snow, then moves or distributes them according to the effect's controls. Some tools create and fit their own domain; others use a mesh you supply.
Surface or object effect	A selected mesh, surface or emitter	Adds Geometry Nodes to scatter, deform or grow geometry on that object, often controlled by masks, distance and density.
Draw tool	A stroke or a sequence of clicks in the viewport	Converts your gesture into a curve or controller that the generator follows, as with drawn flight paths, moss or tracks.
Shader	A selected object and its material	Adds or changes material nodes to create wear, grime, leaks, droplets and similar appearances; geometry stays separate unless the effect explicitly includes a modifier.

Example — Airplanes Generator. Choose **Airplanes** → **Airplanes Generator** from the browser or configured quick menu. The operator adds a box-shaped domain, fits it around the scene with extra space on each side, and raises it above the scene. The mouse wheel changes the airplane amount during the initial overlay; click or press a key to finish. The effect places and moves airplanes at different heights inside this area. Assign an airplane collection and enable **Use Instances** to show plane models. Enable **Use Trail** to add their moving trails. The trails follow the generated motion and are governed by width, length, softness and fade settings. Select the domain afterward to tune the path-like motion, altitude range and trail appearance in **Effect Settings**. **Simple Draw** is a separate Airplanes action: it turns a viewport stroke into a smoothed flight-path curve at a fixed altitude; it does not perform the box-domain setup.

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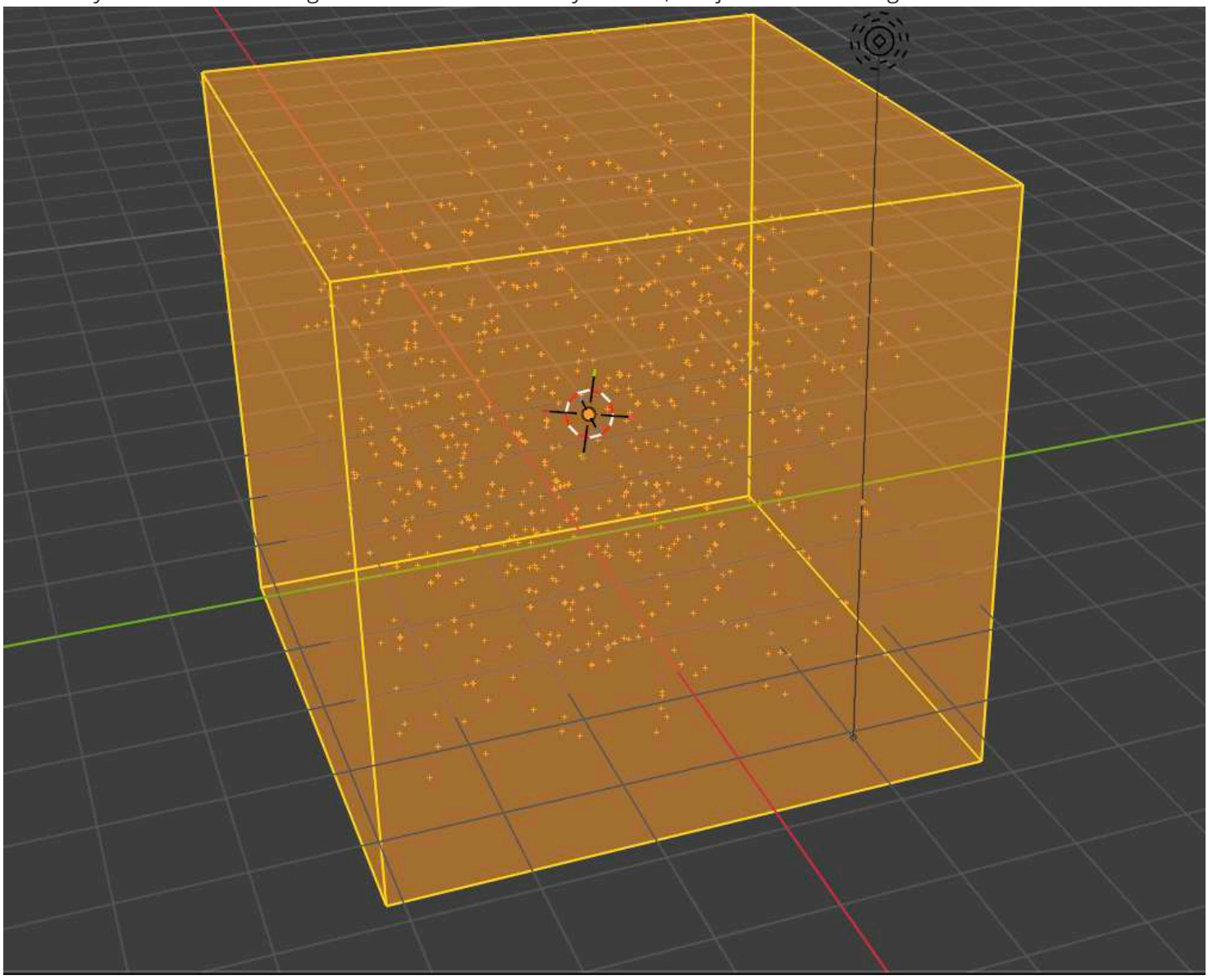
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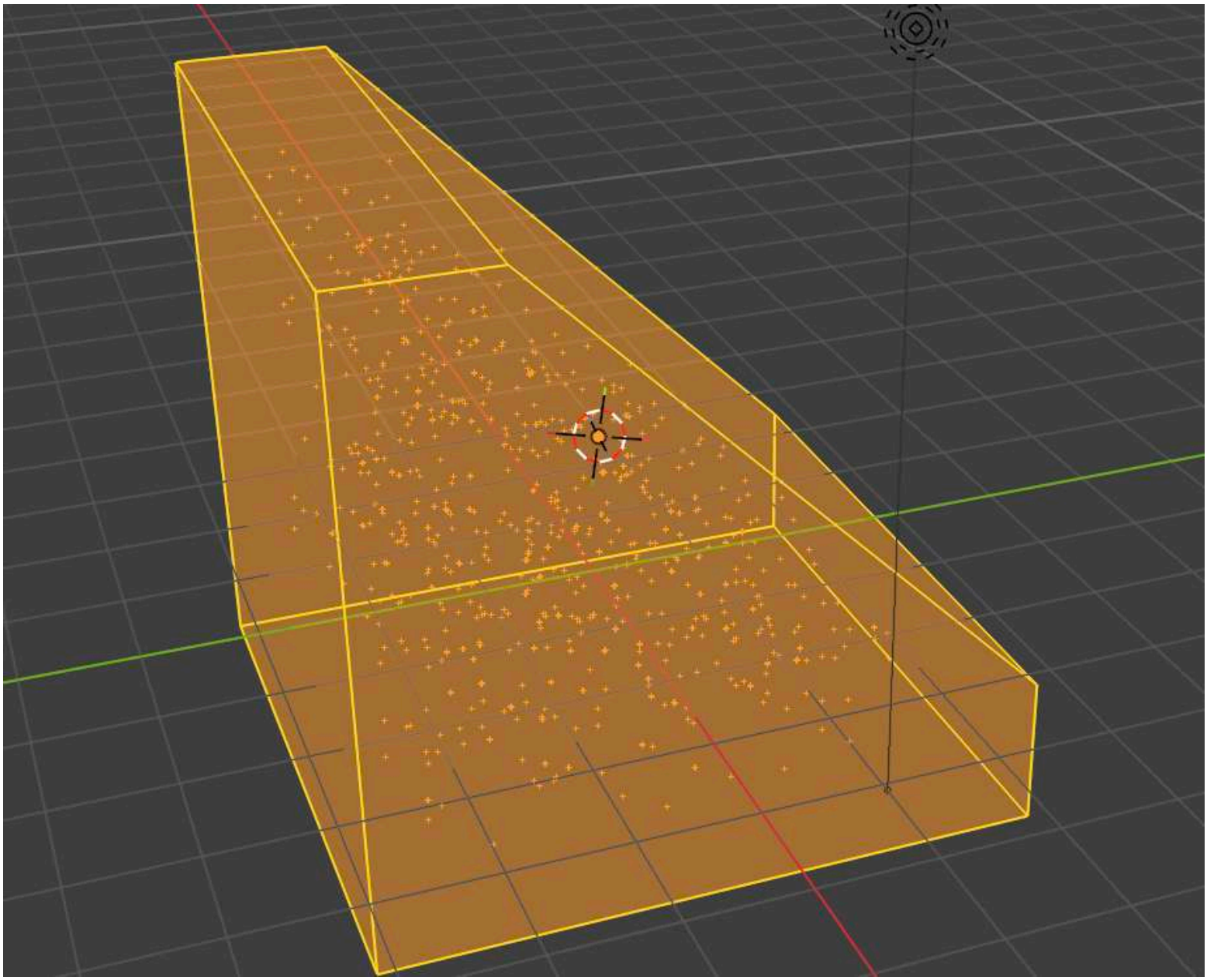
1. Ashes in Domain

Fills a closed mesh area with moving ash. Use the mesh to define where the ash can appear.

Setup and behavior notes

- Use a closed mesh for reliable interior classification. Thin or separated regions may need more Sampling Passes or a larger Batch Multiplier to reach Count.
- Boundary fade is measured against the actual boundary surface, not just the bounding box.





Practical workflow — Ashes in Domain

1. Add **Ashes in Domain** at the 3D Cursor. In **Edit Mode**, reshape its closed domain mesh to the intended region. It can be narrow, tapered, or asymmetrical rather than a box—for example, to keep ash concentrated above a campfire.
2. Select an **Ash Type** for the intended look. Adjust **Count**, **Vertical Speed**, **Wind Direction**, and **Wind Strength** to establish the flow. **Wind Variation** breaks up synchronized movement; **Flow Speed** serves as a convenient overall animation-speed multiplier.
3. Add **Turbulence** and a restrained amount of **Fine Turbulence** for irregular motion. Excessive fine turbulence can make particles visibly jitter. **Swirl Strength** adds circulation; **Individual Variation** prevents particles from moving identically. Adjust the **Instances** controls for size variation and **Spin Speed** for tumbling pieces.

The **Custom Mesh Domain** sampling and edge controls are advanced settings; adjust them if a complex or thin domain cannot be populated reliably. See the corresponding setting descriptions below before increasing sampling passes or batch sizes, which can cost memory.

Settings - Main Flow

Ash Type

Selects a discrete processing mode. Available choices: `Hot - Emissive`, `Dry`, `Mixed`, `Mixed Low-poly`.

- **Hot - Emissive:** Shows the glowing ash style.
- **Dry:** Shows the dry ash style.
- **Mixed:** Shows a mix of ash styles.
- **Mixed Low-poly:** Shows the simpler mixed ash style.

Count

Requested interior population. Sampling attempts are finite; sparse meshes may need more passes. Sets how many elements the effect tries to create.

- **Lowest value:** Creates few elements, or none when zero is allowed.
- **Highest value:** Creates more elements, though masks or spacing may limit how many fit.

Seed

Changes the random arrangement. The same seed gives the same result when the other settings stay the same.

- **Lowest value:** Selects one repeatable arrangement.
- **Highest value:** Selects a different repeatable arrangement.

Vertical Speed

Adds signed vertical travel.

- **Lowest value:** Negative values move particles downward.
- **Highest value:** Positive values move particles upward; zero removes this vertical contribution.

Wind Direction

Points the wind in a chosen direction. Wind Strength controls how hard it blows.

- **Zero or negative components:** Zero adds no direction. Negative components point along the negative axes.
- **Positive or larger components:** Positive components point along positive axes. Use the Strength setting to change the effect's intensity.

Wind Strength

Controls how strongly this movement affects the effect.

- **Lowest value:** Zero removes this movement. Negative values, if available, reverse its direction.
- **Highest value:** Makes the movement or deformation more noticeable.

Wind Variation

Controls how much individual elements differ from one another.

- **Lowest value:** Makes the elements more uniform.
- **Highest value:** Makes the elements differ more from one another.

Flow Speed

Controls the speed of the animated movement.

- **Lowest value:** Zero freezes the time-driven part; static offsets and other independent animation may remain.
- **Highest value:** Advances procedural motion much faster.

Turbulence

Turbulence Strength

Controls how strongly this movement affects the effect.

- **Lowest value:** Zero removes this movement. Negative values, if available, reverse its direction.
- **Highest value:** Makes the movement or deformation more noticeable.

Turbulence Scale

Controls the size of the visible pattern without changing the size of the individual objects.

- **Lowest value:** Creates broader, smoother patterns.
- **Highest value:** Creates smaller, more closely spaced details.

Turbulence Speed

Controls how quickly this part of the effect changes over time.

- **Lowest value:** Zero stops change in this contribution; a static displacement or rotation may remain. Negative values reverse time where supported.
- **Highest value:** Produces rapid changes or rotation; may appear abrupt at the current frame rate.

Fine Turbulence

Controls how strongly this movement affects the effect.

- **Lowest value:** Zero removes this movement. Negative values, if available, reverse its direction.
- **Highest value:** Makes the movement or deformation more noticeable.

Fine Turbulence Speed

Controls how quickly this part of the effect changes over time.

- **Lowest value:** Zero stops change in this contribution; a static displacement or rotation may remain. Negative values reverse time where supported.
- **Highest value:** Produces rapid changes or rotation; may appear abrupt at the current frame rate.

Swirl Strength

Controls how strongly this movement affects the effect.

- **Lowest value:** Zero removes this movement. Negative values, if available, reverse its direction.
- **Highest value:** Makes the movement or deformation more noticeable.

Individual Variation

Controls how much individual elements differ from one another.

- **Lowest value:** Makes the elements more uniform.
- **Highest value:** Makes the elements differ more from one another.

Collision

Instances

Instance Scale

Changes the size of the generated elements.

- **Lowest value:** Zero collapses the affected geometry. Negative values, if accepted, can mirror it.
- **Highest value:** Makes the affected geometry much larger; this is a visual scale and is not automatically a physics or collision scale.

Scale Variation

Controls how much individual elements differ from one another.

- **Lowest value:** Makes the elements more uniform.
- **Highest value:** Makes the elements differ more from one another.

Spin Speed

Controls how quickly this part of the effect changes over time.

- **Lowest value:** Zero stops change in this contribution; a static displacement or rotation may remain. Negative values reverse time where supported.
- **Highest value:** Produces rapid changes or rotation; may appear abrupt at the current frame rate.

Custom Mesh Domain

Scale Near Edges

Turns this option on or off.

- **Off:** Keeps the base particle scale near domain boundaries.
- **On:** Shrinks particles near the actual boundary surface using Edge Fade Distance.

Edge Fade Distance

Distance to the actual boundary surface over which ash scales from zero to full size. Sets the distance over which particles shrink or fade near the associated surface.

- **Lowest value:** Produces a narrow transition, retaining full size until close to the surface.
- **Highest value:** Produces a broad transition, so particles start shrinking farther away and may stay small throughout a small domain.

Sampling Passes

Maximum rejection-sampling passes. Raise for very thin or widely separated mesh regions. Limits repeated attempts to populate the interior of a closed custom domain.

- **Lowest value:** One pass gives fewer opportunities to fill thin or difficult regions.
- **Highest value:** More passes improve the chance of reaching Count in difficult domains, at a higher evaluation cost.

Batch Multiplier

Candidate batch size relative to the missing population. Higher values cost more memory. Scales the number of candidate interior samples relative to the missing population.

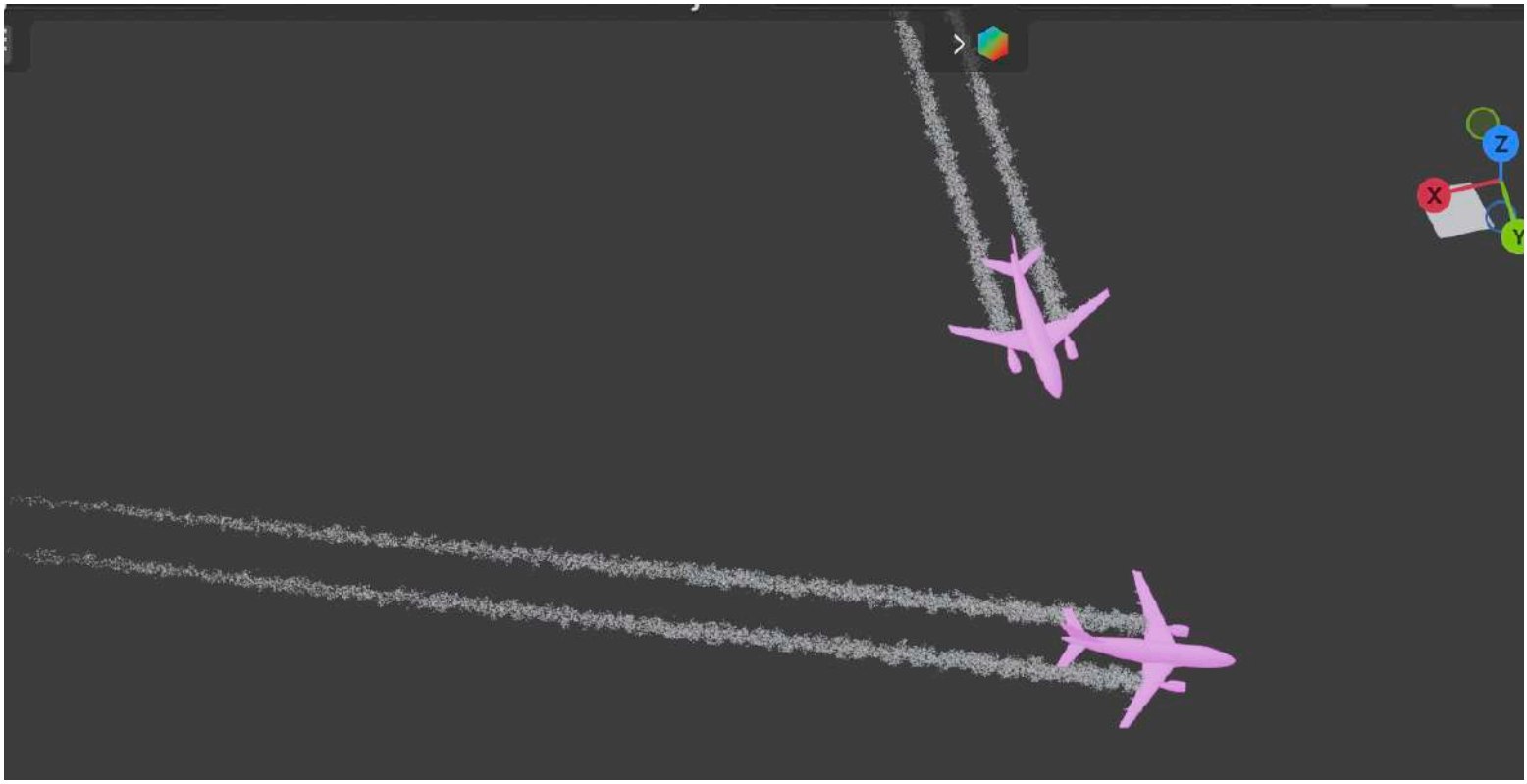
- **Lowest value:** Uses small candidate batches and less temporary memory.
- **Highest value:** Tests larger batches and may improve fill, using more temporary memory.

2. Airplanes

Both the domain generator and the drawn-flight-path tool belong to **Airplanes**. Use the domain for procedural flights in an elevated region, or draw a specific route when you need direct control over an airplane's path.

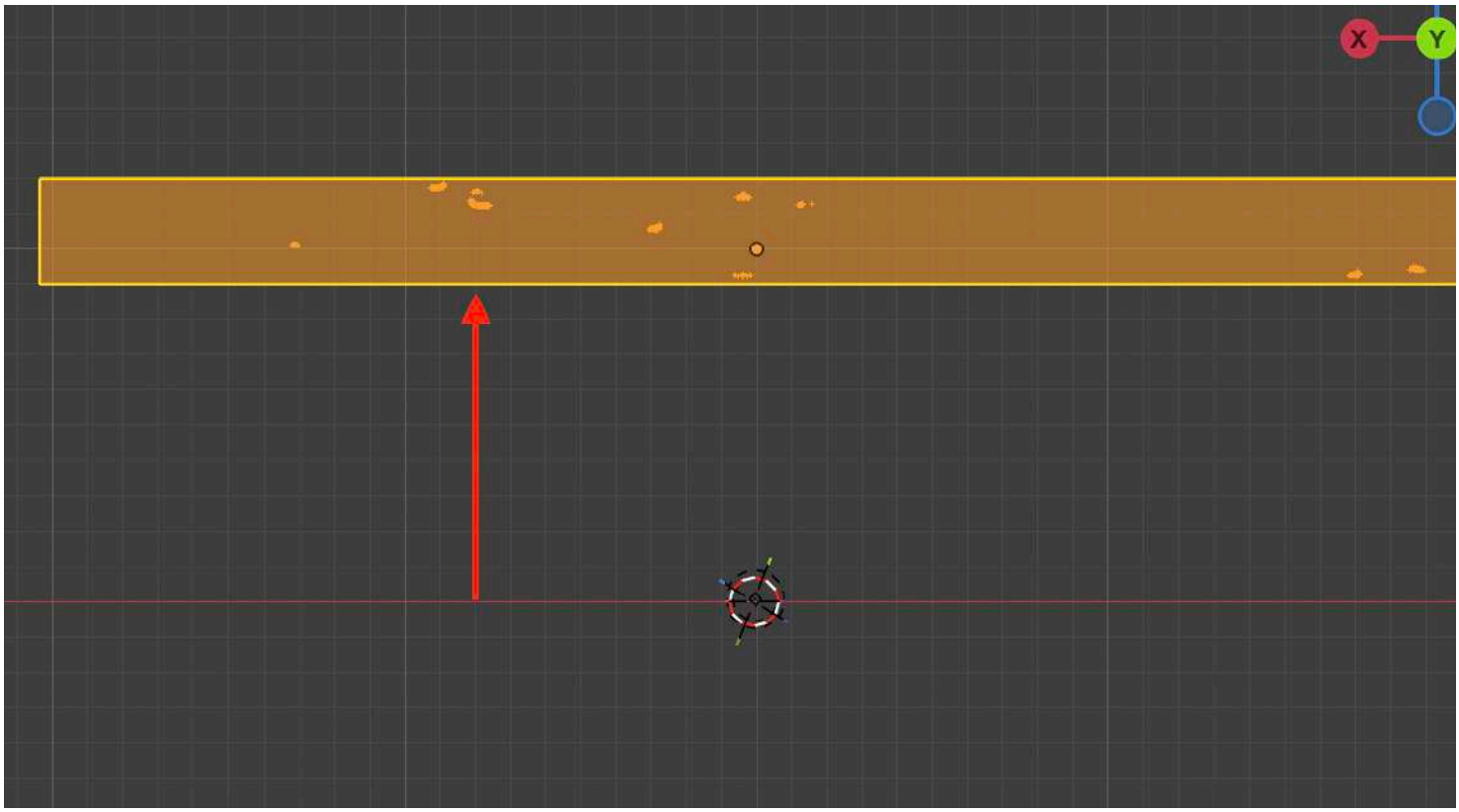
Airplanes — Domain Generator

Places airplanes in an elevated box above the scene. Set their height, direction, speed, and trails with the controls below.



Setup and behavior notes

- The minimum and maximum values for Spawn Height and Initial Trail Length are sorted internally, so reversed endpoints are normalized.
- Initial trail length is capped by Max Trail Length.
- **Domain is automatically placed at 1000m above Z=0**
-



Practical workflow — airplane domain and contrails

The domain generator places airplanes at an elevated level (approximately **1,000 m** in the tutorial setup). Adjust **Count** and the **Spawn Height Min/Max** interval to distribute planes across different altitudes and reduce overlap. Use **Flight Direction**, **Randomize Direction %**, and **Cruise Speed** for the overall movement; rotate or resize the domain if you need to change the scene-wide arrangement.

A trail is a generated ribbon with a procedural material, rather than a physical smoke simulation. **Trail Width** determines its thickness, **Max Trail Length** limits its length, and **Trail Fade Power** changes how quickly the ribbon becomes transparent along the trail. Adjust **Trail Follow Softness** to make the trailing end respond more gradually or more abruptly. **Initial Trail Length Min/Max** can start with a visible trail; these settings may use large scene-scale distances. If increasing initial length appears to do nothing, check whether **Max Trail Length** is clipping it. The contrail's final appearance also depends on its material.

Mode-specific settings

Count

Sets how many elements the effect tries to create.

- **Lowest value:** Creates few elements, or none when zero is allowed.
- **Highest value:** Creates more elements, though masks or spacing may limit how many fit.

Seed

Changes the random arrangement. The same seed gives the same result when the other settings stay the same.

- **Lowest value:** Selects one repeatable arrangement.
- **Highest value:** Selects a different repeatable arrangement.

Spawn Height Min

Sets an endpoint of the airplane spawn-height interval. The two values can be entered in either order.

- **Lowest value:** Moves this endpoint to a lower altitude.
- **Highest value:** Moves this endpoint to a higher altitude.

Spawn Height Max

Sets an endpoint of the airplane spawn-height interval. The two values can be entered in either order.

- **Lowest value:** Moves this endpoint to a lower altitude.
- **Highest value:** Moves this endpoint to a higher altitude.

Flight Direction

Points the airplanes in the direction they should fly. Cruise Speed controls how fast they move.

- **Zero or negative components:** Zero adds no direction. Negative components point along the negative axes.
- **Positive or larger components:** Positive components point along positive axes. Use the Strength setting to change the effect's intensity.

Randomize Direction %

Sets the percentage of elements selected for the named behavior.

- **Lowest value:** At 0%, selects none.
- **Highest value:** At 100%, selects all eligible elements.

Trail Follow Softness

Widens the transition where the trail tail begins following after the length limit is reached.

- **Lowest value:** Zero produces a sharper transition.
- **Highest value:** Produces a broader, softer follow transition.

Initial Trail Length Min

Sets a visible minimum or initial trail length, as named.

- **Lowest value:** Zero removes that minimum or initial length.
- **Highest value:** Creates longer initial or minimum streaks; the maximum trail limit can clamp the result.

Initial Trail Length Max

Sets a visible minimum or initial trail length, as named.

- **Lowest value:** Zero removes that minimum or initial length.
- **Highest value:** Creates longer initial or minimum streaks; the maximum trail limit can clamp the result.

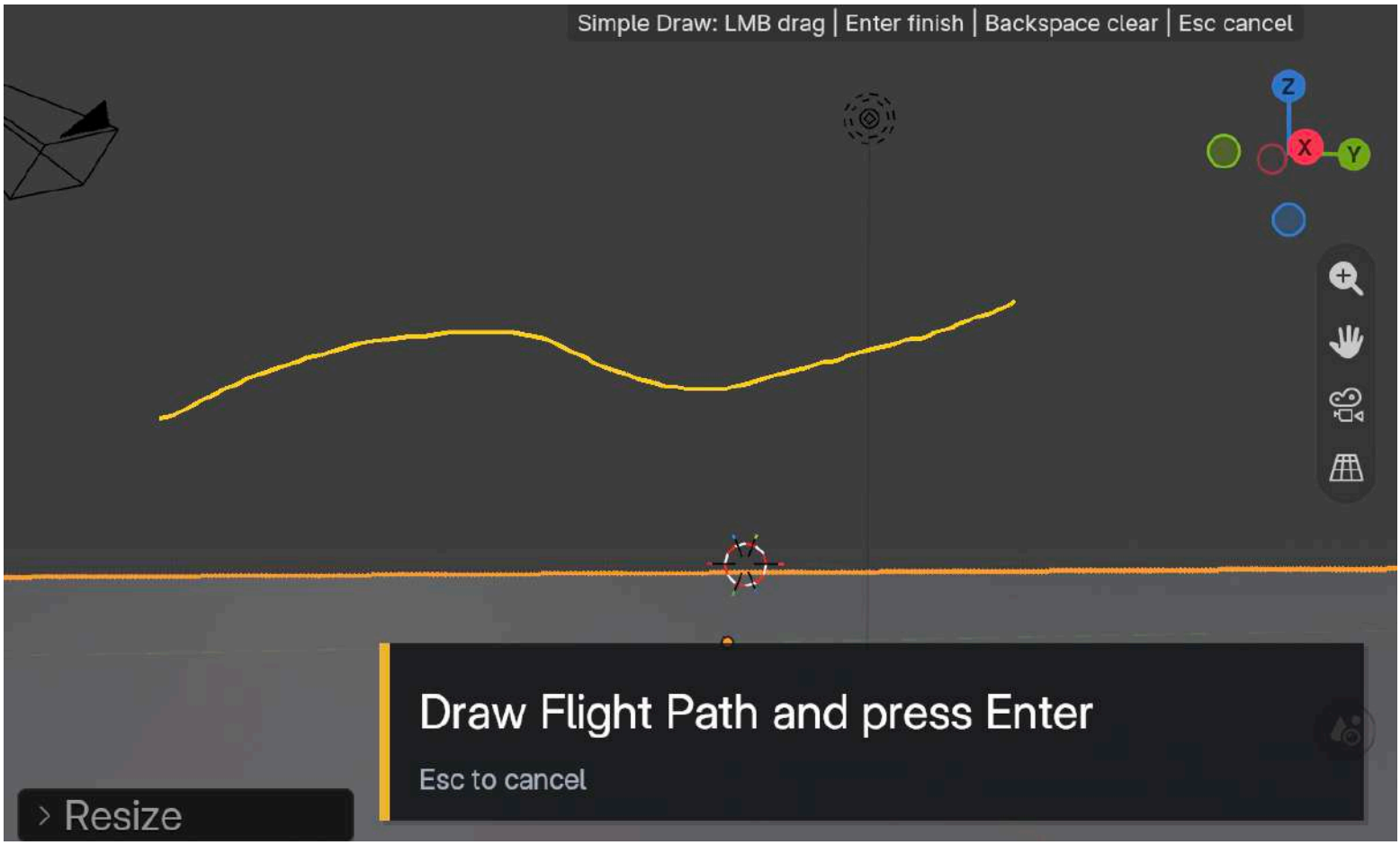
Scale Variation

Controls how much individual elements differ from one another.

- **Lowest value:** Makes the elements more uniform.
- **Highest value:** Makes the elements differ more from one another.

Airplanes — Draw Flight Path

Moves one airplane along a drawn curve, with an optional trail that follows its route.



Setup and behavior notes

- The airplane stops at the end of the path. Draw a longer curve if it needs to travel farther.
- Trail fade also depends on the trail material. Check the rendered result as you adjust these controls.
- Instance Scale changes the airplane model, while Point Radius controls the point output.

Practical workflow — manually drawn airplane

Choose **Airplane — Draw Flight Path** and draw a curve in the viewport. In the **Advanced Draw** placement mode, confirm the drawing and press **Enter**; the airplane is placed along the resulting curve at an initial elevated position. Keep the path above the ground and avoid drawing at negative or insufficient heights in modes that impose altitude restrictions. For direct control over the shape, use **Simple Draw**, then edit the resulting curve manually.

For sharp corners, select the curve in **Edit Mode**, run **Subdivide** on the affected segments, and then **Smooth** until the transition is gradual. Excessive subdivision is unnecessary; smooth flight paths matter more than point count. Adjust **Cruise Speed**, **Position Offset**, the airplane collection, and scale to match the shot. For visible contrails, configure **Trail Width**, **Max Trail Length**, and the available trail fading controls.

Mode-specific settings

Position Offset

Offsets the starting position along the flight path before time-driven movement.

- **Lowest value:** Moves toward the path start or an earlier phase, subject to clamping or looping logic.
- **Highest value:** Moves toward the end or a later phase, subject to clamping or looping logic.

Airplanes — Shared Settings

The following controls are available in both modes.

Cruise Speed

Controls how fast the elements move.

- **Lowest value:** Produces the slowest allowed travel; zero removes this travel contribution but other motion may remain.
- **Highest value:** Produces faster travel; high values cover more distance per frame and may require finer simulation steps.

Trail Fade Power

Controls how quickly the airplane trail fades toward its end.

- **Lowest value:** Keeps most of the trail visible, with a short fade near its end.
- **Highest value:** Concentrates the visible trail close to the airplane; the trail length stays the same.

Trail Width

Sets the width of the generated airplane trail ribbon.

- **Lowest value:** Produces a very narrow ribbon.
- **Highest value:** Produces a very wide ribbon; does not lengthen it.

Max Trail Length

Caps the visible trail length.

- **Lowest value:** Keeps only a very short trail.
- **Highest value:** Permits a much longer trail; available history or path length can still limit the result.

Airplane Collection

Collection supplying the airplane geometry used when Use Instances is enabled.

- **Unassigned or empty:** Leave this unassigned and any effect that needs this source may disappear.
- **Assigned or named:** Choose the object, collection, material, or named mask that should drive the effect.

Instance Scale

Changes the size of the generated elements.

- **Lowest value:** Zero collapses the affected geometry. Negative values, if accepted, can mirror it.
- **Highest value:** Makes the affected geometry much larger; this is a visual scale and is not automatically a physics or collision scale.

Point Radius

Changes the visible thickness of the elements.

- **Lowest value:** Produces very small or thin elements; zero can collapse them if allowed.
- **Highest value:** Produces large or thick elements; this may overlap surrounding geometry.

3. Dripping

Creates dripping water from a curve or mesh surface. You can adjust falling drops, wind, contact with other objects, and splashes.

Practical workflow — Dripping

Choose **Dripping**, then draw or assign the curve along which drops should originate. The generator distributes falling droplets along that path. Set **Fall Height** to reach the intended landing surface; a value of zero eliminates the fall distance. When changing the height, also review **Speed** so drops reach the surface at the intended time. **Distance Between Droplets** controls how many emission points fit on the path: wider spacing usually means fewer drops. In versions showing **Random Spawn** → **Spawn Interval**, increasing that interval likewise produces fewer generated drops.

Customize the procedural drop silhouette with **Droplet Radius**, tip length, tail-stretch timing, and size randomness. Use the path-noise controls for turbulent trajectories and the **Wind** controls to push falling water sideways. To detect contact, assign a **Collision Collection** and adjust **Collision Margin**; if contacts are missed, try a modestly larger margin and visible droplet radius. Secondary contact droplets or splashes can be configured with the splash controls. Edit the generated water material through the material assigned to the effect.

Settings

Realize Instances

Turns this option on or off.

- **Off:** Preserves instances in the visible result.
- **On:** Converts generated copies into mesh geometry for further effects or export. This can slow down the scene. Press - CONVERT TO MESH to make in as baked object - it will became static

Distance Between Droplets

Sets minimum spacing between emission points.

- **Lowest value:** Allows points to be close together, increasing crowding and potential overlap.
- **Highest value:** Separates points over a much wider area, usually reducing how many fit on the emitter.

Material

Choose the material used by the generated elements.

- **Unassigned or empty:** Leave this unassigned and any effect that needs this source may disappear.
- **Assigned or named:** Choose the object, collection, material, or named mask that should drive the effect.

Fall Height

Controls the height of the area where elements appear or move.

- **Lowest value:** Gives a short travel span; signed negative values are not a useful falling height.
- **Highest value:** Gives a tall travel span and can increase flight duration or population.

Direction

Sets which way the elements move. Use the related Strength or Speed control to change the intensity.

- **Zero or negative components:** Zero adds no direction. Negative components point along the negative axes.
- **Positive or larger components:** Positive components point along positive axes. Check the preview to see how strongly the direction affects motion.

Speed

Controls how fast the elements move.

- **Lowest value:** Produces the slowest allowed travel; zero removes this travel contribution but other motion may remain.

- **Highest value:** Produces faster travel; high values cover more distance per frame and may require finer simulation steps.

Speed Curve

Controls the emitter movement along its source curve in Dripping.

- **Lowest value:** Zero stops this curve-travel component.
- **Highest value:** Moves the emission pattern along the curve more quickly.

Shrink

Sets the start threshold of the normalized droplet-size remapping.

- **Lowest value:** At 0, uses the full normalized progress range for scaling.
- **Highest value:** At 1, collapses the remapping interval; most earlier progress is clamped to the small end.

Strength

Controls how strongly this movement affects the effect.

- **Lowest value:** Zero removes this movement. Negative values, if available, reverse its direction.
- **Highest value:** Makes the movement or deformation more noticeable.

W

Shifts the animation or pattern to a different starting point.

- **Lowest value:** Uses an earlier or negative phase or a different noise slice.
- **Highest value:** Uses a later or positive phase or a different noise slice; does not inherently increase detail.

W Speed

Controls how quickly this part of the effect changes over time.

- **Lowest value:** Zero stops change in this contribution; a static displacement or rotation may remain. Negative values reverse time where supported.
- **Highest value:** Produces rapid changes or rotation; may appear abrupt at the current frame rate.

Similarity

Changes how similar the movement patterns are across individual droplets.

- **Lowest value:** At zero, the droplets follow a more similar pattern.
- **Highest value:** Increases phase differences between elements, reducing their shared noise motion; it is not a spatial frequency control.

Scale

Changes the size of the generated elements.

- **Lowest value:** Zero collapses the affected geometry. Negative values, if accepted, can mirror it.
- **Highest value:** Makes the affected geometry much larger; this is a visual scale and is not automatically a physics or collision scale.

Detail

Adds higher-frequency layers to the named noise texture.

- **Lowest value:** Uses only the base noise layer.
- **Highest value:** Adds finer noise detail and may increase evaluation cost.

Droplet Radius

Changes the visible thickness of the elements.

- **Lowest value:** Produces very small or thin elements; zero can collapse them if allowed.
- **Highest value:** Produces large or thick elements; this may overlap surrounding geometry.

Tip Length

Controls how far the droplet tip or tail extends.

- **Lowest value:** Zero gives a rounder shape with no added tail length.
- **Highest value:** Produces a long, narrow trailing extension.

Randomize Tip Length Value

Adds stable per-droplet variation to Tip Length. 0 keeps all droplets identical; 0.3 gives approximately 30% in either direction variation. Controls how much individual elements differ from one another.

- **Lowest value:** Makes the elements more uniform.
- **Highest value:** Makes the elements differ more from one another.

Tip Stretch Time

Sets the time for the procedural trailing tip to grow to its full length.

- **Lowest value:** The tip reaches full extension quickly.
- **Highest value:** The tip grows slowly; short-lived droplets may never reach full extension.

Tip Sharpness

Controls tapering toward the procedural droplet tip.

- **Lowest value:** Gives a blunt or rounded trailing end.
- **Highest value:** Gives a sharper, narrower tip; built-in limits may prevent a fully collapsed tip.

Droplet Segments

Controls how smooth or detailed this part of the effect appears.

- **Lowest value:** Shows a simpler, rougher shape.
- **Highest value:** Shows a smoother shape, which can slow down the scene.

Droplet Rings

Controls how smooth or detailed this part of the effect appears.

- **Lowest value:** Shows a simpler, rougher shape.
- **Highest value:** Shows a smoother shape, which can slow down the scene.

Min Scale

Sets one end of the size range. Keep the minimum below the maximum.

- **Lowest value:** Sets that endpoint to a very small size; zero allows collapsed elements and negative scale can mirror them.
- **Highest value:** Sets that endpoint to a larger size, allowing or forcing larger elements depending on the other endpoint.

Max Scale

Sets one end of the size range. Keep the minimum below the maximum.

- **Lowest value:** Sets that endpoint to a very small size; zero allows collapsed elements and negative scale can mirror them.
- **Highest value:** Sets that endpoint to a larger size, allowing or forcing larger elements depending on the other endpoint.

Noise Range

Sets a threshold or range used when converting noise to random scale.

- **Lowest value:** A narrow range makes the remapping more selective.
- **Highest value:** At 1, uses the full supported range for the scale remapping.

Noise Scale

Controls the size of the visible pattern without changing the size of the individual objects.

- **Lowest value:** Creates broader, smoother patterns.
- **Highest value:** Creates smaller, more closely spaced details.

Wind Direction

Points the wind in a chosen direction. Wind Strength controls how hard it blows.

- **Zero or negative components:** Zero adds no direction. Negative components point along the negative axes.
- **Positive or larger components:** Positive components point along positive axes. Use the Strength setting to change the effect's intensity.

Wind Strength

Controls how strongly this movement affects the effect.

- **Lowest value:** Zero removes this movement. Negative values, if available, reverse its direction.
- **Highest value:** Makes the movement or deformation more noticeable.

Collision Collection

Choose the objects that this effect should collide with or avoid.

- **Unassigned or empty:** Leave this unassigned and any effect that needs this source may disappear.
- **Assigned or named:** Choose the object, collection, material, or named mask that should drive the effect.

Collision Margin

Sets additional surface clearance, particle contact size or contact padding as named.

- **Lowest value:** Allows movement or rendering close to the surface, with less padding.
- **Highest value:** Creates a larger gap or contact region; excessive padding can prevent passage through small spaces.

Splash Count

Sets secondary droplet count per primary impact.

- **Lowest value:** Zero creates no secondary droplets.
- **Highest value:** Creates the maximum secondary count per impact, increasing output cost.

Splash Speed

Sets the outgoing secondary-particle speed; in the surface-drop simulation it is the tangential component.

- **Lowest value:** Zero removes this outward or tangential speed contribution; lift and gravity can still move splashes.
- **Highest value:** Throws secondary droplets farther or faster.

Splash Lift

Sets the velocity away from the contacted surface normal.

- **Lowest value:** Zero removes the outward-normal launch component.
- **Highest value:** Launches splash droplets strongly away from the surface.

Splash Lifetime

Controls particle lifetime or one endpoint of its randomized lifetime.

- **Lowest value:** Shortens visible particle persistence; zero can remove the lifetime where accepted.
- **Highest value:** Keeps particles alive longer, which can raise the visible population. Check the generator notes for time units.

Splash Radius

Changes the visible thickness of the elements.

- **Lowest value:** Produces very small or thin elements; zero can collapse them if allowed.
- **Highest value:** Produces large or thick elements; this may overlap surrounding geometry.

Splash Gravity

Sets the world-axis splash gravity vector. Dripping converts it into emitter coordinates.

- **Zero or negative components:** Zero disables splash gravity. Negative Z accelerates downward.
- **Positive or larger components:** Positive Z accelerates upward; larger vector magnitude gives stronger acceleration.

Splash Drag

Damps particle velocity and, where implemented, couples it to wind.

- **Lowest value:** Low drag preserves momentum longer.
- **Highest value:** Strong drag quickly damps relative movement; particles respond more strongly to the wind coupling.

Splash Seed

Changes the random arrangement. The same seed gives the same result when the other settings stay the same.

- **Lowest value:** Selects one repeatable arrangement.
- **Highest value:** Selects a different repeatable arrangement.

Max Splash Particles

Limits how many elements can be active or retained.

- **Lowest value:** Restricts the layer to the smallest permitted budget; zero prevents that layer when supported.
- **Highest value:** Allows more particles or stored samples, increasing memory use; does not create particles by itself.

4. Surface Droplets

Combines small condensation droplets, moving surface heads, wet-trail history, detached procedural drips and one-shot secondary splashes.

Setup and behavior notes

- **Scatter** and **Draw** add the same Droplets Geometry Nodes modifier. **Draw** immediately starts **Weight Paint** on the active object's vertex-group mask; **Scatter** starts with automatic distribution. The painted mask can also be configured later under **Paint Mask**.
- On import, choose whether to **Enable Simulation**. With simulation, long droplets move, deposit wet trails and may detach as falling drips. Without simulation, use the effect as a static surface-droplet layer.
- In **Effect Settings**, use the effect's presets, **Bake Droplets** and the control for deleting an existing bake. Bake when you want to reuse calculated animation without evaluating the full simulation again.
- **Disable Dynamics | Skip Simulation** stops dynamic movement. In contrast, switching **Enable Spawning New Simulated Droplets** off keeps existing moving droplets active while preventing *new simulated droplets* from being emitted. This spawning toggle does not freeze droplets and is separate from the static small-droplet layer.
- **Max Surface Droplets** is a performance cap on admission of new surface droplets; it does not remove droplets that are already alive. The initial small layer is bounded separately, with initial long heads retained in addition. If this cap is too low for a large object, coverage can appear cut off or confined to rectangular regions; raise it carefully if needed.
- **Fast Mode** is the recommended default. Disabling Fast Mode without enabling Accurate Mode produces the smoother intermediate-quality result; **Weld Head to Trail** improves the head/trail seam in that mode. **Accurate Mode** uses volume-based remeshing and can require substantially more memory and processing time.
- **Prevent Surface Overlap** gives conservative tangent-plane clearance rather than a guarantee for every nearby surface. Signed **Normal Offset** values can intentionally embed droplets in the mesh.
- **Enable Surface Collision** redirects *surface-attached* moving droplets around the assigned colliders. **Enable Drip Collision** detects impacts after drips detach: without **Collision Splashes**, impacting drips are removed; with splashes enabled, each primary impact can emit secondary icosphere particles. Splashes are one-shot and do not recursively spawn more splashes.
- **Vertex Group** uses **Weight Paint**, not vertex-color paint. If a vertex group is required but missing or unpainted, droplets may not spawn in the intended areas.

Practical workflow — simulated and painted droplets

1. Import and preview. Select the target mesh, choose **Droplets** → **Scatter** for automatic distribution or **Droplets** → **Draw** to paint a spawn region immediately. The two choices use the same underlying modifier. At the import prompt, enable simulation if you need motion and falling drips; otherwise choose the static setup. Play the timeline from the simulation start to check the moving long droplets and their wet trails. Use the effect's bake controls once the result is ready to cache.

2. Choose simulation behavior and limits. Disable Dynamics | Skip Simulation is useful for static shots because droplet movement no longer needs to be evaluated. For a moving scene, keep dynamics active and use **Enable Spawning New Simulated Droplets** to decide whether additional simulated heads appear over time. With the switch **Off**, the already-created heads can still move and leave trails; with it **On**, new heads can spawn according to **Spawn Rate**. **Max Surface Droplets** protects large scenes from excessive population. If you see hard or rectangular boundaries to the

scatter distribution, the cap may be too low; increase it within your performance budget.

3. Select an appropriate quality mode. Keep **Fast Mode** enabled for ordinary shots. Close up, its lightweight geometry may show a seam between a droplet head and its wet trail. Disable Fast Mode while leaving **Accurate Mode** off for the intermediate mode, where the shape is smoother; **Weld Head to Trail**, enabled by default in this mode, can improve the join. For an extreme close-up, enable **Accurate Mode** to remesh the result through a volume. Increase **Quality** only as needed: the tutorial demonstrates increasing it from approximately **1,000** to **7,000**, but high values can significantly affect responsiveness and memory.

4. Distribute simulated and small droplets. The **Simulated Droplets** group controls moving, trail-producing heads: **Amount** is the initial count, **Spawn Rate** the number added during animation when new spawning is enabled, **Minimum Distance** the separation, and **Seed** the repeatable variation. **Initial Clump Scale** changes the grouping of the starting spawn pattern. **Initial Trail Length** determines how much wet history appears on the initial frame, and **Initial Trail Samples** controls its smoothness; the trails subsequently follow the **Wet Trails** drying settings. The **Small Droplets** group has its own **Amount**, **Minimum Distance**, **Seed**, and **Size Variation** for the small scattered layer.

5. Fix positioning and size. Surface Clearance is the shared clearance adjustment. Fine-tune **Small Normal Offset** and **Long Normal Offset** to move each layer along the target normals if droplets intersect or float above the surface. **Simulated Droplets Head Scale** changes the visible size of moving heads; review the head-to-trail proportions before using very large values. **Flatten Droplets** adjusts how flat the droplets sit against the surface. **Uniform Random Scale**, **Randomize Scale XYZ**, and the small-droplet scale interval vary their sizes and proportions without changing their movement physics.

6. Animate surface flow. Increase **Simulation Substeps** when fast movement or contacts need better temporal accuracy; higher values cost performance. **Gravity** drives downward movement. **Path Meander** adds *coherent lateral wandering* rather than unrelated random kicks every frame; zero gives straight paths, whereas larger values create curved trails. A small nonzero value (for example, **0.01**) can avoid an overly mechanical appearance. **Path Wavelength** determines whether the bends are short and frequent (low values) or broad and gentle (high values). **Flow Response** controls how quickly a droplet adjusts toward its requested surface velocity, and **Maximum Surface Speed** caps that speed. Use **Wind** and **Wind Strength** to push attached droplets and affect airborne drips.

7. Grow new heads naturally. Under **Head Growth**, **Growth Time** determines how long a new droplet takes to reach full size, so it does not suddenly appear at its final scale. **Growth Distance** adds travel-distance-based growth; **0** disables the distance condition and leaves time-based growth. **Start Scale** defines the size at the moment a head and its trail form. **Trail Follow Head** changes how older trail samples are sized: toward **1**, the entire trail follows the current head scale; toward **0**, each sample preserves the head size it had when deposited, producing a thin beginning and thicker sections closer to a growing head.

8. Control wet trails and painted emission. Trail Disappear Speed changes how quickly deposited streaks are removed, **Trail Hold Time** controls the delay before drying begins, and **Trail Fade Time** sets the duration of the smooth drying transition. **Max Trail Points** is a storage safeguard; if it is too low, trails may be truncated. To localize the entire effect, assign a **Vertex Group** under **Paint Mask** and paint its weights on the active object; the Draw import mode simply starts this process for you.

9. Enable falling drips. Activate **Dripping** to let surface droplets detach and fall when they lose contact with the geometry. **Edge Detach Distance** belongs to the **Dripping** controls: a small tolerance allows detachment at slight bumps or edges, while a larger one keeps heads attached more readily. **Dripping Radius**, tip length/shape and **Drip Tip Stretch Time** govern the procedural airborne droplet geometry. **Drip Air Drag** damps velocity and couples it to wind, **Drip Lifetime** controls how long airborne particles remain, and **Max Dripping Particles** protects performance.

10. Add collisions, splashes and materials. Choose the relevant **Drip Colliders** collection. **Enable Surface Collision** makes attached simulated droplets reroute around those obstacles. **Enable Drip Collision** handles airborne impacts: when **Collision Splashes** is off, colliding drips disappear; enable it to create one-shot secondary splashes. Use **Splash Count**, **Splash Speed**, **Splash Lift**, **Splash Lifetime**, and **Splash Radius** to control their number, outward travel, launch, lifespan and icosphere size. For alternate liquid looks, enable **Use Custom Material** and select the material in the revealed **Custom Material** field.

Troubleshooting. If droplets stop filling a large object's surface or rectangular scatter boundaries are visible, check **Max Surface Droplets**. If Fast Mode shows a head/trail seam, try the intermediate quality mode and **Weld Head to Trail** before selecting the much heavier Accurate Mode. If trails are unexpectedly short, review **Trail Hold Time**, **Trail Fade Time**, **Trail Disappear Speed** and **Max Trail Points**. If moving heads detach at tiny surface irregularities, raise **Edge Detach Distance**. If collision impacts create no splashes, confirm both **Enable Drip Collision** and **Collision Splashes** and check the assigned collider collection.

Settings

Disable Dynamics | Skip Simulation

Disables the dynamic part of the droplet effect so the droplets remain static. Unlike the new-spawning toggle, this stops movement rather than only preventing emission.

- **Off:** Allows simulated surface droplets to move, leave trails and detach when other relevant features are enabled.
- **On:** Skips droplet dynamics for a static result; useful for still renders and faster previews.

Master Seed

Changes the random arrangement. The same seed gives the same result when the other settings stay the same.

- **Lowest value:** Selects one repeatable arrangement.
- **Highest value:** Selects a different repeatable arrangement.

Enable Spawning New Simulated Droplets

Allows the simulation to generate additional moving droplets over time according to Spawn Rate. This is the renamed, positively worded control; it is separate from Disable Dynamics.

- **Off:** Stops the generation of *new simulated droplets* while previously created droplets continue moving when dynamics are enabled.
- **On:** Enables new simulated-droplet spawning according to the configured Spawn Rate.

Fast Mode

Controls whether the inexpensive procedural droplet geometry is used. In extreme close-ups, Fast Mode can show a seam where the rounded head meets its trail.

- **Off:** Uses the smoother intermediate mode when Accurate Mode is also disabled; permits the Weld Head to Trail improvement.
- **On:** Uses the faster, lightweight default geometry, recommended for most shots.

Accurate Mode

Enables a volume-to-mesh remeshing workflow that blends the rounded head and wet trail more smoothly. This is mainly intended for extreme close-up shots.

- **Off:** Uses either Fast Mode or the smoother intermediate procedural mode.
- **On:** Uses the expensive volume-based mode whose resolution is controlled by Quality.

Quality

Sets the resolution of the volume-based remeshing when Accurate Mode is enabled. For very close shots, the tutorial demonstrates increasing the default value of around 1,000 to around 7,000, but high values can be expensive.

- **Lowest value:** Gives a coarser result that may not smoothly merge the head and trail.
- **Highest value:** Produces finer remeshing at a substantially higher memory and evaluation cost.

Weld Head to Trail

Improves the join between the rounded simulated-droplet head and the trail in the intermediate-quality mode, when Fast Mode is disabled. It is enabled by default in that mode.

- **Off:** Leaves the intermediate geometry without this additional seam improvement.
- **On:** Welds the head to its trail more precisely, reducing visible joins.

Initial Trail Length

Backward surface-traced wet length already present on the simulation start frame. Sets a visible minimum or initial trail length, as named.

- **Lowest value:** Zero removes that minimum or initial length.
- **Highest value:** Creates longer initial or minimum streaks; the maximum trail limit can clamp the result.

Initial Trail Samples

Number of surface projection steps in the initial wet history. Controls how smooth or detailed this part of the effect appears.

- **Lowest value:** Shows a simpler, rougher shape.
- **Highest value:** Shows a smoother shape, which can slow down the scene.

Max Surface Droplets

Limits admission of new surface droplets. Already living droplets are not deleted to satisfy this budget. The initial small layer is bounded separately; initial long heads are retained in addition to that layer. Limits how many elements can be active or retained.

- **Lowest value:** Restricts the layer to the smallest permitted budget; zero prevents that layer when supported.
- **Highest value:** Allows more particles or stored samples, increasing memory use; does not create particles by itself.

Prevent Surface Overlap

Conservative tangent-plane clearance before signed artistic offset. Does not guarantee clearance against every nearby surface.

- **Off:** Omits the automatic conservative surface-clearance adjustment.
- **On:** Adds conservative tangent-plane clearance before the signed artistic Normal Offsets; it does not guarantee clearance against every nearby surface.

Surface Clearance

Automatic positive clearance. Sets additional surface clearance, particle contact size or contact padding as named.

- **Lowest value:** Allows movement or rendering close to the surface, with less padding.
- **Highest value:** Creates a larger gap or contact region; excessive padding can prevent passage through small spaces.

Small Normal Offset

Signed normal offset for static or small droplets. Negative intentionally embeds drops. Moves the associated geometry along the surface normal without changing its emission mask.

- **Lowest value:** Negative values intentionally embed it into the surface. Zero adds no artistic offset.
- **Highest value:** Positive values lift it away from the surface.

Long Normal Offset

Signed normal offset for trails and their heads, outside the simulation. Moves the associated geometry along the surface normal without changing its emission mask.

- **Lowest value:** Negative values intentionally embed it into the surface. Zero adds no artistic offset.
- **Highest value:** Positive values lift it away from the surface.

Droplets Scale

Overall scale multiplier for droplets. Changes the size of the generated elements.

- **Lowest value:** Zero collapses the affected geometry. Negative values, if accepted, can mirror it.
- **Highest value:** Makes the affected geometry much larger; this is a visual scale and is not automatically a physics or collision scale.

Small Droplets Scale Multiplier

Additional scale multiplier applied only to small droplets. Changes the size of the generated elements.

- **Lowest value:** Zero collapses the affected geometry. Negative values, if accepted, can mirror it.
- **Highest value:** Makes the affected geometry much larger; this is a visual scale and is not automatically a physics or collision scale.

Flatten Droplets

Flattens droplets against the surface. Scales droplet thickness along its surface-normal axis.

- **Lowest value:** Produces flatter droplets; zero collapses thickness where allowed.
- **Highest value:** Produces thicker or taller droplets. At 1, the normal-axis scale is uncompressed; values above 1 stretch it.

Simulated Droplets Head Scale

Visible size of rounded long-droplet heads. 1 = original, 2 = double. Does not change trail thickness, small droplets, airborne drip tips or physics. Scales only the visible rounded heads of long surface droplets. It does not change trail thickness, small droplets, airborne tips or physics.

- **Lowest value:** At 0, hides or collapses the rounded heads.
- **Highest value:** At 10, makes heads ten times the baseline size.

Uniform Random Scale

Turns this option on or off.

- **Off:** Uses the other scale mode.
- **On:** Uses a uniform scale value per element from the random interval.

Scale

Changes the size of the generated elements.

- **Lowest value:** Zero collapses the affected geometry. Negative values, if accepted, can mirror it.
- **Highest value:** Makes the affected geometry much larger; this is a visual scale and is not automatically a physics or collision scale.

Scale Min

Sets one end of the size range. Keep the minimum below the maximum.

- **Lowest value:** Sets that endpoint to a very small size; zero allows collapsed elements and negative scale can mirror them.
- **Highest value:** Sets that endpoint to a larger size, allowing or forcing larger elements depending on the other endpoint.

Scale Max

Sets one end of the size range. Keep the minimum below the maximum.

- **Lowest value:** Sets that endpoint to a very small size; zero allows collapsed elements and negative scale can mirror them.
- **Highest value:** Sets that endpoint to a larger size, allowing or forcing larger elements depending on the other endpoint.

Simulation Substeps

Substeps for surface motion and ballistic dripping. Splits each frame step into smaller surface-motion and ballistic integration steps.

- **Lowest value:** One step is cheaper but less accurate for fast movement or contact.
- **Highest value:** Sixteen substeps improve temporal resolution at a higher simulation cost.

Gravity

Gravity strength used by moving surface droplets and airborne drips. Controls the downward gravity contribution.

- **Lowest value:** Zero removes this gravity contribution; wind and initial velocity may still move particles.
- **Highest value:** Produces stronger downward acceleration or motion.

Path Meander

Coherent lateral variation, not independent frame-by-frame random kicks. Controls the strength of the small details.

- **Lowest value:** Zero removes this variation.
- **Highest value:** Makes the variation more pronounced; large displacement values may overwhelm the base shape or path.

Path Wavelength

Spatial size of path variations. Sets the spatial length scale of lateral surface-flow variations.

- **Lowest value:** Produces tightly spaced direction changes.
- **Highest value:** Produces broad, slowly varying bends along the surface.

Flow Response

Velocity relaxation rate per second. Sets the rate at which surface-droplet velocity relaxes toward its requested flow velocity.

- **Lowest value:** Low values respond slowly and preserve more previous velocity.
- **Highest value:** High values follow the requested velocity quickly.

Maximum Surface Speed

Caps requested surface-flow speed of droplets attached to the surface; it is not an airborne-drip speed cap.

- **Lowest value:** Keeps surface motion very slow.

- **Highest value:** Allows much faster surface flow.

Growth Time

Time available for a newly spawned simulated-droplet head to grow from its starting size to its full size.

- **Lowest value:** Reaches full size almost immediately.
- **Highest value:** Grows more gradually, reducing sudden full-size spawning.

Growth Distance

Distance a new head must travel to reach full size. Used alongside time-based head growth; zero disables the distance-based contribution.

- **Lowest value:** At 0, leaves growth governed by Growth Time.
- **Highest value:** Requires more travel before reaching full size.

Start Scale

Initial size of the new droplet head and trail at the moment the head forms.

- **Lowest value:** Starts the new head and trail very small.
- **Highest value:** Starts them closer to their eventual full size, reducing visible growth.

Trail Follow Head

Sets how recorded trail samples respond to head growth.

- **Lowest value:** At 0, each trail sample preserves the head scale it had when deposited; older samples remain thin.
- **Highest value:** At 1, the whole trail follows the current head scale, thickening together with the growing head.

Wind

World-axis wind velocity, in scene units per second.

- **Zero or negative components:** Zero adds no direction. Negative components point along the negative axes.
- **Positive or larger components:** Positive components point along positive axes. Check the preview to see how strongly the direction affects motion.

Wind Strength

Wind influence; surface-film response is intentionally weaker than airborne response. Controls how strongly this movement affects the effect.

- **Lowest value:** Zero removes this movement. Negative values, if available, reverse its direction.
- **Highest value:** Makes the movement or deformation more noticeable.

Trail Disappear Speed

Multiplies the legacy wet-trail disappearance rate.

- **Lowest value:** Zero removes the speed contribution to drying.
- **Highest value:** Makes trails disappear faster; Trail Fade Time still affects their fade.

Trail Hold Time

Seconds before deposited trail samples start drying. Sets how long deposited trails or landed particles persist before fading.

- **Lowest value:** Starts drying or removal almost immediately.
- **Highest value:** Keeps the deposited effect visible much longer.

Trail Fade Time

Seconds of smooth drying. Legacy disappearance speed scales this, independently of Gravity. Sets the duration of the scale or drying fade.

- **Lowest value:** The effect disappears quickly.
- **Highest value:** The effect disappears gradually over a longer interval.

Max Trail Points

Safety cap on stored wet history; lower values may shorten trails. Limits how many elements can be active or retained.

- **Lowest value:** Restricts the layer to the smallest permitted budget; zero prevents that layer when supported.
- **Highest value:** Allows more particles or stored samples, increasing memory use; does not create particles by itself.

Vertex Group

Exact vertex-group name whose weights control droplet emission.

- **Unassigned or empty:** Leave this unassigned and any effect that needs this source may disappear.
- **Assigned or named:** Choose the object, collection, material, or named mask that should drive the effect.

Edge Detach Distance

Tolerance for a moving surface droplet to remain attached when it encounters an edge or cannot closely follow the surface. This control is in the Dripping section.

- **Lowest value:** Even small changes, bumps or projection errors can cause droplets to detach into airborne drips.
- **Highest value:** Allows larger surface deviations before detachment, keeping droplets attached more readily.

Dripping Radius

Changes the visible thickness of the elements.

- **Lowest value:** Produces very small or thin elements; zero can collapse them if allowed.
- **Highest value:** Produces large or thick elements; this may overlap surrounding geometry.

Drip Tip Length

Tail length in droplet radii, matching the procedural Dripping-style droplet shape. Controls how far the droplet tip or tail extends.

- **Lowest value:** Zero gives a rounder shape with no added tail length.
- **Highest value:** Produces a long, narrow trailing extension.

Drip Tip Randomness

Stable per-droplet variation of Tip Length. Controls how much individual elements differ from one another.

- **Lowest value:** Makes the elements more uniform.
- **Highest value:** Makes the elements differ more from one another.

Drip Tip Stretch Time

Seconds after detachment for the procedural tail to reach full length. Sets the time for the procedural trailing tip to grow to its full length.

- **Lowest value:** The tip reaches full extension quickly.
- **Highest value:** The tip grows slowly; short-lived droplets may never reach full extension.

Drip Tip Sharpness

Tapers only the trailing hemisphere toward the procedural tip. Controls tapering toward the procedural droplet tip.

- **Lowest value:** Gives a blunt or rounded trailing end.
- **Highest value:** Gives a sharper, narrower tip; built-in limits may prevent a fully collapsed tip.

Drip Segments

UV-sphere segment count for each procedural dripping droplet. Controls how smooth or detailed this part of the effect appears.

- **Lowest value:** Shows a simpler, rougher shape.
- **Highest value:** Shows a smoother shape, which can slow down the scene.

Drip Rings

UV-sphere ring count for each procedural dripping droplet. Controls how smooth or detailed this part of the effect appears.

- **Lowest value:** Shows a simpler, rougher shape.
- **Highest value:** Shows a smoother shape, which can slow down the scene.

Drip Air Drag

World-velocity damping rate and wind coupling, per second. Damps particle velocity and, where implemented, couples it to wind.

- **Lowest value:** Low drag preserves momentum longer.
- **Highest value:** Strong drag quickly damps relative movement; particles respond more strongly to the wind coupling.

Drip Lifetime

Finite flight duration; replaces the original local Z=-0.5 deletion plane. Controls particle lifetime or one endpoint of its randomized lifetime.

- **Lowest value:** Shortens visible particle persistence; zero can remove the lifetime where accepted.
- **Highest value:** Keeps particles alive longer, which can raise the visible population. Check the generator notes for time units.

Max Dripping Particles

Limits how many elements can be active or retained.

- **Lowest value:** Restricts the layer to the smallest permitted budget; zero prevents that layer when supported.
- **Highest value:** Allows more particles or stored samples, increasing memory use; does not create particles by itself.

Enable Surface Collision

Avoid Drip Colliders while attached droplets move over the emitter. Collider contact also promotes touched static small droplets into dynamic falling droplets.

- **Off:** Attached droplets ignore the optional surface collider-avoidance mode.
- **On:** Avoids Drip Colliders on the surface; collider contact can promote touched small static droplets into dynamic falling drops.

Enable Drip Collision

Detects contact between detached airborne drips and the selected collider collection. When Collision Splashes is disabled, drips are removed on collision; when enabled, impacts can emit splash particles.

- **Off:** Falling drips do not use the optional collider-impact behavior.
- **On:** Detects airborne drip collisions and removes or splashes the drips depending on the Collision Splashes setting.

Collision Splashes

Controls whether impacts detected by Enable Drip Collision emit secondary particles instead of only removing the impacting drip.

- **Off:** Colliding airborne drips are removed without secondary splash particles.
- **On:** A detected impact can generate a single burst of splash particles, configured by Splash Count, Splash Speed, Splash Lift, Splash Lifetime and Splash Radius.

Drip Colliders

Collection of meshes used by both attached-drop avoidance and airborne drip or splash collision. Exclude the generator host.

- **Unassigned or empty:** Leave this unassigned and any effect that needs this source may disappear.
- **Assigned or named:** Choose the object, collection, material, or named mask that should drive the effect.

Collision Margin

Additional swept-contact padding. Sets additional surface clearance, particle contact size or contact padding as named.

- **Lowest value:** Allows movement or rendering close to the surface, with less padding.
- **Highest value:** Creates a larger gap or contact region; excessive padding can prevent passage through small spaces.

Splash Count

Secondary particles per primary impact; never recursively emit more splashes. Sets secondary droplet count per primary impact.

- **Lowest value:** Zero creates no secondary droplets.
- **Highest value:** Creates the maximum secondary count per impact, increasing output cost.

Splash Speed

Base tangential splash velocity. Sets the outgoing secondary-particle speed; in the surface-drop simulation it is the tangential component.

- **Lowest value:** Zero removes this outward or tangential speed contribution; lift and gravity can still move splashes.
- **Highest value:** Throws secondary droplets farther or faster.

Splash Lift

Outward normal splash velocity. Sets the velocity away from the contacted surface normal.

- **Lowest value:** Zero removes the outward-normal launch component.

- **Highest value:** Launches splash droplets strongly away from the surface.

Splash Lifetime

Controls particle lifetime or one endpoint of its randomized lifetime.

- **Lowest value:** Shortens visible particle persistence; zero can remove the lifetime where accepted.
- **Highest value:** Keeps particles alive longer, which can raise the visible population. Check the generator notes for time units.

Splash Radius

Icosphere particle radius. Changes the visible thickness of the elements.

- **Lowest value:** Produces very small or thin elements; zero can collapse them if allowed.
- **Highest value:** Produces large or thick elements; this may overlap surrounding geometry.

Splash Drag

Damps particle velocity and, where implemented, couples it to wind.

- **Lowest value:** Low drag preserves momentum longer.
- **Highest value:** Strong drag quickly damps relative movement; particles respond more strongly to the wind coupling.

Max Splash Particles

Limits how many elements can be active or retained.

- **Lowest value:** Restricts the layer to the smallest permitted budget; zero prevents that layer when supported.
- **Highest value:** Allows more particles or stored samples, increasing memory use; does not create particles by itself.

Droplet Material

Source material for the normal droplet-material mode.

- **Unassigned or empty:** Leave this unassigned and any effect that needs this source may disappear.
- **Assigned or named:** Choose the object, collection, material, or named mask that should drive the effect.

Use Custom Material

Switches from the supplied droplet material to the material chosen in the Custom Material input, which is exposed in the effect panel when this option is enabled.

- **Off:** Uses the source or default material mode.
- **On:** Uses the assigned custom material for the applicable output.

Custom Material

Override material selected when Use Custom Material is enabled.

- **Unassigned or empty:** Leave this unassigned and any effect that needs this source may disappear.
- **Assigned or named:** Choose the object, collection, material, or named mask that should drive the effect.

Realize Instances

Turns generated droplets into mesh geometry when another effect or export requires it.

- **Off:** Preserves instances in the visible result.

- **On:** Converts generated copies into mesh geometry for further effects or export. This can slow down the scene.

After setting it to ON - press - "Convert to mesh" to turn generator into 3d mesh

5. Dust on Object

Adds dust and fine fibers to a surface. Adjust their coverage, size, and placement with the controls below.

Setup and behavior notes

- Several dust layers have their own Density and Scale settings. Adjust the layer whose appearance you want to change.
- Master Density adds layer density; Layer density + Master density. master density can be -X values
- Remove from Vertical uses world slope: Vertical Angle 0° permits horizontal surfaces; 90° permits vertical slopes up to that threshold.
- Draw Only is an exclusive painted-area mode. Keep Surface Mask Inside determines whether the ordinary masks still apply within that area.
- Each fiber layer has its own switch. Enable the layers you want to see.

Practical workflow — Fast Dust and Detailed Dust

Choose **Fast Dust** for lightweight textured-plane instances or **Detailed Dust** for low-poly three-dimensional dust pieces. During import, the optional **Dust Material** adds a gray dust appearance to the target; it is the standard Dust material, not **Pro Dust**. On large targets, enable **Fast Mode** to preview point proxies instead of detailed instances and avoid heavy viewport evaluation. Disable it to inspect the finished dust geometry.

Start with **Master Density** and **Height Sensivity** (the UI spelling): increasing height sensitivity confines dust more strongly to upward-facing areas. **Remove from Vertical** plus **Vertical Angle** excludes steep faces, useful for keeping dust on the tops of shelves or ledges. The asset includes small dust particles and multiple fibers/hairs; tweak each layer's density, thickness, and size separately. If long fibers are invisible, increase their size range, **Strand Size**, or **Hair Thickness**.

For manual placement, use **Draw / Erase Dust** and paint the resulting vertex-group mask on a sufficiently subdivided target. **Enable Draw Only** isolates the painted area; **Keep Surface Mask Inside** determines whether orientation and other ordinary masks also affect that area. Fine-tune **Noise Mask Scale** and **Noise Mask Contrast** for less uniform dust coverage.

Settings

Fast Mode

Turns this option on or off.

- **Off:** Uses the regular generated shapes mode.
- **On:** Uses the simplified preview or fast mode; the exact representation depends on the generator.

Master Density

An overall density adjustment shared across dust-particle and fiber layers; layer-specific Density values provide additional control.

- **Lowest value:** Negative values reduce the combined density; around **−1** the generator can produce no dust.
- **Highest value:** Increases the combined dust/fiber coverage, subject to masks and evaluation cost.

Height Sensivity

Sets the lower threshold of the blurred upward-surface mask remapped into dust coverage.

- **Lowest value:** At 0, admits a wider range of the orientation mask.
- **Highest value:** At 1, makes the mask interval collapse toward the fully upward value, heavily restricting coverage.

Remove from Vertical

Exclude surface points whose world slope exceeds Vertical Angle.

- **Off:** Keeps eligible points on steep surfaces.
- **On:** Filters points according to the generator-specific normal or slope threshold.

Vertical Angle (degrees)

Maximum allowed slope: 0 = horizontal; 90 = vertical. Steeper surfaces are excluded. Sets the maximum allowed world-space slope when Remove from Vertical is enabled.

- **Lowest value:** At 0°, retains horizontal surfaces only within the comparison tolerance.
- **Highest value:** At 90°, admits slopes through vertical; steeper or downward surfaces are still subject to the signed-normal logic.

Seed

Changes the random arrangement. The same seed gives the same result when the other settings stay the same.

- **Lowest value:** Selects one repeatable arrangement.
- **Highest value:** Selects a different repeatable arrangement.

Material

Choose the material used by the generated elements.

- **Unassigned or empty:** Leave this unassigned and any effect that needs this source may disappear.
- **Assigned or named:** Choose the object, collection, material, or named mask that should drive the effect.

Density

Controls how densely the elements appear.

- **Lowest value:** Zero suppresses this emission layer. Negative density is not a meaningful negative particle count.
- **Highest value:** Creates a denser population, subject to spacing, masks and particle budgets; increases evaluation cost.

Dust Scale Max

Sets one end of the size range. Keep the minimum below the maximum.

- **Lowest value:** Sets that endpoint to a very small size; zero allows collapsed elements and negative scale can mirror them.
- **Highest value:** Sets that endpoint to a larger size, allowing or forcing larger elements depending on the other endpoint.

Dust Scale Min

Sets one end of the size range. Keep the minimum below the maximum.

- **Lowest value:** Sets that endpoint to a very small size; zero allows collapsed elements and negative scale can mirror them.
- **Highest value:** Sets that endpoint to a larger size, allowing or forcing larger elements depending on the other

endpoint.

Noise Mask Scale

Scales the spatial coordinates used by the named noise.

- **Lowest value:** Low positive values make broad features; zero makes spatial coordinates constant.
- **Highest value:** Makes fine, tightly spaced noise features; negative values reverse coordinate scale.

Noise Mask Contrast

Changes the remapping contrast of the procedural distribution mask.

- **Lowest value:** Produces a softer or less selective mask.
- **Highest value:** Produces stronger contrast or selection. Exact occupied area also depends on the mask thresholds and offset.

Density

Controls how densely the elements appear.

- **Lowest value:** Zero suppresses this emission layer. Negative density is not a meaningful negative particle count.
- **Highest value:** Creates a denser population, subject to spacing, masks and particle budgets; increases evaluation cost.

Hair Thickness

Changes the visible thickness of the elements.

- **Lowest value:** Produces very small or thin elements; zero can collapse them if allowed.
- **Highest value:** Produces large or thick elements; this may overlap surrounding geometry.

Hair Resolution

Controls how smooth or detailed this part of the effect appears.

- **Lowest value:** Shows a simpler, rougher shape.
- **Highest value:** Shows a smoother shape, which can slow down the scene.

Noisify Scale

Controls the size of the visible pattern without changing the size of the individual objects.

- **Lowest value:** Creates broader, smoother patterns.
- **Highest value:** Creates smaller, more closely spaced details.

Scale Min

Sets one end of the size range. Keep the minimum below the maximum.

- **Lowest value:** Sets that endpoint to a very small size; zero allows collapsed elements and negative scale can mirror them.
- **Highest value:** Sets that endpoint to a larger size, allowing or forcing larger elements depending on the other endpoint.

Scale Max

Sets one end of the size range. Keep the minimum below the maximum.

- **Lowest value:** Sets that endpoint to a very small size; zero allows collapsed elements and negative scale can mirror them.
- **Highest value:** Sets that endpoint to a larger size, allowing or forcing larger elements depending on the other endpoint.

Enable Procedural Hair

Third dust layer: procedural hair strands Turns this option on or off.

- **Off:** Omits this dust or hair layer.
- **On:** Adds this dust or hair layer, subject to its density, scale and masks.

Strand Size

Overall strand size in world units Changes the size of the generated elements.

- **Lowest value:** Zero collapses the affected geometry. Negative values, if accepted, can mirror it.
- **Highest value:** Makes the affected geometry much larger; this is a visual scale and is not automatically a physics or collision scale.

Profile Resolution

Sides of the strand cross section Controls how smooth or detailed this part of the effect appears.

- **Lowest value:** Shows a simpler, rougher shape.
- **Highest value:** Shows a smoother shape, which can slow down the scene.

Tip Taper

Part of the strand over which it thins to the tip Sets the portion of each procedural hair strand that thins toward its tip.

- **Lowest value:** Limits taper to a small end region.
- **Highest value:** Extends taper over a larger portion of the strand.

Curl Noise Scale

Controls the size of the visible pattern without changing the size of the individual objects.

- **Lowest value:** Creates broader, smoother patterns.
- **Highest value:** Creates smaller, more closely spaced details.

Curl Noise Strength

How strongly the strand curls Controls how strongly this movement affects the effect.

- **Lowest value:** Zero removes this movement. Negative values, if available, reverse its direction.
- **Highest value:** Makes the movement or deformation more noticeable.

Curl Noise Detail

Adds higher-frequency layers to the named noise texture.

- **Lowest value:** Uses only the base noise layer.
- **Highest value:** Adds finer noise detail and may increase evaluation cost.

Smooth Strands

Smooth NURBS strands instead of polylines Turns this option on or off.

- **Off:** Uses polyline strands.
- **On:** Uses smoother NURBS strands.

Density

Controls how densely the elements appear.

- **Lowest value:** Zero suppresses this emission layer. Negative density is not a meaningful negative particle count.
- **Highest value:** Creates a denser population, subject to spacing, masks and particle budgets; increases evaluation cost.

Scale Min

Sets one end of the size range. Keep the minimum below the maximum.

- **Lowest value:** Sets that endpoint to a very small size; zero allows collapsed elements and negative scale can mirror them.
- **Highest value:** Sets that endpoint to a larger size, allowing or forcing larger elements depending on the other endpoint.

Scale Max

Sets one end of the size range. Keep the minimum below the maximum.

- **Lowest value:** Sets that endpoint to a very small size; zero allows collapsed elements and negative scale can mirror them.
- **Highest value:** Sets that endpoint to a larger size, allowing or forcing larger elements depending on the other endpoint.

Enable Draw Only

Dust ONLY where the Draw Only vertex group is painted - replaces every other mask Turns this option on or off.

- **Off:** Uses the normal combined dust masks.
- **On:** Restricts dust to the Draw Only Vertex Group, replacing other masks unless Keep Surface Mask Inside changes that behavior.

Draw Only Vertex Group

Weight Paint vertex group defining the exclusive dust area.

- **Unassigned or empty:** Leave this unassigned and any effect that needs this source may disappear.
- **Assigned or named:** Choose the object, collection, material, or named mask that should drive the effect.

Draw Mask Vertex Group

Weight Paint vertex group adding dust density.

- **Unassigned or empty:** Leave this unassigned and any effect that needs this source may disappear.
- **Assigned or named:** Choose the object, collection, material, or named mask that should drive the effect.

Keep Surface Mask Inside

Controls whether the Height, Draw, and Erase masks also apply inside the painted area.

- **Off:** Uses the Draw Only painted weight alone inside the selected area.
- **On:** Also applies the Height, Draw, and Erase masks inside the painted area.

6. Falling Sparkles

Creates falling sparkles from an emitter. Control how many appear, how they move, and how they react to other objects.

Setup and behavior notes

- Continuous Spawn uses Spawn Rate per simulation frame, not per second.
- Collision Flow shrinks points toward zero size at collider surfaces. Crossing to the back side of a collider face permanently removes them from simulation state; face-normal orientation matters.

Practical workflow — Dynamic / Falling Sparkles

Place the 3D Cursor at the origin of the sparks and add the effect; the emitter aligns to the normal of a nearby surface. Choose a one-time emission or **Continuous Spawn**, then specify the **Start Frame**. Confirm the current frame or select another frame when prompted. Play the timeline through the start frame so the simulation has a chance to evaluate.

Use **Spawn Rate** for repeated emission, then **Density**, **Scale**, **Scale Min/Max**, and **Age Min/Max** for population, size, and lifetime variation. **Velocity** controls the initial launch; **Spread** changes a tight jet into a wider burst. **Turbulence** introduces scattered flight paths. In the simplified physics workflow, **Weight** is the main vertical force: positive values make sparks fall, while negative values make them rise. **Direction** and **Random Movement** modify the motion further. Assign a collider where sparks should disappear on contact rather than pass through the object; the procedural appearance controls adjust strand density, thickness, and length.

Settings

Seed

Changes the random arrangement. The same seed gives the same result when the other settings stay the same.

- **Lowest value:** Selects one repeatable arrangement.
- **Highest value:** Selects a different repeatable arrangement.

Start Frame

Sets the timeline frame when emission starts.

- **Lowest value:** Begins earlier on the timeline.
- **Highest value:** Begins later on the timeline; no start event occurs before that frame.

Emitter

Object whose evaluated surface emits spark particles.

- **Unassigned or empty:** Leave this unassigned and any effect that needs this source may disappear.
- **Assigned or named:** Choose the object, collection, material, or named mask that should drive the effect.

Continuous Spawn

Keep emitting new spark particles after Continuous Start Frame. When disabled, the original one-shot Start Frame behavior is preserved.

- **Off:** Uses the original one-shot Start Frame emission.
- **On:** Emits additional batches from Continuous Start Frame onward, scaled by Spawn Rate.

Continuous Start Frame

First frame on which continuous emission becomes active. Sets the timeline frame when emission starts.

- **Lowest value:** Begins earlier on the timeline.
- **Highest value:** Begins later on the timeline; no start event occurs before that frame.

Spawn Rate

Continuous emission amount per simulation frame. 1.0 emits one full Density batch every frame; 0.5 emits half that density; 2.0 emits twice it. Multiplies the continuous emission batch per simulation frame. A value of 1 emits one Density batch each frame.

- **Lowest value:** Zero emits no new continuous particles.
- **Highest value:** Requests many Density batches each frame, increasing particle count and simulation cost.

Density

Controls how densely the elements appear.

- **Lowest value:** Zero suppresses this emission layer. Negative density is not a meaningful negative particle count.
- **Highest value:** Creates a denser population, subject to spacing, masks and particle budgets; increases evaluation cost.

Scale

Changes the size of the generated elements.

- **Lowest value:** Zero collapses the affected geometry. Negative values, if accepted, can mirror it.
- **Highest value:** Makes the affected geometry much larger; this is a visual scale and is not automatically a physics or collision scale.

Scale Min

Sets one end of the size range. Keep the minimum below the maximum.

- **Lowest value:** Sets that endpoint to a very small size; zero allows collapsed elements and negative scale can mirror them.
- **Highest value:** Sets that endpoint to a larger size, allowing or forcing larger elements depending on the other endpoint.

Scale Max

Sets one end of the size range. Keep the minimum below the maximum.

- **Lowest value:** Sets that endpoint to a very small size; zero allows collapsed elements and negative scale can mirror them.
- **Highest value:** Sets that endpoint to a larger size, allowing or forcing larger elements depending on the other endpoint.

Age Min

Sets one endpoint of the randomized spark lifetime in simulation steps or frames. Stored age increments by 1 per simulation step.

- **Lowest value:** A small or zero lifetime removes sparks rapidly after emission.
- **Highest value:** A longer step lifetime keeps sparks alive longer and increases retained simulation population.

Age Max

Sets one endpoint of the randomized spark lifetime in simulation steps or frames. Stored age increments by 1 per simulation step.

- **Lowest value:** A small or zero lifetime removes sparks rapidly after emission.
- **Highest value:** A longer step lifetime keeps sparks alive longer and increases retained simulation population.

Velocity

Controls the initial movement speed of falling sparks.

- **Lowest value:** Zero removes the initial velocity contribution.
- **Highest value:** Launches sparks faster; gravity and turbulence still affect their trajectories.

Spread

Controls how widely the elements spread through the scene.

- **Lowest value:** Zero or a low value concentrates elements close to the guide, center or base direction.
- **Highest value:** Spreads elements farther apart or across a wider region; domain clipping can limit the result.

Turbulence

Controls how strongly this movement affects the effect.

- **Lowest value:** Zero removes this movement. Negative values, if available, reverse its direction.
- **Highest value:** Makes the movement or deformation more noticeable.

Turbulence scale

Controls the size of the visible pattern without changing the size of the individual objects.

- **Lowest value:** Creates broader, smoother patterns.
- **Highest value:** Creates smaller, more closely spaced details.

Weight

Weights the falling-spark movement or gravity contribution.

- **Lowest value:** Zero removes the weighted contribution.
- **Highest value:** Increases the weighted contribution; the resulting trajectory also depends on Gravity and Velocity.

Direction

Sets which way the elements move. Use the related Strength or Speed control to change the intensity.

- **Zero or negative components:** Zero adds no direction. Negative components point along the negative axes.
- **Positive or larger components:** Positive components point along positive axes. Check the preview to see how strongly the direction affects motion.

Random Movement

Controls how strongly this movement affects the effect.

- **Lowest value:** Zero removes this movement. Negative values, if available, reverse its direction.
- **Highest value:** Makes the movement or deformation more noticeable.

Collision

Enable smooth shrinking near colliders and permanently delete spark points after they cross to the back side of a collider face.

- **Off:** Bypasses the optional collider mode.
- **On:** Uses the assigned collider geometry for the generator-specific collision or fade behavior.

Colliders

Choose the objects that particles should collide with.

- **Unassigned or empty:** Leave this unassigned and any effect that needs this source may disappear.
- **Assigned or named:** Choose the object, collection, material, or named mask that should drive the effect.

Collision Flow

Smooth fade distance before the collider surface. At the surface scale reaches zero; after crossing to a backface the point is removed from the simulation state. Sets the distance over which particles shrink or fade near the associated surface.

- **Lowest value:** Produces a narrow transition, retaining full size until close to the surface.
- **Highest value:** Produces a broad transition, so particles start shrinking farther away and may stay small throughout a small domain.

Spark Thickness

Diameter of the procedural spark streak before the existing Scale and minimum and maximum variation are applied. Changes the visible thickness of the elements.

- **Lowest value:** Produces very small or thin elements; zero can collapse them if allowed.
- **Highest value:** Produces large or thick elements; this may overlap surrounding geometry.

Trail Stretch

Multiplier for the physical distance travelled by the particle. Higher values make faster sparks leave longer streaks. Scales the extra streak or droplet elongation produced by motion.

- **Lowest value:** Zero removes motion-driven elongation.
- **Highest value:** Makes faster elements visibly longer, subject to the maximum length or stretch cap.

Trail Follow Frames

For this many simulation frames the far end stays approximately behind while the spark stretches. Afterwards the tail follows the head. Sets the number of simulation frames during which the streak tail remains approximately behind its head.

- **Lowest value:** The tail starts following sooner.
- **Highest value:** The tail remains behind longer before following.

Minimum Trail Length

Minimum visible spark streak length when motion is very slow. Sets a visible minimum or initial trail length, as named.

- **Lowest value:** Zero removes that minimum or initial length.
- **Highest value:** Creates longer initial or minimum streaks; the maximum trail limit can clamp the result.

Maximum Trail Length

Safety cap for very fast particles so a single spark cannot create an excessively long streak. Caps the visible trail length.

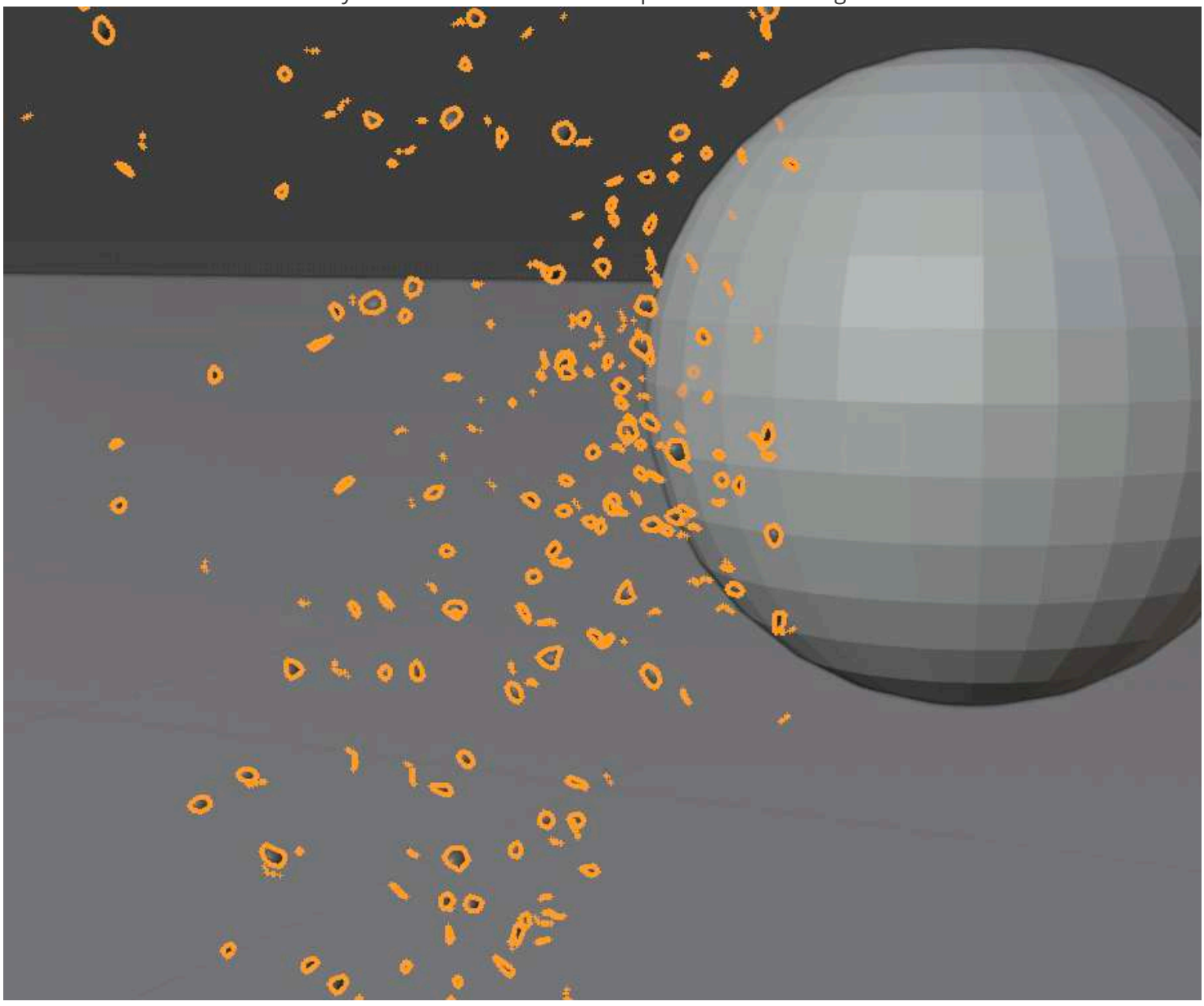
- **Lowest value:** Keeps only a very short trail.
- **Highest value:** Permits a much longer trail; available history or path length can still limit the result.

7. Flying Ashes — Emitter

Emits flying ash from a surface. Control its amount, movement, fade, and interaction with nearby objects.

Setup and behavior notes

- Count is converted into emitter-surface density from face area; it is a target live population rather than a guaranteed count of exactly N particles on every frame.
- Lifetime is a repeating travel cycle. Fade Time hides particle birth and death recycling.
- Size Collision changes the collider response. Avoidance Strength = 0 still retains solid surface contacts in reflect mode.
- Choose a built-in style with Ash Type.
- ! Collision is made mathematically and is not simulated - so expect some flickering



Practical workflow — emitter-based ashes

Use **Flying Ashes — Emitter** when ash should travel outward from a particular source instead of being distributed inside a volume. Its default source is a circle, but you can assign your own emitter geometry. Adjust **Lifetime** for how long a particle exists and **Fade Time** for how gradually it appears or vanishes. The motion and instance controls are similar to the domain version: upward speed, wind, turbulence, size variation, and spin.

For obstacle avoidance, first create and assign a collection of collider meshes. Only then adjust the available avoidance controls. **Avoidance Distance** sets how early particles respond; collision clearance/radius determines how close they can approach; **Avoidance Strength** affects deflection; a look-ahead setting, where available, anticipates obstacles farther ahead. If there is an option to remove particles inside collider meshes, enable it to avoid visible intersection when steering alone is insufficient. Avoidance is a procedural approximation and should be checked at the final shot scale.

Settings

Ash Type

Selects a discrete processing mode. Available choices: `Hot - Emissive`, `Dry`, `Mixed`, `Mixed Low-poly`.

- **Hot - Emissive:** Shows the glowing ash style.
- **Dry:** Shows the dry ash style.
- **Mixed:** Shows a mix of ash styles.
- **Mixed Low-poly:** Shows the simpler mixed ash style.

Count

Target live particle population. Converted to surface density from the emitter face area. Sets how many elements the effect tries to create.

- **Lowest value:** Creates few elements, or none when zero is allowed.
- **Highest value:** Creates more elements, though masks or spacing may limit how many fit.

Seed

Changes the random arrangement. The same seed gives the same result when the other settings stay the same.

- **Lowest value:** Selects one repeatable arrangement.
- **Highest value:** Selects a different repeatable arrangement.

Vertical Speed

Adds signed vertical travel.

- **Lowest value:** Negative values move particles downward.
- **Highest value:** Positive values move particles upward; zero removes this vertical contribution.

Wind Direction

Points the wind in a chosen direction. Wind Strength controls how hard it blows.

- **Zero or negative components:** Zero adds no direction. Negative components point along the negative axes.
- **Positive or larger components:** Positive components point along positive axes. Use the Strength setting to change the effect's intensity.

Wind Strength

Controls how strongly this movement affects the effect.

- **Lowest value:** Zero removes this movement. Negative values, if available, reverse its direction.
- **Highest value:** Makes the movement or deformation more noticeable.

Wind Variation

Controls how much individual elements differ from one another.

- **Lowest value:** Makes the elements more uniform.
- **Highest value:** Makes the elements differ more from one another.

Flow Speed

Controls the speed of the animated movement.

- **Lowest value:** Zero freezes the time-driven part; static offsets and other independent animation may remain.
- **Highest value:** Advances procedural motion much faster.

Turbulence Strength

Controls how strongly this movement affects the effect.

- **Lowest value:** Zero removes this movement. Negative values, if available, reverse its direction.
- **Highest value:** Makes the movement or deformation more noticeable.

Turbulence Scale

Controls the size of the visible pattern without changing the size of the individual objects.

- **Lowest value:** Creates broader, smoother patterns.
- **Highest value:** Creates smaller, more closely spaced details.

Turbulence Speed

Controls how quickly this part of the effect changes over time.

- **Lowest value:** Zero stops change in this contribution; a static displacement or rotation may remain. Negative values reverse time where supported.
- **Highest value:** Produces rapid changes or rotation; may appear abrupt at the current frame rate.

Fine Turbulence

Controls how strongly this movement affects the effect.

- **Lowest value:** Zero removes this movement. Negative values, if available, reverse its direction.
- **Highest value:** Makes the movement or deformation more noticeable.

Fine Turbulence Speed

Controls how quickly this part of the effect changes over time.

- **Lowest value:** Zero stops change in this contribution; a static displacement or rotation may remain. Negative values reverse time where supported.
- **Highest value:** Produces rapid changes or rotation; may appear abrupt at the current frame rate.

Swirl Strength

Controls how strongly this movement affects the effect.

- **Lowest value:** Zero removes this movement. Negative values, if available, reverse its direction.
- **Highest value:** Makes the movement or deformation more noticeable.

Individual Variation

Controls how much individual elements differ from one another.

- **Lowest value:** Makes the elements more uniform.
- **Highest value:** Makes the elements differ more from one another.

Instance Collection

This setting currently has no visible effect.

- **Unassigned or empty:** No visible change in this version.
- **Assigned or named:** No visible change in this version.

Instance Scale

Changes the size of the generated elements.

- **Lowest value:** Zero collapses the affected geometry. Negative values, if accepted, can mirror it.
- **Highest value:** Makes the affected geometry much larger; this is a visual scale and is not automatically a physics or collision scale.

Scale Variation

Controls how much individual elements differ from one another.

- **Lowest value:** Makes the elements more uniform.
- **Highest value:** Makes the elements differ more from one another.

Spin Speed

Controls how quickly this part of the effect changes over time.

- **Lowest value:** Zero stops change in this contribution; a static displacement or rotation may remain. Negative values reverse time where supported.
- **Highest value:** Produces rapid changes or rotation; may appear abrupt at the current frame rate.

Point Radius

Changes the visible thickness of the elements.

- **Lowest value:** Produces very small or thin elements; zero can collapse them if allowed.
- **Highest value:** Produces large or thick elements; this may overlap surrounding geometry.

Lifetime

Travel cycle length. Particles fade out and restart from the emitter face. Controls particle lifetime or one endpoint of its randomized lifetime.

- **Lowest value:** Shortens visible particle persistence; zero can remove the lifetime where accepted.
- **Highest value:** Keeps particles alive longer, which can raise the visible population. Check the generator notes for time units.

Fade Time

Smooth birth and death scale fade used to hide particle recycling. Sets the duration of the scale or drying fade.

- **Lowest value:** The effect disappears quickly.
- **Highest value:** The effect disappears gradually over a longer interval.

8. Flying Birds

Choose **Domain** to generate flight paths across an area, or **Draw** to guide a flock along a manually created curve. Both modes use the same core bird and formation settings.

Flying Birds — Domain

Fills a scene area with flying birds. Change their paths, formation, and models to shape the flock. Spawn the on the 25meters height

Setup and behavior notes

- Total requested birds are main guides × Amount of Birds per Curve, before path filtering and other output conditions.
- Domain generation uses the mesh bounding box. It is not guaranteed to avoid empty regions of a concave mesh.
- POI placement changes a fraction of existing guide anchors; it does not add extra paths.
- Formation Randomness is stable slot jitter, not per-frame random noise.
- You can assign up to eight different Bird Variant objects.

Practical workflow — Flying Birds (domain)

Add the automatic domain to calculate an area around the scene and raise the birds above it (the example adds roughly **25 m**). A manual domain can instead be resized or edited to confine the flight. The generator creates **flight paths first**, then places birds on them: **Amount of Curves** sets main path count, while **Amount of Birds per Curve** changes how many birds are distributed along each one. Enable path/debug visualization when you need to inspect the layout.

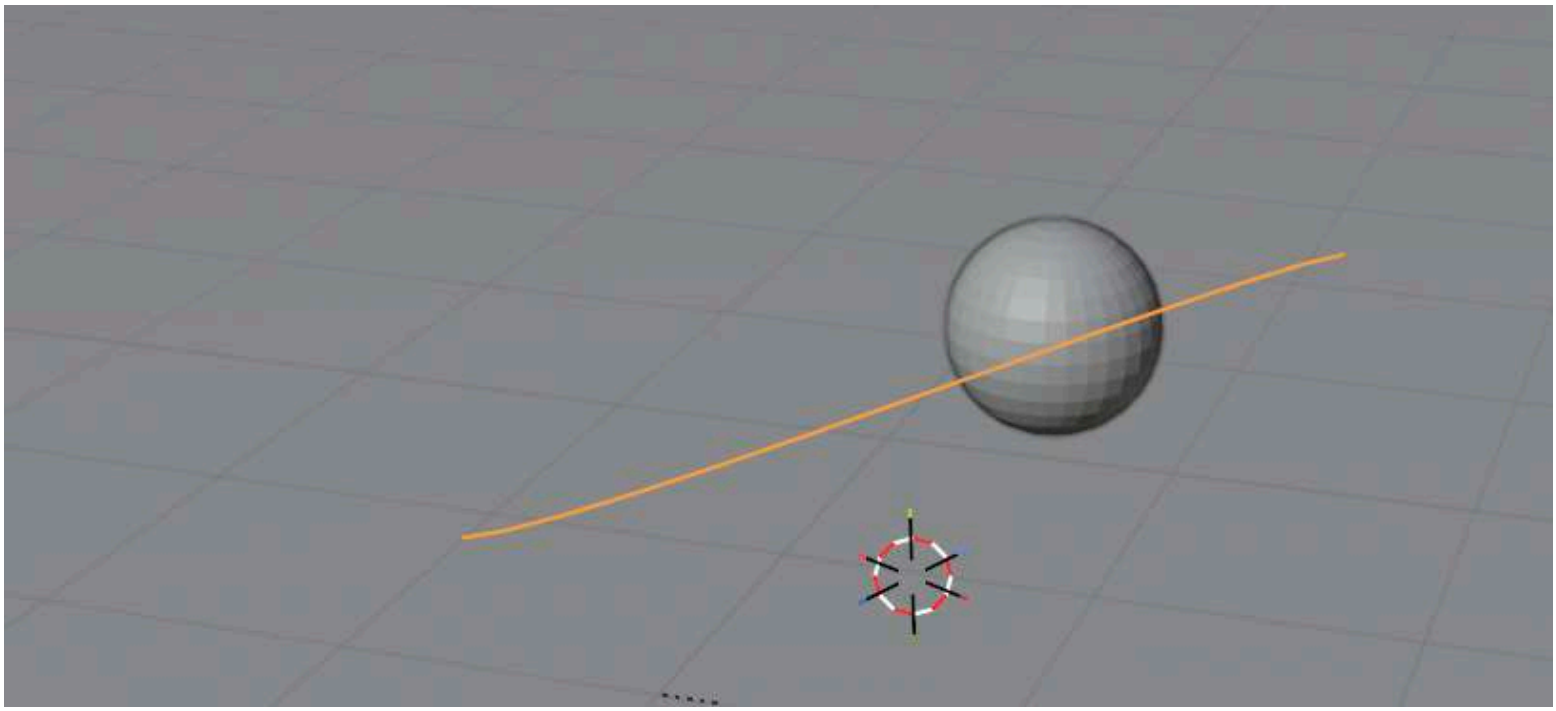
Point of Interest is an Empty used to concentrate flights through a chosen part of the sky. During creation, place the 3D Cursor and confirm with **Enter**, then move the Empty to refine the area. **POI Probability**, **POI Distance**, and **POI Falloff** control how often and how closely flight paths respond. A strong POI influence together with a directed-flight setting can make paths converge into nearly the same line; temporarily disable the POI to inspect the unmodified base-direction behavior. **Random Direction Probability** blends randomized and specified directions; rotating the domain is another easy way to change the overall flow.

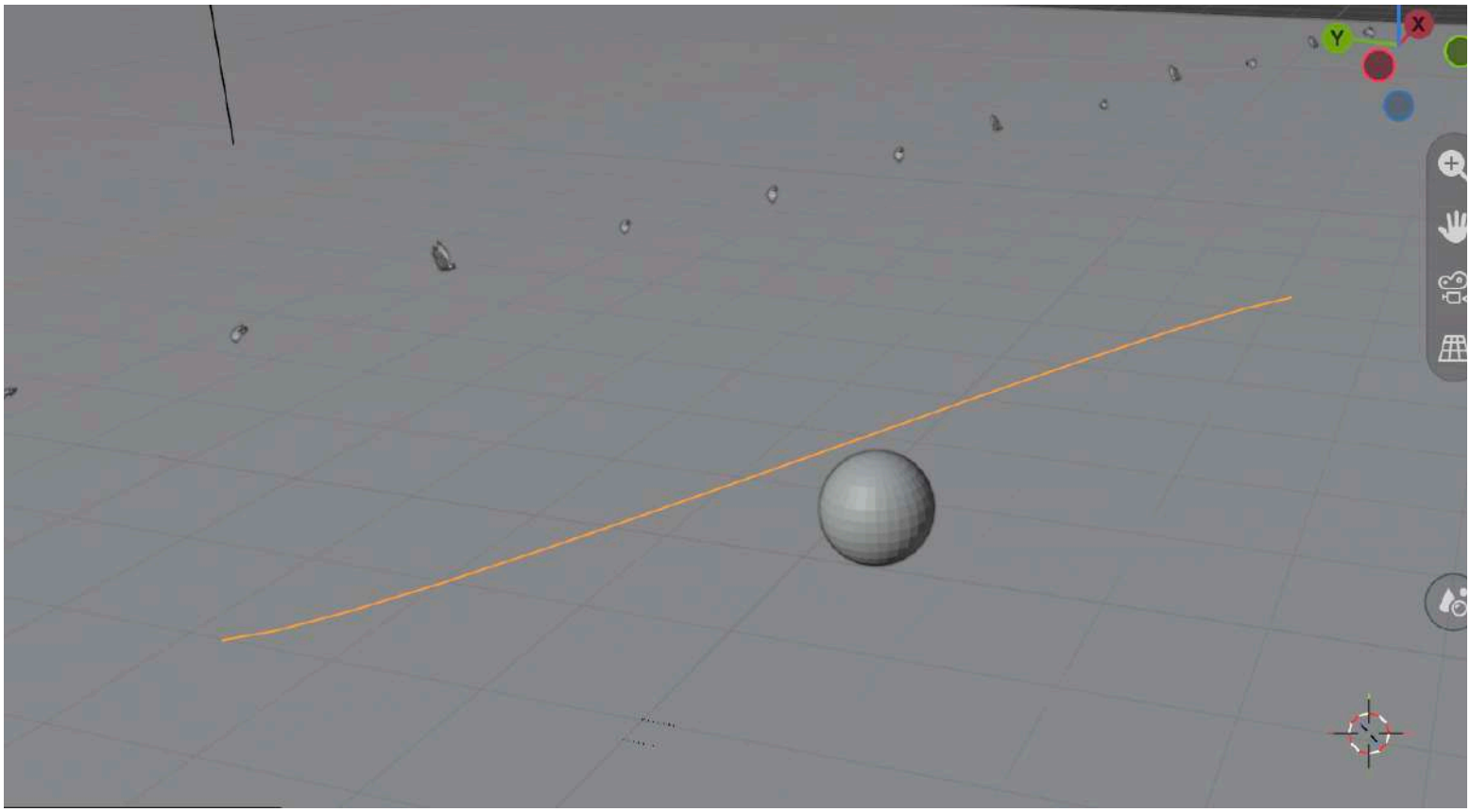
Move Speed Min/Max sets speed variation, **Birds Position** offsets initial placement along each curve, and **Looped** decides whether a bird returns to the path start after leaving. If Looping seems to have no effect in the tutorial preset, set **Birds Position** from its negative start offset (for example, **−500**) back to **0** to expose the difference. Disable **Looped** for a single flyby in which birds do not return. **Constant Speed on All Curves** keeps speed comparable across paths of different lengths; otherwise path length can change perceived velocity.

Birds Spreadness, **Spread Paths**, and **Outside Flock Probability/Range** control separation and outliers. **Bird Formations** supports a loose default, flocks, and organized arrangements such as lines or V-shaped formations. Adjust formation spacing, angle, and randomness together: a little randomness prevents an unnaturally rigid formation. The generator is procedural and does not require simulation baking.

Flying Birds — Draw

Draw flight paths in the viewport, then use the bird controls to choose how the flock follows them.
! Important
Make sure that curve is big enough to spawn birds, if birds are not spawning just scale up your curve path





Practical workflow — drawn bird paths

Choose the drawn-birds action and create the desired flight curve directly in the viewport rather than letting the domain generate the entire route. Assign the bird asset and adjust **Amount of Birds per Curve**, position, **Looped**, flock spread, outside-flock probability, and formation settings in the same way as the domain version. Use drawn paths for a deliberate fly-through or a particular flock silhouette.

If no birds appear, check that the curve is not extremely small or short. Scaling up the curve is often sufficient. The same procedural, non-baked movement system powers both bird generators.

Flying Birds — Shared Settings

The following controls are available in both modes.

Hide Domain Bounds

Turns this option on or off.

- **Off:** Allows domain-bound visualization.
- **On:** Hides the domain bounds; does not remove the domain input from flight calculations.

Spawn in Domain

On: generate fixed-count paths inside the input mesh bounding box. Off: use Guide Curve Object.

- **Off:** Uses Guide Curve Object as the raw manual guide source.
- **On:** Generates fixed-count guide paths from the input mesh bounding box.

Guide Curve Object

Raw manual flight guide curve, used when Spawn in Domain is off. Do not assign the generator host.

- **Unassigned or empty:** Leave this unassigned and any effect that needs this source may disappear.
- **Assigned or named:** Choose the object, collection, material, or named mask that should drive the effect.

Amount of Curves

Number of main guide paths in domain mode. Independent of domain volume. Sets how many elements the effect tries to create.

- **Lowest value:** Creates few elements, or none when zero is allowed.
- **Highest value:** Creates more elements, though masks or spacing may limit how many fit.

Minimal Path Length

Delete domain-generated flight paths whose spline length is shorter than this value. Set to 0 to disable filtering. Filters out generated domain flight paths shorter than this length.

- **Lowest value:** At 0, disables the minimum-length filter.
- **Highest value:** Deletes more short paths and can remove all guides if the threshold exceeds their lengths.

Amount of Birds per Curve

Each main guide generates this many individual bird paths; total is guides multiplied by this value. Sets how many elements the effect tries to create.

- **Lowest value:** Creates few elements, or none when zero is allowed.
- **Highest value:** Creates more elements, though masks or spacing may limit how many fit.

Seed

Changes the random arrangement. The same seed gives the same result when the other settings stay the same.

- **Lowest value:** Selects one repeatable arrangement.
- **Highest value:** Selects a different repeatable arrangement.

Base Direction

Base XY heading in domain-local coordinates.

- **Zero or negative components:** Zero adds no direction. Negative components point along the negative axes.
- **Positive or larger components:** Positive components point along positive axes. Check the preview to see how strongly the direction affects motion.

Random Direction Probability

Fraction of guides with a random XY heading. Zero: base heading. One: all headings randomized; never changes with time. Sets the fraction of guides and birds selected for the named behavior.

- **Lowest value:** At 0, no elements use that behavior.
- **Highest value:** At 1, all eligible elements use that behavior; selection is stable rather than new random selection every frame.

Point of Interest

Mesh or Empty whose origin is the focus position, including XYZ.

- **Unassigned or empty:** Leave this unassigned and any effect that needs this source may disappear.
- **Assigned or named:** Choose the object, collection, material, or named mask that should drive the effect.

POI Probability

Allocates round(Amount of Curves * Probability) guide anchors to the focus region. Does not add paths. Sets the fraction of guides and birds selected for the named behavior.

- **Lowest value:** At 0, no elements use that behavior.
- **Highest value:** At 1, all eligible elements use that behavior; selection is stable rather than new random selection every frame.

POI Distance

Maximum spawn-anchor distance around the POI origin before clipping to domain bounds. Controls how widely the elements spread through the scene.

- **Lowest value:** Zero or a low value concentrates elements close to the guide, center or base direction.
- **Highest value:** Spreads elements farther apart or across a wider region; domain clipping can limit the result.

POI Falloff

Zero: uniform volume in the POI ball. Higher values concentrate anchors toward the center. Concentrates generated guide anchors toward the point of interest.

- **Lowest value:** At 0, distributes uniformly by volume within the POI ball.
- **Highest value:** Concentrates anchors strongly near its center.

Move Speed Max

Controls how fast the elements move.

- **Lowest value:** Produces the slowest allowed travel; zero removes this travel contribution but other motion may remain.
- **Highest value:** Produces faster travel; high values cover more distance per frame and may require finer simulation steps.

Move Speed Min

Controls how fast the elements move.

- **Lowest value:** Produces the slowest allowed travel; zero removes this travel contribution but other motion may remain.
- **Highest value:** Produces faster travel; high values cover more distance per frame and may require finer simulation

steps.

Birds Position

Offsets the starting position along the flight path before time-driven movement.

- **Lowest value:** Moves toward the path start or an earlier phase, subject to clamping or looping logic.
- **Highest value:** Moves toward the end or a later phase, subject to clamping or looping logic.

Looped

Turns this option on or off.

- **Off:** Uses the nonlooping position mode.
- **On:** Repeats travel along the flight paths.

Constant Speed on All Curves

Keep physical travel speed independent of curve length. When disabled, the original setup scales travel so longer paths can move faster. Move Speed minimum and maximum still control the base speed range.

- **Off:** Uses the legacy curve-length-dependent travel scaling.
- **On:** Keeps physical travel speed independent of curve length; the minimum and maximum speed range remains active.

Birds Spreadness

Controls how widely the elements spread through the scene.

- **Lowest value:** Zero or a low value concentrates elements close to the guide, center or base direction.
- **Highest value:** Spreads elements farther apart or across a wider region; domain clipping can limit the result.

Spread Paths

Controls how widely the elements spread through the scene.

- **Lowest value:** Zero or a low value concentrates elements close to the guide, center or base direction.
- **Highest value:** Spreads elements farther apart or across a wider region; domain clipping can limit the result.

Outside Flock Probability

Sets the fraction of guides and birds selected for the named behavior.

- **Lowest value:** At 0, no elements use that behavior.
- **Highest value:** At 1, all eligible elements use that behavior; selection is stable rather than new random selection every frame.

Outside Flock Range

Controls how widely the elements spread through the scene.

- **Lowest value:** Zero or a low value concentrates elements close to the guide, center or base direction.
- **Highest value:** Spreads elements farther apart or across a wider region; domain clipping can limit the result.

Bird Formations

Selects a discrete processing mode. Available choices: `Default`, `Flock`, `Formation`.

- **Default:** Uses the original spread-based individual paths.

- **Flock:** Uses a rounded flock volume controlled by Flock Radius and Height.
- **Formation:** Uses the slot-based formation selected by Formation Type.

Formation Type

Selects a discrete processing mode. Available choices: `Line`, `V Formation`, `Echelon`.

- **Line:** Places formation slots in a line.
- **V Formation:** Places slots in two wings around a leading slot.
- **Echelon:** Places slots along a one-sided diagonal.

Flock Radius

Horizontal radius of a round flock around each guide. Flock mode only. Controls how widely the elements spread through the scene.

- **Lowest value:** Zero or a low value concentrates elements close to the guide, center or base direction.
- **Highest value:** Spreads elements farther apart or across a wider region; domain clipping can limit the result.

Flock Height

Vertical half-height of the flock. Zero gives a flat round flock. Controls how widely the elements spread through the scene.

- **Lowest value:** Zero or a low value concentrates elements close to the guide, center or base direction.
- **Highest value:** Spreads elements farther apart or across a wider region; domain clipping can limit the result.

Formation Spacing

Distance between neighboring slots. More birds make Line, V, or Echelon larger. Controls how widely the elements spread through the scene.

- **Lowest value:** Zero or a low value concentrates elements close to the guide, center or base direction.
- **Highest value:** Spreads elements farther apart or across a wider region; domain clipping can limit the result.

Formation Angle

Angle of each V wing or the echelon relative to the flight direction. Sets the angle of a V wing or echelon relative to travel direction.

- **Lowest value:** Makes wings align more closely with the forward direction.
- **Highest value:** Opens the formation angle farther, within the allowed angular range.

Formation Randomness

Stable slot variation as a fraction of Formation Spacing, not frame noise. Controls how much individual elements differ from one another.

- **Lowest value:** Makes the elements more uniform.
- **Highest value:** Makes the elements differ more from one another.

Instance Scale

Changes the size of the generated elements.

- **Lowest value:** Zero collapses the affected geometry. Negative values, if accepted, can mirror it.
- **Highest value:** Makes the affected geometry much larger; this is a visual scale and is not automatically a physics or collision scale.

Instance Scale Randomness

Controls how much individual elements differ from one another.

- **Lowest value:** Makes the elements more uniform.
- **Highest value:** Makes the elements differ more from one another.

9. Flying Dust

Fills a closed area with moving dust. Use the presets or adjust wind, drift, and density yourself.

Setup and behavior notes

- Interior rejection sampling uses the actual closed domain. Count is a target, and finite sampling may underfill difficult shapes.
- Collision Collection drives shrinking by proximity; do not interpret Fade Distance as solid-body bounce strength.

Practical workflow — Fly Dust

Automatic Domain uses scene dimensions to create the particle region; **3D Cursor** creates a ready-made region at the cursor that you can resize and reshape. Choose a custom dust material through the first material input, or edit the one imported with the generator. Set population and size variation first, then use **Flow Speed** as a convenient master animation-speed control. **Vertical Drift** moves particles downward with negative values and upward with positive values. Wind direction can be set in the inputs or with the **Wind Direction** viewport gizmo; wind variation keeps motion from looking synchronized.

Air Turbulence bends trajectories with a procedural field; a smaller turbulence scale produces tighter changes. **Brownian Motion** adds continuously varying, small random jitter (the artistic equivalent of tiny dust particles being jostled in air). Larger Brownian values combined with slow overall motion create especially noticeable floating and wandering. For elongated instances, **Align to Flow** follows the travel direction; **Tumble Amount/Speed** changes the random tumbling, while **Spin Speed** adds continuous rotation.

Assign collision geometry when particles should shrink to nothing near nearby objects instead of passing through them. Edge fading can soften the boundary of the dust domain; disable it if you need a hard edge. **For the final render, enable camera Depth of Field** and place the focus distance appropriately: out-of-focus foreground/background particles create a more convincing suspended-dust look than uniformly sharp specks.

Settings

Count

Sets how many elements the effect tries to create.

- **Lowest value:** Creates few elements, or none when zero is allowed.
- **Highest value:** Creates more elements, though masks or spacing may limit how many fit.

Seed

Changes the random arrangement. The same seed gives the same result when the other settings stay the same.

- **Lowest value:** Selects one repeatable arrangement.
- **Highest value:** Selects a different repeatable arrangement.

Point Radius

Changes the visible thickness of the elements.

- **Lowest value:** Produces very small or thin elements; zero can collapse them if allowed.
- **Highest value:** Produces large or thick elements; this may overlap surrounding geometry.

Size Variation

Controls how much individual elements differ from one another.

- **Lowest value:** Makes the elements more uniform.
- **Highest value:** Makes the elements differ more from one another.

Clumping

Weights the nonuniform dust-spawn distribution.

- **Lowest value:** At 0, removes the extra clumping contribution.
- **Highest value:** At 1, applies full clumping influence, producing more uneven density.

Flow Speed

Controls the speed of the animated movement.

- **Lowest value:** Zero freezes the time-driven part; static offsets and other independent animation may remain.
- **Highest value:** Advances procedural motion much faster.

Vertical Drift

Adds signed vertical travel.

- **Lowest value:** Negative values move particles downward.
- **Highest value:** Positive values move particles upward; zero removes this vertical contribution.

Wind Direction

Sets which way the elements move. Use the related Strength or Speed control to change the intensity.

- **Zero or negative components:** Zero adds no direction. Negative components point along the negative axes.
- **Positive or larger components:** Positive components point along positive axes. Check the preview to see how strongly the direction affects motion.

Wind Strength

Controls how strongly this movement affects the effect.

- **Lowest value:** Zero removes this movement. Negative values, if available, reverse its direction.
- **Highest value:** Makes the movement or deformation more noticeable.

Wind Variation

Controls how much individual elements differ from one another.

- **Lowest value:** Makes the elements more uniform.
- **Highest value:** Makes the elements differ more from one another.

Allow Moving Outside Boundaries

Turns this option on or off.

- **Off:** Applies the domain boundary confinement or wrapping mode.

- **On:** Allows the motion mode to move outside the domain bounds.

Turbulence Strength

Controls how strongly this movement affects the effect.

- **Lowest value:** Zero removes this movement. Negative values, if available, reverse its direction.
- **Highest value:** Makes the movement or deformation more noticeable.

Turbulence Scale

Controls the size of the visible pattern without changing the size of the individual objects.

- **Lowest value:** Creates broader, smoother patterns.
- **Highest value:** Creates smaller, more closely spaced details.

Turbulence Speed

Controls how quickly this part of the effect changes over time.

- **Lowest value:** Zero stops change in this contribution; a static displacement or rotation may remain. Negative values reverse time where supported.
- **Highest value:** Produces rapid changes or rotation; may appear abrupt at the current frame rate.

Brownian Motion

Controls how strongly this movement affects the effect.

- **Lowest value:** Zero removes this movement. Negative values, if available, reverse its direction.
- **Highest value:** Makes the movement or deformation more noticeable.

Brownian Speed

Controls how quickly this part of the effect changes over time.

- **Lowest value:** Zero stops change in this contribution; a static displacement or rotation may remain. Negative values reverse time where supported.
- **Highest value:** Produces rapid changes or rotation; may appear abrupt at the current frame rate.

Instance Collection

Choose the collection of objects to display in this effect.

- **Unassigned or empty:** Leave this unassigned and any effect that needs this source may disappear.
- **Assigned or named:** Choose the object, collection, material, or named mask that should drive the effect.

Instance Scale

Changes the size of the generated elements.

- **Lowest value:** Zero collapses the affected geometry. Negative values, if accepted, can mirror it.
- **Highest value:** Makes the affected geometry much larger; this is a visual scale and is not automatically a physics or collision scale.

Align to Flow

Blends orientation or direction toward the named reference.

- **Lowest value:** At 0, leaves the unblended base orientation or direction.

- **Highest value:** At 1, applies the full normal, upright, and flow alignment.

Tumble Amount

Sets the amplitude of instance tumbling.

- **Lowest value:** Zero removes the tumble offset.
- **Highest value:** At about pi radians, permits large orientation swings.

Tumble Speed

Controls how quickly this part of the effect changes over time.

- **Lowest value:** Zero stops change in this contribution; a static displacement or rotation may remain. Negative values reverse time where supported.
- **Highest value:** Produces rapid changes or rotation; may appear abrupt at the current frame rate.

Spin Speed

Controls how quickly this part of the effect changes over time.

- **Lowest value:** Zero stops change in this contribution; a static displacement or rotation may remain. Negative values reverse time where supported.
- **Highest value:** Produces rapid changes or rotation; may appear abrupt at the current frame rate.

Collision Collection

Choose the objects that this effect should collide with or avoid.

- **Unassigned or empty:** Leave this unassigned and any effect that needs this source may disappear.
- **Assigned or named:** Choose the object, collection, material, or named mask that should drive the effect.

Fade Distance

Sets the distance over which particles shrink or fade near the associated surface.

- **Lowest value:** Produces a narrow transition, retaining full size until close to the surface.
- **Highest value:** Produces a broad transition, so particles start shrinking farther away and may stay small throughout a small domain.

Edge Fade Distance

Distance to the actual custom-domain surface over which Dust scales from zero to full size. Sets the distance over which particles shrink or fade near the associated surface.

- **Lowest value:** Produces a narrow transition, retaining full size until close to the surface.
- **Highest value:** Produces a broad transition, so particles start shrinking farther away and may stay small throughout a small domain.

10. Crawling Insects

Places insects on a selected surface and moves them across it. Add wandering, grouping, attraction, or avoidance as needed.

Setup and behavior notes

- Enabling Alignment or Cohesion is not enough when its strength is zero; raise the strength to obtain a response.

- Neighbor Radius is a two-grid local averaging cell size, not an exact radius-based neighbor query.```

Practical workflow — Crawling Insects

Select a mesh, import an insect asset (for example, ants), and assign it to **Target**. **Amount**, **Insect Scale**, **Random Scale**, and **Seed** control the initial population and appearance. Use the **Paint Vertex Group** button beside **Spawn Zone** to create and paint a mask on the target; insects then originate only in the painted areas. **Fast Mode** previews them as points when working with dense populations.

Tune **Crawl Speed**, **Random Speed**, and **Motion Paths Scale** for general movement. Small path scales create quick, sharp turns; larger scales produce broader movement. **Random Movement** adds natural irregularity; enable the wall-removal option to exclude steep surfaces and **Edge Stabilizer** when intricate geometry causes jitter. **Alignment** moves neighbors in parallel (none at the lowest value, tight parallel streams at the highest); **Cohesion** draws neighboring insects together, using **Neighbor Radius** to set the neighborhood. **Wander**, **Pause**, and **Roam** vary direction, rest time, and periodically selected destinations. Roam arrival distance decides how close a destination must be before it counts as reached.

Avoidance pushes insects away from assigned objects; provide a nonempty collection before enabling it, since an empty source can stall the behavior. **Attraction** pulls them toward objects. **Attraction Distance** is the desired stand-off distance **from the object's surface**; **Attraction Target Distance** is how far away insects can **detect the target**. The optional guidance-priority control weights attraction, avoidance, and curve following against ordinary movement behaviors.

Directed vs precise curve following: assign or draw a curve, set **Following Strength** and **Detection Distance**, and enable **Move in Curve Direction**. With **Precise Curve Follow?** off, the curve supplies general directional guidance; insects can wander away and the final curve segment indicates their eventual travel direction. Enable **Precise Curve Follow?** to make them follow the **actual curve shape**, including bends. **Lane Width** sets how far they can drift from the path, and **Return Strength** draws them back when they leave it. To create an organized ant trail, paint a narrow Spawn Zone first, draw a directional curve along it, enable precise following, then adjust lane width and return strength until the stream looks natural.

Settings

Target

Mesh whose evaluated surface receives the generated effect.

- **Unassigned or empty:** Leave this unassigned and any effect that needs this source may disappear.
- **Assigned or named:** Choose the object, collection, material, or named mask that should drive the effect.

Amount

Surface density on Target (): Amount x 100 x Crawling Density Multiplier. Controls how densely the elements appear.

- **Lowest value:** Zero suppresses this emission layer. Negative density is not a meaningful negative particle count.
- **Highest value:** Creates a denser population, subject to spacing, masks and particle budgets; increases evaluation cost.

Insect Scale

Changes the size of the generated elements.

- **Lowest value:** Zero collapses the affected geometry. Negative values, if accepted, can mirror it.
- **Highest value:** Makes the affected geometry much larger; this is a visual scale and is not automatically a physics or collision scale.

Random Scale

Controls how much individual elements differ from one another.

- **Lowest value:** Makes the elements more uniform.
- **Highest value:** Makes the elements differ more from one another.

Seed

Changes the random arrangement. The same seed gives the same result when the other settings stay the same.

- **Lowest value:** Selects one repeatable arrangement.
- **Highest value:** Selects a different repeatable arrangement.

Fast Mode

Turns this option on or off.

- **Off:** Uses the regular generated shapes mode.
- **On:** Uses the simplified preview or fast mode; the exact representation depends on the generator.

Fast Mode Point Size

Changes the visible thickness of the elements.

- **Lowest value:** Produces very small or thin elements; zero can collapse them if allowed.
- **Highest value:** Produces large or thick elements; this may overlap surrounding geometry.

Crawl Speed

Controls how fast the elements move.

- **Lowest value:** Produces the slowest allowed travel; zero removes this travel contribution but other motion may remain.
- **Highest value:** Produces faster travel; high values cover more distance per frame and may require finer simulation steps.

Random Speed

Controls how much individual elements differ from one another.

- **Lowest value:** Makes the elements more uniform.
- **Highest value:** Makes the elements differ more from one another.

Motion Paths Scale

Controls the size of the visible pattern without changing the size of the individual objects.

- **Lowest value:** Creates broader, smoother patterns.
- **Highest value:** Creates smaller, more closely spaced details.

Speed Variation

Controls how much individual elements differ from one another.

- **Lowest value:** Makes the elements more uniform.
- **Highest value:** Makes the elements differ more from one another.

Random Movement Frequency

Controls how frequently the named cyclic or random movement changes.

- **Lowest value:** Zero removes time progression of this component.
- **Highest value:** Produces more frequent oscillation or changes.

Sensivity

Adjusts this effect's response. If changing it makes no visible difference, leave it unchanged.

- **Lowest value:** Uses the lower threshold or response. Signed extremes are technical limits and not a recommended physical setting.
- **Highest value:** Uses the higher threshold or response. This control needs the corresponding option enabled-specific interpretation; see the generator notes.

Enable Alignment

Turns this option on or off.

- **Off:** Bypasses the named steering module.
- **On:** Evaluates the named steering module. Its strength or amount must also be above zero for a response.

Alignment

Set the strength above zero after enabling. Controls how strongly this setting changes direction or rotation.

- **Lowest value:** Zero removes this response even if its enable switch is on.
- **Highest value:** Gives this response a stronger influence; it can dominate the base movement.

Enable Cohesion

Turns this option on or off.

- **Off:** Bypasses the named steering module.
- **On:** Evaluates the named steering module. Its strength or amount must also be above zero for a response.

Cohesion

Set the strength above zero after enabling. Controls how strongly this setting changes direction or rotation.

- **Lowest value:** Zero removes this response even if its enable switch is on.
- **Highest value:** Gives this response a stronger influence; it can dominate the base movement.

Neighbor Radius

Donor two-grid local averaging cell size; not an exact radius query. Sets the size of the local neighborhood used by alignment or cohesion. In crawling insects it is a two-grid averaging cell size, not an exact radial query.

- **Lowest value:** Uses a very local neighborhood.
- **Highest value:** Combines information across a larger neighborhood and can make movement more globally coordinated.

Wander Strength

Controls how strongly this setting changes direction or rotation.

- **Lowest value:** Zero removes this response even if its enable switch is on.
- **Highest value:** Gives this response a stronger influence; it can dominate the base movement.

Wander Scale

Controls the size of the visible pattern without changing the size of the individual objects.

- **Lowest value:** Creates broader, smoother patterns.
- **Highest value:** Creates smaller, more closely spaced details.

Wander Rate

Controls how quickly this part of the effect changes over time.

- **Lowest value:** Zero stops change in this contribution; a static displacement or rotation may remain. Negative values reverse time where supported.
- **Highest value:** Produces rapid changes or rotation; may appear abrupt at the current frame rate.

Pause Amount

Weights the amount of stop or pause behavior.

- **Lowest value:** Zero removes pause influence.
- **Highest value:** At 1, gives the strongest supported pause influence.

Roam Strength

Controls how strongly this setting changes direction or rotation.

- **Lowest value:** Zero removes this response even if its enable switch is on.
- **Highest value:** Gives this response a stronger influence; it can dominate the base movement.

Roam Interval

Sets how often roaming targets are refreshed.

- **Lowest value:** Repicks targets frequently.
- **Highest value:** Keeps a roaming destination for a long time.

Roam Arrival Distance

Repick distance, decoupled from the removed self-separation module. Sets the arrival or stand-off distance to the destination or attractor.

- **Lowest value:** Allows movement close to the target before arrival or stand-off.
- **Highest value:** Stops or repicks farther away from the target.

Avoiders

Collection whose objects repel crawling insects.

- **Unassigned or empty:** Leave this unassigned and any effect that needs this source may disappear.
- **Assigned or named:** Choose the object, collection, material, or named mask that should drive the effect.

Avoidance Strength

Controls how strongly this setting changes direction or rotation.

- **Lowest value:** Zero removes this response even if its enable switch is on.
- **Highest value:** Gives this response a stronger influence; it can dominate the base movement.

Avoidance Distance

Controls how far away this setting begins to affect the result.

- **Lowest value:** Restricts the response to a small neighborhood; zero can remove its usable range.
- **Highest value:** Extends the response over a much larger area.

Random Distance

Controls how much individual elements differ from one another.

- **Lowest value:** Makes the elements more uniform.
- **Highest value:** Makes the elements differ more from one another.

Avoidance Speed Min

Controls how fast the elements move.

- **Lowest value:** Produces the slowest allowed travel; zero removes this travel contribution but other motion may remain.
- **Highest value:** Produces faster travel; high values cover more distance per frame and may require finer simulation steps.

Avoidance Speed Max

Controls how fast the elements move.

- **Lowest value:** Produces the slowest allowed travel; zero removes this travel contribution but other motion may remain.
- **Highest value:** Produces faster travel; high values cover more distance per frame and may require finer simulation steps.

Attractors

Collection whose objects attract crawling insects.

- **Unassigned or empty:** Leave this unassigned and any effect that needs this source may disappear.
- **Assigned or named:** Choose the object, collection, material, or named mask that should drive the effect.

Attraction Strength

Controls how strongly this setting changes direction or rotation.

- **Lowest value:** Zero removes this response even if its enable switch is on.
- **Highest value:** Gives this response a stronger influence; it can dominate the base movement.

Attraction Target Distance

Detection range around the attractor. Insects begin responding to the target once they are close enough to detect it.

- **Lowest value:** Only insects very close to the target detect its attraction.
- **Highest value:** Insects can detect and move toward the target from farther away.

Attraction Distance

Desired stand-off distance from the **surface of the attraction object**; it controls how closely insects approach before treating the goal as reached.

- **Lowest value:** Insects can approach the attractor's surface closely.
- **Highest value:** Insects maintain a larger distance from the object.

Assign Curve

Curve object for the optional crawling curve-follow behavior.

- **Unassigned or empty:** Leave this unassigned and any effect that needs this source may disappear.
- **Assigned or named:** Choose the object, collection, material, or named mask that should drive the effect.

Following Strength

Controls how strongly this setting changes direction or rotation.

- **Lowest value:** Zero removes this response even if its enable switch is on.
- **Highest value:** Gives this response a stronger influence; it can dominate the base movement.

Detection Distance

Controls how far away this setting begins to affect the result.

- **Lowest value:** Restricts the response to a small neighborhood; zero can remove its usable range.
- **Highest value:** Extends the response over a much larger area.

Move in Curve Direction

Turns this option on or off.

- **Off:** Uses curve proximity without the forward tangent-travel option.
- **On:** Uses the curve tangent to guide forward movement.

Precise Curve Follow?

Makes insects follow the actual shape of the assigned curve instead of treating its tangent mainly as a movement-direction guide. When enabled, **Lane Width** and **Return Strength** become available.

- **Off:** Uses looser curve-based directional guidance; insects can wander away from the path.
- **On:** Keeps insects following the path and its turns more closely, subject to lane width and return strength.

Lane Width

Available when **Precise Curve Follow?** is enabled. Defines the width of the corridor around the path inside which insects can travel.

- **Lowest value:** Creates a narrow stream close to the curve.
- **Highest value:** Allows a wider lane and more side-to-side separation.

Return Strength

Available when **Precise Curve Follow?** is enabled. Determines how strongly insects are guided back toward the curve when they drift outside the preferred path.

- **Lowest value:** Allows more gradual returns and wider deviations.
- **Highest value:** Pulls deviating insects back toward the curve more strongly.

Enable Locked Motion

Turns this option on or off.

- **Off:** Uses the ordinary motion mode.
- **On:** Selects the surface-locked motion mode in the crawling simulator.

Instances

Collection containing the insect models placed on crawling points.

- **Unassigned or empty:** Leave this unassigned and any effect that needs this source may disappear.
- **Assigned or named:** Choose the object, collection, material, or named mask that should drive the effect.

11. Insect Swarm

Creates a flying insect swarm. Choose a preset or draw paths and adjust how the insects fly.

Setup and behavior notes

- Use Curve Draw switches to Flight Path. The path-spread or noise controls apply to this mode.
- Collision smoothing changes adjusted trajectories; Collision Radius and clearance are distinct from Point Radius and model size.

Practical workflow — Insect Swarm

Choose an insect type and place the swarm domain. The domain is editable: change its shape, even to an Icosphere, to influence where flight paths begin. Use **Proxy View** for point-only preview in heavy scenes. Turn on **Show Trajectory Curves** to see the procedural paths, then adjust **Curve Resolution** (smoothness), **Trajectory Radius** (spread), **Trajectory Complexity**, and **Vertical Motion**. Disable **Allow Moving Outside Boundaries** if the paths must stay confined. **Curve Deform Strength/Speed/Scale** add time-evolving noise, so the paths change gradually rather than staying frozen.

Flight Speed, **Individual Variation**, and **Reverse Percentage** control how individuals travel. The hovering options let a percentage of insects pause along a flight, with editable entry time, duration, time variation, and **Hover Micro Motion** to keep them from freezing unnaturally. For collisions, assign a collider collection and inspect trajectory debug view: the system **bends paths around objects**, not rigid-body-simulates each insect, so rare intersections may remain. **Avoidance Distance/Strength** control steering, while **Collision Smooth Iterations** and **Collision Smoothness** remove sharp corners from corrected paths.

Use **Curve Draw / Flight Path** to concentrate the swarm along a user-created curve instead of the original domain arrangement. After drawing, check the curve origin and position (use **Set Origin to Geometry** if needed), then adjust path spread and noise.

Settings

Proxy View

Turns this option on or off.

- **Off:** Uses normal swarm output.
- **On:** Uses the swarm proxy preview mode.

Control Menu

Selects a discrete processing mode. Available choices: `Bees`, `Flies`, `Mosquitoes`, `Moths`.

- **Bees:** Selects the Bees parameter preset.
- **Flies:** Selects the Flies parameter preset.
- **Mosquitoes:** Selects the Mosquitoes parameter preset.
- **Moths:** Selects the Moths parameter preset.

Count

Sets how many elements the effect tries to create.

- **Lowest value:** Creates few elements, or none when zero is allowed.
- **Highest value:** Creates more elements, though masks or spacing may limit how many fit.

Seed

Changes the random arrangement. The same seed gives the same result when the other settings stay the same.

- **Lowest value:** Selects one repeatable arrangement.
- **Highest value:** Selects a different repeatable arrangement.

Curve Resolution

Controls how smooth or detailed this part of the effect appears.

- **Lowest value:** Shows a simpler, rougher shape.
- **Highest value:** Shows a smoother shape, which can slow down the scene.

Dependency: Use Preset Physics must be Off for this manual input to control the ACTIVE mode.

Trajectory Radius

Controls how widely the elements spread through the scene.

- **Lowest value:** Zero or a low value concentrates elements close to the guide, center or base direction.
- **Highest value:** Spreads elements farther apart or across a wider region; domain clipping can limit the result.

Dependency: Use Preset Physics must be Off for this manual input to control the ACTIVE mode.

Trajectory Complexity

Increases the complexity of procedural insect trajectories.

- **Lowest value:** Zero gives the simplest underlying trajectory.
- **Highest value:** Produces more complex deviations and turns.

Dependency: Use Preset Physics must be Off for this manual input to control the ACTIVE mode.

Vertical Motion

Weights the vertical component of insect trajectories.

- **Lowest value:** Zero suppresses the trajectory vertical component.
- **Highest value:** Adds larger vertical excursions.

Dependency: Use Preset Physics must be Off for this manual input to control the ACTIVE mode.

Allow Moving Outside Boundaries

Turns this option on or off.

- **Off:** Applies the domain boundary confinement or wrapping mode.
- **On:** Allows the motion mode to move outside the domain bounds.

Curve Deform Strength

Controls how strongly this movement affects the effect.

- **Lowest value:** Zero removes this movement. Negative values, if available, reverse its direction.

- **Highest value:** Makes the movement or deformation more noticeable.

Dependency: Use Preset Physics must be Off for this manual input to control the ACTIVE mode.

Curve Deform Speed

Controls how quickly this part of the effect changes over time.

- **Lowest value:** Zero stops change in this contribution; a static displacement or rotation may remain. Negative values reverse time where supported.
- **Highest value:** Produces rapid changes or rotation; may appear abrupt at the current frame rate.

Dependency: Use Preset Physics must be Off for this manual input to control the ACTIVE mode.

Curve Deform Scale

Controls the size of the visible pattern without changing the size of the individual objects.

- **Lowest value:** Creates broader, smoother patterns.
- **Highest value:** Creates smaller, more closely spaced details.

Dependency: Use Preset Physics must be Off for this manual input to control the ACTIVE mode.

Flight Speed

Controls how fast the elements move.

- **Lowest value:** Produces the slowest allowed travel; zero removes this travel contribution but other motion may remain.
- **Highest value:** Produces faster travel; high values cover more distance per frame and may require finer simulation steps.

Dependency: Use Preset Physics must be Off for this manual input to control the ACTIVE mode.

Individual Variation

Controls how much individual elements differ from one another.

- **Lowest value:** Makes the elements more uniform.
- **Highest value:** Makes the elements differ more from one another.

Dependency: Use Preset Physics must be Off for this manual input to control the ACTIVE mode.

Reverse Percentage

Sets the percentage of elements selected for the named behavior.

- **Lowest value:** At 0%, selects none.
- **Highest value:** At 100%, selects all eligible elements.

Dependency: Use Preset Physics must be Off for this manual input to control the ACTIVE mode.

Add Hovering Points

Turns this option on or off.

- **Off:** Uses uninterrupted traversal for the hovering module.
- **On:** Enables the selected stop-and-go or hovering behavior.

Dependency: Use Preset Physics must be Off for this manual input to control the ACTIVE mode.

Hovering Percentage

Sets the percentage of elements selected for the named behavior.

- **Lowest value:** At 0%, selects none.
- **Highest value:** At 100%, selects all eligible elements.

Dependency: Use Preset Physics must be Off for this manual input to control the ACTIVE mode.

Hovering Move Time

Sets the length of the movement or hovering part of the stop-and-go cycle, as named.

- **Lowest value:** Shortens this part of the cycle.
- **Highest value:** Lengthens this part of the cycle.

Dependency: Use Preset Physics must be Off for this manual input to control the ACTIVE mode.

Hovering Duration

Sets the length of the movement or hovering part of the stop-and-go cycle, as named.

- **Lowest value:** Shortens this part of the cycle.
- **Highest value:** Lengthens this part of the cycle.

Dependency: Use Preset Physics must be Off for this manual input to control the ACTIVE mode.

Hovering Time Variation

Controls how much individual elements differ from one another.

- **Lowest value:** Makes the elements more uniform.
- **Highest value:** Makes the elements differ more from one another.

Dependency: Use Preset Physics must be Off for this manual input to control the ACTIVE mode.

Hover Micro Motion

Controls the strength of the small details.

- **Lowest value:** Zero removes this variation.
- **Highest value:** Makes the variation more pronounced; large displacement values may overwhelm the base shape or path.

Dependency: Use Preset Physics must be Off for this manual input to control the ACTIVE mode.

Hover Micro Speed

Controls how quickly this part of the effect changes over time.

- **Lowest value:** Zero stops change in this contribution; a static displacement or rotation may remain. Negative values reverse time where supported.
- **Highest value:** Produces rapid changes or rotation; may appear abrupt at the current frame rate.

Dependency: Use Preset Physics must be Off for this manual input to control the ACTIVE mode.

Collision Collection

Choose the objects that this effect should collide with or avoid.

- **Unassigned or empty:** Leave this unassigned and any effect that needs this source may disappear.

- **Assigned or named:** Choose the object, collection, material, or named mask that should drive the effect.

Collision Clearance

Sets additional surface clearance, particle contact size or contact padding as named.

- **Lowest value:** Allows movement or rendering close to the surface, with less padding.
- **Highest value:** Creates a larger gap or contact region; excessive padding can prevent passage through small spaces.

Dependency: Use Preset Physics must be Off for this manual input to control the ACTIVE mode.

Avoidance Distance

Controls how far away this setting begins to affect the result.

- **Lowest value:** Restricts the response to a small neighborhood; zero can remove its usable range.
- **Highest value:** Extends the response over a much larger area.

Dependency: Use Preset Physics must be Off for this manual input to control the ACTIVE mode.

Avoidance Strength

Controls how strongly this setting changes direction or rotation.

- **Lowest value:** Zero removes this response even if its enable switch is on.
- **Highest value:** Gives this response a stronger influence; it can dominate the base movement.

Dependency: Use Preset Physics must be Off for this manual input to control the ACTIVE mode.

Collision Smooth Iterations

Sets the number of smoothing passes on collision-adjusted paths.

- **Lowest value:** Zero leaves the collision-adjusted path unsmoothed.
- **Highest value:** Uses many smoothing passes, increasing cost and softening local path changes.

Dependency: Use Preset Physics must be Off for this manual input to control the ACTIVE mode.

Collision Smoothness

Weights smoothing of collision-adjusted paths.

- **Lowest value:** Zero leaves the unsmoothed path.
- **Highest value:** At 1, applies the full smoothing result.

Dependency: Use Preset Physics must be Off for this manual input to control the ACTIVE mode.

Collision Detail

Controls how smooth or detailed this part of the effect appears.

- **Lowest value:** Shows a simpler, rougher shape.
- **Highest value:** Shows a smoother shape, which can slow down the scene.

Dependency: Use Preset Physics must be Off for this manual input to control the ACTIVE mode.

Instance Collection

Choose the collection of objects to display in this effect.

- **Unassigned or empty:** Leave this unassigned and any effect that needs this source may disappear.

- **Assigned or named:** Choose the object, collection, material, or named mask that should drive the effect.

Instance Scale

Changes the size of the generated elements.

- **Lowest value:** Zero collapses the affected geometry. Negative values, if accepted, can mirror it.
- **Highest value:** Makes the affected geometry much larger; this is a visual scale and is not automatically a physics or collision scale.

Point Radius

Changes the visible thickness of the elements.

- **Lowest value:** Produces very small or thin elements; zero can collapse them if allowed.
- **Highest value:** Produces large or thick elements; this may overlap surrounding geometry.

Keep Upright

Turns this option on or off.

- **Off:** Uses the freely oriented flight rotation.
- **On:** Selects the stabilized upright orientation mode.

Show Trajectory Curves

Turns this option on or off.

- **Off:** Omits guide or trajectory visualization.
- **On:** Displays the procedural flight paths.

Show Tangent Debug

Turns this option on or off.

- **Off:** Omits orientation markers.
- **On:** Shows markers representing flight-path tangent directions.

Debug Arrow Length

Scales debug markers or text or arrow length.

- **Lowest value:** Makes debug indicators small.
- **Highest value:** Makes debug indicators large; does not change the actual flight motion.

Debug Arrow Width

Changes the visible thickness of the elements.

- **Lowest value:** Produces very small or thin elements; zero can collapse them if allowed.
- **Highest value:** Produces large or thick elements; this may overlap surrounding geometry.

Flight Path

Curve object used by the drawn swarm flight mode.

- **Unassigned or empty:** Leave this unassigned and any effect that needs this source may disappear.
- **Assigned or named:** Choose the object, collection, material, or named mask that should drive the effect.

Path Spread

Controls how widely the elements spread through the scene.

- **Lowest value:** Zero or a low value concentrates elements close to the guide, center or base direction.
- **Highest value:** Spreads elements farther apart or across a wider region; domain clipping can limit the result.

Path Noise Strength

Controls how strongly this movement affects the effect.

- **Lowest value:** Zero removes this movement. Negative values, if available, reverse its direction.
- **Highest value:** Makes the movement or deformation more noticeable.

Path Noise Scale

Controls the size of the visible pattern without changing the size of the individual objects.

- **Lowest value:** Creates broader, smoother patterns.
- **Highest value:** Creates smaller, more closely spaced details.

Path Noise Speed

Controls how quickly this part of the effect changes over time.

- **Lowest value:** Zero stops change in this contribution; a static displacement or rotation may remain. Negative values reverse time where supported.
- **Highest value:** Produces rapid changes or rotation; may appear abrupt at the current frame rate.

12. Moisture Specks

Adds small water specks to a selected surface. Control their density, spacing, and size.

Setup and behavior notes

- Assign Target to provide the surface on which specks appear.
- Density is the Poisson maximum density; Min Distance can reduce the actual number of accepted points.
- Flatness controls normal-axis thickness, not overall radius.

Practical workflow — Moisture Specks

Select the target surface and import **Moisture Specks** to distribute small wet droplets without running a fluid simulation. Begin with **Density** and droplet **Radius Min/Max**, then use **Flatness** and **Stretch** to control how each droplet sits on the surface. **Cluster Scale**, **Coverage**, and **Softness** create groups and variations rather than evenly spaced dots. **Upward Bias** can favor surfaces facing up. This lightweight layer is useful for condensation and scattered moisture detail.

Settings

Target

Mesh whose evaluated surface receives the generated effect.

- **Unassigned or empty:** Leave this unassigned and any effect that needs this source may disappear.
- **Assigned or named:** Choose the object, collection, material, or named mask that should drive the effect.

Seed

Changes the random arrangement. The same seed gives the same result when the other settings stay the same.

- **Lowest value:** Selects one repeatable arrangement.
- **Highest value:** Selects a different repeatable arrangement.

Density

Controls how densely the elements appear.

- **Lowest value:** Zero suppresses this emission layer. Negative density is not a meaningful negative particle count.
- **Highest value:** Creates a denser population, subject to spacing, masks and particle budgets; increases evaluation cost.

Min Distance

Sets minimum spacing between emission points.

- **Lowest value:** Allows points to be close together, increasing crowding and potential overlap.
- **Highest value:** Separates points over a much wider area, usually reducing how many fit on the emitter.

Radius Min

Sets an endpoint of the random droplet radius range.

- **Lowest value:** Allows tiny or zero-radius droplets at this endpoint.
- **Highest value:** Allows larger droplets at this endpoint. The other endpoint and Size Bias still determine the distribution.

Radius Max

Sets an endpoint of the random droplet radius range.

- **Lowest value:** Allows tiny or zero-radius droplets at this endpoint.
- **Highest value:** Allows larger droplets at this endpoint. The other endpoint and Size Bias still determine the distribution.

Size Bias

Raises a normalized random value to a power before mapping it into the radius range.

- **Lowest value:** At 1, radii have an unbiased linear distribution between Radius Min and Radius Max.
- **Highest value:** Concentrates most droplets near Radius Min, leaving fewer large droplets.

Flatness

Scales droplet thickness along its surface-normal axis.

- **Lowest value:** Produces flatter droplets; zero collapses thickness where allowed.
- **Highest value:** Produces thicker or taller droplets. At 1, the normal-axis scale is uncompressed; values above 1 stretch it.

Stretch

Adds elongation to moisture-speck shapes.

- **Lowest value:** Zero removes additional elongation.
- **Highest value:** Produces more elongated specks.

Irregularity

Controls shape irregularity of moisture specks.

- **Lowest value:** Produces more regular shapes.
- **Highest value:** Produces the strongest supported irregularity.

Cluster Scale

Controls the size of the visible pattern without changing the size of the individual objects.

- **Lowest value:** Creates broader, smoother patterns.
- **Highest value:** Creates smaller, more closely spaced details.

Coverage

Sets the coverage threshold of the procedural clustering mask.

- **Lowest value:** Leaves little or no covered area.
- **Highest value:** Allows coverage across most or all of the target, subject to other masks.

Softness

Sets the width of the moisture-coverage transition.

- **Lowest value:** Produces sharply bounded patches.
- **Highest value:** Produces broad, gradual transitions.

Upward Bias

Weights moisture emission toward upward-facing normals.

- **Lowest value:** At 0, applies no upward preference.
- **Highest value:** At 1, strongly favors upward-facing surfaces.

Resolution

Controls how smooth or detailed this part of the effect appears.

- **Lowest value:** Shows a simpler, rougher shape.
- **Highest value:** Shows a smoother shape, which can slow down the scene.

13. Moss — Draw & Scatter

This is the **Geometry Nodes moss scatter generator**, with automatic **Scatter** and hand-directed **Draw** modes. It is separate from the regular **Moss Shader**, which uses a geometry-derived surface mask and a material.

Moss — Scatter

Scatters moss over source geometry using a procedural noise or vertex-group mask. Supports steep-surface filtering, upright alignment, animation, wind and proximity-driven rotation.

Setup and behavior notes

- Add Scatter Moss to a selected mesh. The separate Random Color effect has its own controls.
- Masking chooses Noise or Vertex Group. Noise-only settings do not replace the named group when Vertex Group is selected.

- Collision Rotation is proximity-based instance rotation rather than rigid-body simulation.

Practical workflow — Moss Scatter

Moss — Scatter uses the same basic controls as Moss Draw but populates the selected target surface automatically, without drawing a placement curve. Increase **Scatter Density** for stronger coverage, use procedural noise or a **Vertex Group Mask** to keep moss only in selected areas, and set **Upright Alignment** for upward or normal-facing growth. Optional animated moss and simple wind add movement, while the moss collider collection adjusts instance rotations around nearby objects. Animated and static instances may have similar viewport costs; test the selected instance collection at your final density. For very dense Cycles moss, check the **Transparent Bounces** troubleshooting advice in Moss Draw.

Moss — Draw

Projects drawn-curve influence onto Target Object and scatters moss inside a Brush Radius falloff. Uses the same coverage, wind and collision-rotation controls as Scatter Moss.

Setup and behavior notes

- The Draw Moss operator requires an active mesh target, assigns it to Target Object and enters Curve Draw.
- Brush Radius controls the influence distance around the guide. Normal Offset moves generated placement along the target normal.
- Upright Alignment changes instance orientation; it does not itself delete steep-surface points.
- Collision Rotation bends or rotates nearby moss rather than running a rigid-body contact simulation.

Practical workflow — Moss Draw

Select a target, choose **Moss — Draw**, and draw strokes directly onto its surface. Press **Enter** to finish; the generator uses the strokes as curves and scatters moss around them. **Brush Radius** controls the influence around each stroke, not a conventional Blender sculpt brush. If instances are embedded in the geometry, increase **Normal Offset** slightly to shift their roots along the surface normal.

Use **Scatter Density** and instance-scale controls for coverage. **Upright Alignment** blends from surface-normal alignment to upward-facing plants; with a value of **1**, the instances face upward regardless of a sloping host face. **Exclude Vertical Surfaces** removes moss from steep faces. Choose procedural **Coverage** noise or a painted **Vertex Group Mask** to decide exactly where it grows. **Enable Animated Moss** uses preanimated instances; **Enable Simple Wind** adds directional movement.

For contact with other geometry, create the moss collider collection and add meshes to it. Nearby moss **rotates** to bend around the objects rather than undergoing full physical displacement. Reduce the collision distance for small setups and soften the effect with collision strength. **Cycles tip:** extremely dense moss with many overlapping transparent surfaces may develop black artifacts; increase **Render Properties** → **Light Paths** → **Transparent Bounces** until they disappear, balancing that against render time.

Mode-specific settings

Target Object

Mesh whose surface receives the drawn moss.

- **Unassigned or empty:** Leave this unassigned and any effect that needs this source may disappear.
- **Assigned or named:** Choose the object, collection, material, or named mask that should drive the effect.

Brush Radius

Sets the distance falloff around the drawn moss curve on Target Object.

- **Lowest value:** Creates a narrow painted strip.

- **Highest value:** Creates a broad strip around the guide and may cover most of a small target.

Normal Offset

Moves the associated geometry along the surface normal without changing its emission mask.

- **Lowest value:** Negative values intentionally embed it into the surface. Zero adds no artistic offset.
- **Highest value:** Positive values lift it away from the surface.

Moss — Draw & Scatter — Shared Settings

The following controls are available in both modes.

Scatter Density

Controls how densely the elements appear.

- **Lowest value:** Zero suppresses this emission layer. Negative density is not a meaningful negative particle count.
- **Highest value:** Creates a denser population, subject to spacing, masks and particle budgets; increases evaluation cost.

Instance Scale

Changes the size of the generated elements.

- **Lowest value:** Zero collapses the affected geometry. Negative values, if accepted, can mirror it.
- **Highest value:** Makes the affected geometry much larger; this is a visual scale and is not automatically a physics or collision scale.

Instance Scale Min

Sets one end of the size range. Keep the minimum below the maximum.

- **Lowest value:** Sets that endpoint to a very small size; zero allows collapsed elements and negative scale can mirror them.
- **Highest value:** Sets that endpoint to a larger size, allowing or forcing larger elements depending on the other endpoint.

Upright Alignment

Blends orientation or direction toward the named reference.

- **Lowest value:** At 0, leaves the unblended base orientation or direction.
- **Highest value:** At 1, applies the full normal, upright, and flow alignment.

Exclude Vertical Surfaces

Turns this option on or off.

- **Off:** Keeps eligible points on steep surfaces.
- **On:** Filters points according to the generator-specific normal or slope threshold.

Masking

Selects a discrete processing mode. Available choices: `Noise`, `Vertex Group`.

- **Noise:** Uses the procedural noise coverage mask.
- **Vertex Group:** Uses the weights read from Vertex Group Mask.

Coverage Seed

Shifts the animation or pattern to a different starting point.

- **Lowest value:** Uses an earlier or negative phase or a different noise slice.
- **Highest value:** Uses a later or positive phase or a different noise slice; does not inherently increase detail.

Coverage Scale

Controls the size of the visible pattern without changing the size of the individual objects.

- **Lowest value:** Creates broader, smoother patterns.
- **Highest value:** Creates smaller, more closely spaced details.

Coverage Contrast

Changes the remapping contrast of the procedural distribution mask.

- **Lowest value:** Produces a softer or less selective mask.
- **Highest value:** Produces stronger contrast or selection. Exact occupied area also depends on the mask thresholds and offset.

Coverage Offset

Adds an offset to the procedural mask remap thresholds.

- **Lowest value:** Moves thresholds downward, accepting more of the noise field.
- **Highest value:** Moves thresholds upward, accepting less of the noise field.

Vertex Group Mask

Exact vertex-group name used by moss when Masking is Vertex Group.

- **Unassigned or empty:** Leave this unassigned and any effect that needs this source may disappear.
- **Assigned or named:** Choose the object, collection, material, or named mask that should drive the effect.

Enable Animated Moss

Turns this option on or off.

- **Off:** Uses the static moss output mode.
- **On:** Uses the animated moss output mode.

Enable Simple Wind

Turns this option on or off.

- **Off:** Bypasses simple moss wind deformation.
- **On:** Adds directional moss wind deformation using strength, speed and scale.

Wind Strength

Controls how strongly this movement affects the effect.

- **Lowest value:** Zero removes this movement. Negative values, if available, reverse its direction.
- **Highest value:** Makes the movement or deformation more noticeable.

Wind Speed

Controls how quickly this part of the effect changes over time.

- **Lowest value:** Zero stops change in this contribution; a static displacement or rotation may remain. Negative values reverse time where supported.
- **Highest value:** Produces rapid changes or rotation; may appear abrupt at the current frame rate.

Wind Scale

Controls the size of the visible pattern without changing the size of the individual objects.

- **Lowest value:** Creates broader, smoother patterns.
- **Highest value:** Creates smaller, more closely spaced details.

Wind Direction

Sets which way the elements move. Use the related Strength or Speed control to change the intensity.

- **Zero or negative components:** Zero adds no direction. Negative components point along the negative axes.
- **Positive or larger components:** Positive components point along positive axes. Check the preview to see how strongly the direction affects motion.

Colliders

Choose the objects that particles should collide with.

- **Unassigned or empty:** Leave this unassigned and any effect that needs this source may disappear.
- **Assigned or named:** Choose the object, collection, material, or named mask that should drive the effect.

Collision Distance

Controls how far away this setting begins to affect the result.

- **Lowest value:** Restricts the response to a small neighborhood; zero can remove its usable range.
- **Highest value:** Extends the response over a much larger area.

Collision Strength

Controls how strongly this setting changes direction or rotation.

- **Lowest value:** Zero removes this response even if its enable switch is on.
- **Highest value:** Gives this response a stronger influence; it can dominate the base movement.

14. Procedural Rain

Creates animated rain inside a chosen area, with options for hits and splashes.

Setup and behavior notes

- The rain fills the area defined by the generator. Scrub the timeline to preview its movement.
- Drop Radius and Streak Length both change how long a rain streak looks. Adjust them together.
- Floor splash controls and collider controls are separate. A floor impact can exist without an assigned collider collection.
- Procedural splashes are rendering unstable if cycles Motion Blur Is enabled

Practical workflow — Procedural Rain

Add **Procedural Rain** at the 3D Cursor. It generates animated rain without simulation baking. **Intensity** determines rainfall density and **Max Drops** caps the population to protect viewport and render performance. Adjust fall height, speed, wind, and streak length to fit the shot. Assign **Colliders** so drops stop passing visibly through designated surfaces. Enable floor or object splash options to generate secondary splashes at impact. This is a fast environmental rain effect, separate from the more advanced Simulated Droplets system.

Settings

Intensity

Controls how densely the elements appear.

- **Lowest value:** Zero suppresses this emission layer. Negative density is not a meaningful negative particle count.
- **Highest value:** Creates a denser population, subject to spacing, masks and particle budgets; increases evaluation cost.

Max Drops

Limits how many elements can be active or retained.

- **Lowest value:** Restricts the layer to the smallest permitted budget; zero prevents that layer when supported.
- **Highest value:** Allows more particles or stored samples, increasing memory use; does not create particles by itself.

Fall Height

Controls the height of the area where elements appear or move.

- **Lowest value:** Gives a short travel span; signed negative values are not a useful falling height.
- **Highest value:** Gives a tall travel span and can increase flight duration or population.

Drop Speed

Controls how fast the elements move.

- **Lowest value:** Produces the slowest allowed travel; zero removes this travel contribution but other motion may remain.
- **Highest value:** Produces faster travel; high values cover more distance per frame and may require finer simulation steps.

Speed Variation

Controls how much individual elements differ from one another.

- **Lowest value:** Makes the elements more uniform.
- **Highest value:** Makes the elements differ more from one another.

Wind

Sets which way the elements move. Use the related Strength or Speed control to change the intensity.

- **Zero or negative components:** Zero adds no direction. Negative components point along the negative axes.
- **Positive or larger components:** Positive components point along positive axes. Check the preview to see how strongly the direction affects motion.

Seed

Changes the random arrangement. The same seed gives the same result when the other settings stay the same.

- **Lowest value:** Selects one repeatable arrangement.
- **Highest value:** Selects a different repeatable arrangement.

Streak Length

Sets the rain streak shape contribution. The longitudinal scale mode also uses Drop Radius.

- **Lowest value:** At 0, removes the extra streak-length contribution.
- **Highest value:** Creates longer streaks; it does not increase fall speed.

Drop Radius

Changes the visible thickness of the elements.

- **Lowest value:** Produces very small or thin elements; zero can collapse them if allowed.
- **Highest value:** Produces large or thick elements; this may overlap surrounding geometry.

Colliders

Choose the objects that particles should collide with.

- **Unassigned or empty:** Leave this unassigned and any effect that needs this source may disappear.
- **Assigned or named:** Choose the object, collection, material, or named mask that should drive the effect.

Splash On Floor

Turns this option on or off.

- **Off:** Does not create floor-triggered rain splashes.
- **On:** Allows splashes at the generator floor plane; Enable Splashes still governs secondary droplets.

Splash Droplets

Sets secondary droplet count per primary impact.

- **Lowest value:** Zero creates no secondary droplets.
- **Highest value:** Creates the maximum secondary count per impact, increasing output cost.

Splash Speed

Sets the outgoing secondary-particle speed; in the surface-drop simulation it is the tangential component.

- **Lowest value:** Zero removes this outward or tangential speed contribution; lift and gravity can still move splashes.
- **Highest value:** Throws secondary droplets farther or faster.

Splash Duration

Controls particle lifetime or one endpoint of its randomized lifetime.

- **Lowest value:** Shortens visible particle persistence; zero can remove the lifetime where accepted.
- **Highest value:** Keeps particles alive longer, which can raise the visible population. Check the generator notes for time units.

Splash Droplet Size

Changes the visible thickness of the elements.

- **Lowest value:** Produces very small or thin elements; zero can collapse them if allowed.

- **Highest value:** Produces large or thick elements; this may overlap surrounding geometry.

Crown

Turns this option on or off.

- **Off:** Omits rain splash crowns.
- **On:** Includes crown meshes at qualifying impacts, scaled by Crown Size.

Crown Size

Scales the rain splash crown mesh.

- **Lowest value:** At 0, collapses the crown.
- **Highest value:** Produces a large crown; secondary droplet size remains separately controlled.

15. Snowfall

Uses source bounds for airborne flakes and an optional resting ground layer. Wind, sway, turbulence, tumbling and proximity fades control the visible movement.

Setup and behavior notes

- Max Flakes caps the underlying airborne population; the additional ground-particle layer can exceed it.
- Ground Particle Multiplier affects only resting flakes.
- Collider Fade Distance controls shrinking near colliders, independently of the base Flake Size.
- This tree generates resting flakes, not a continuous solid snow-cover mesh.

Practical workflow — Snowfall

Snowfall animates procedurally and requires **no simulation cache**. Set **Intensity**, **Fall Height**, **Fall Speed**, speed variation, and flake size to establish the storm. Flutter/sway and turbulence make gentle snowfall turn into increasingly chaotic blowing snow.

Ground and **Colliders** have different jobs. **Ground** is where flakes settle and remain visible as resting particles. Assign the ground collection, then position the generator and set fall height so the snowfall volume reaches those surfaces; if the ground appears to float above the generator's falling region, move the generator downward or increase **Fall Height**. A ground-particle multiplier, where available, increases the amount of already-resting snow. **Ground Rest Time** governs how flakes settle after reaching a surface.

Colliders prevent flakes from visibly passing through other objects, including animated characters or props, by **shrinking flakes to zero** as they approach, rather than calculating heavy physical impacts. Adjust the collider's influence distance and **Melt Time** to control the disappearance. A collider is not automatically a ground accumulator: use the appropriate object source for the desired behavior.

Settings

Intensity

Controls how densely the elements appear.

- **Lowest value:** Zero suppresses this emission layer. Negative density is not a meaningful negative particle count.
- **Highest value:** Creates a denser population, subject to spacing, masks and particle budgets; increases evaluation cost.

Max Flakes

Limits how many elements can be active or retained.

- **Lowest value:** Restricts the layer to the smallest permitted budget; zero prevents that layer when supported.
- **Highest value:** Allows more particles or stored samples, increasing memory use; does not create particles by itself.

Fall Height

Controls the height of the area where elements appear or move.

- **Lowest value:** Gives a short travel span; signed negative values are not a useful falling height.
- **Highest value:** Gives a tall travel span and can increase flight duration or population.

Fall Speed

Controls how fast the elements move.

- **Lowest value:** Produces the slowest allowed travel; zero removes this travel contribution but other motion may remain.
- **Highest value:** Produces faster travel; high values cover more distance per frame and may require finer simulation steps.

Speed Variation

Controls how much individual elements differ from one another.

- **Lowest value:** Makes the elements more uniform.
- **Highest value:** Makes the elements differ more from one another.

Wind

Sets which way the elements move. Use the related Strength or Speed control to change the intensity.

- **Zero or negative components:** Zero adds no direction. Negative components point along the negative axes.
- **Positive or larger components:** Positive components point along positive axes. Check the preview to see how strongly the direction affects motion.

Seed

Changes the random arrangement. The same seed gives the same result when the other settings stay the same.

- **Lowest value:** Selects one repeatable arrangement.
- **Highest value:** Selects a different repeatable arrangement.

Flake Size

Changes the visible thickness of the elements.

- **Lowest value:** Produces very small or thin elements; zero can collapse them if allowed.
- **Highest value:** Produces large or thick elements; this may overlap surrounding geometry.

Size Variation

Controls how much individual elements differ from one another.

- **Lowest value:** Makes the elements more uniform.
- **Highest value:** Makes the elements differ more from one another.

Tumble Speed

Controls how quickly this part of the effect changes over time.

- **Lowest value:** Zero stops change in this contribution; a static displacement or rotation may remain. Negative values reverse time where supported.
- **Highest value:** Produces rapid changes or rotation; may appear abrupt at the current frame rate.

Sway Amount

Controls the strength of the small details.

- **Lowest value:** Zero removes this variation.
- **Highest value:** Makes the variation more pronounced; large displacement values may overwhelm the base shape or path.

Sway Frequency

Controls how frequently the named cyclic or random movement changes.

- **Lowest value:** Zero removes time progression of this component.
- **Highest value:** Produces more frequent oscillation or changes.

Turbulence

Controls how strongly this movement affects the effect.

- **Lowest value:** Zero removes this movement. Negative values, if available, reverse its direction.
- **Highest value:** Makes the movement or deformation more noticeable.

Ground

Collection supplying ground meshes for snow landing.

- **Unassigned or empty:** Leave this unassigned and any effect that needs this source may disappear.
- **Assigned or named:** Choose the object, collection, material, or named mask that should drive the effect.

Ground Rest Time

Sets how long deposited trails or landed particles persist before fading.

- **Lowest value:** Starts drying or removal almost immediately.
- **Highest value:** Keeps the deposited effect visible much longer.

Colliders

Choose the objects that particles should collide with.

- **Unassigned or empty:** Leave this unassigned and any effect that needs this source may disappear.
- **Assigned or named:** Choose the object, collection, material, or named mask that should drive the effect.

Melt Time

Sets the duration of the scale or drying fade.

- **Lowest value:** The effect disappears quickly.
- **Highest value:** The effect disappears gradually over a longer interval.

16. Spray Generator

Distributes emitter points and calculates time-driven spray trajectories with gravity, drag, wind and turbulence. Procedural mesh droplets can stretch, oscillate and wobble.

Setup and behavior notes

- Use Normal blends the emission direction toward surface normals: 0 uses the direction mode; 1 uses normals.
- Droplet Detail is an icosphere subdivision level. It affects mesh droplets, not the point-only representation.
- Adjust Radius by eye in your scene; it may not correspond exactly to a real-world measurement.
- The optional floor controls the spray's lower boundary.
- Use Time Scale to change speed and Time Offset to shift the animation.

Practical workflow — Spray Generator

Use **Spray Generator** where a water stream hits an obstacle or ejects fine secondary droplets. **Density**, **Lifetime**, **Spawn Jitter**, **Speed**, **Spread**, and **Wind** establish the spray volume. Lower downward gravity for slower, longer-lasting airborne particles; increase it for a faster fall. Tail, stretch, and acceleration-dependent shape controls make fast droplets look more like streaks. **Time Scale** can speed up or slow down the overall action. Spray is useful for impact mist or burst effects; it is not the same as the path-based Dripping generator.

Settings

Density

Controls how densely the elements appear.

- **Lowest value:** Zero suppresses this emission layer. Negative density is not a meaningful negative particle count.
- **Highest value:** Creates a denser population, subject to spacing, masks and particle budgets; increases evaluation cost.

Seed

Changes the random arrangement. The same seed gives the same result when the other settings stay the same.

- **Lowest value:** Selects one repeatable arrangement.
- **Highest value:** Selects a different repeatable arrangement.

Lifetime

Controls particle lifetime or one endpoint of its randomized lifetime.

- **Lowest value:** Shortens visible particle persistence; zero can remove the lifetime where accepted.
- **Highest value:** Keeps particles alive longer, which can raise the visible population. Check the generator notes for time units.

Lifetime Random

Controls how much individual elements differ from one another.

- **Lowest value:** Makes the elements more uniform.
- **Highest value:** Makes the elements differ more from one another.

Spawn Jitter

Controls the strength of the small details.

- **Lowest value:** Zero removes this variation.

- **Highest value:** Makes the variation more pronounced; large displacement values may overwhelm the base shape or path.

Direction

Sets which way the elements move. Use the related Strength or Speed control to change the intensity.

- **Zero or negative components:** Zero adds no direction. Negative components point along the negative axes.
- **Positive or larger components:** Positive components point along positive axes. Check the preview to see how strongly the direction affects motion.

Use Normal

Blends orientation or direction toward the named reference.

- **Lowest value:** At 0, leaves the unblended base orientation or direction.
- **Highest value:** At 1, applies the full normal, upright, and flow alignment.

Spread

Controls how widely the elements spread through the scene.

- **Lowest value:** Zero or a low value concentrates elements close to the guide, center or base direction.
- **Highest value:** Spreads elements farther apart or across a wider region; domain clipping can limit the result.

Speed

Controls how fast the elements move.

- **Lowest value:** Produces the slowest allowed travel; zero removes this travel contribution but other motion may remain.
- **Highest value:** Produces faster travel; high values cover more distance per frame and may require finer simulation steps.

Speed Random

Controls how much individual elements differ from one another.

- **Lowest value:** Makes the elements more uniform.
- **Highest value:** Makes the elements differ more from one another.

Gravity

Controls the downward gravity contribution.

- **Lowest value:** Zero removes this gravity contribution; wind and initial velocity may still move particles.
- **Highest value:** Produces stronger downward acceleration or motion.

Drag

Damps particle velocity and, where implemented, couples it to wind.

- **Lowest value:** Low drag preserves momentum longer.
- **Highest value:** Strong drag quickly damps relative movement; particles respond more strongly to the wind coupling.

Wind

Sets which way the elements move. Use the related Strength or Speed control to change the intensity.

- **Zero or negative components:** Zero adds no direction. Negative components point along the negative axes.
- **Positive or larger components:** Positive components point along positive axes. Check the preview to see how strongly the direction affects motion.

Turbulence

Controls how strongly this movement affects the effect.

- **Lowest value:** Zero removes this movement. Negative values, if available, reverse its direction.
- **Highest value:** Makes the movement or deformation more noticeable.

Turbulence Scale

Controls the size of the visible pattern without changing the size of the individual objects.

- **Lowest value:** Creates broader, smoother patterns.
- **Highest value:** Creates smaller, more closely spaced details.

Turbulence Speed

Controls how quickly this part of the effect changes over time.

- **Lowest value:** Zero stops change in this contribution; a static displacement or rotation may remain. Negative values reverse time where supported.
- **Highest value:** Produces rapid changes or rotation; may appear abrupt at the current frame rate.

Radius Min

Sets an endpoint of the random droplet radius range.

- **Lowest value:** Allows tiny or zero-radius droplets at this endpoint.
- **Highest value:** Allows larger droplets at this endpoint. The other endpoint and Size Bias still determine the distribution.

Radius Max

Sets an endpoint of the random droplet radius range.

- **Lowest value:** Allows tiny or zero-radius droplets at this endpoint.
- **Highest value:** Allows larger droplets at this endpoint. The other endpoint and Size Bias still determine the distribution.

Size Bias

Raises a normalized random value to a power before mapping it into the radius range.

- **Lowest value:** At 1, radii have an unbiased linear distribution between Radius Min and Radius Max.
- **Highest value:** Concentrates most droplets near Radius Min, leaving fewer large droplets.

Mesh Droplets

Turns this option on or off.

- **Off:** Outputs the point representation of spray.
- **On:** Builds procedural droplet meshes with shape or detail controls.

Droplet Detail

Controls how smooth or detailed this part of the effect appears.

- **Lowest value:** Shows a simpler, rougher shape.
- **Highest value:** Shows a smoother shape, which can slow down the scene.

Motion Stretch

Scales the extra streak or droplet elongation produced by motion.

- **Lowest value:** Zero removes motion-driven elongation.
- **Highest value:** Makes faster elements visibly longer, subject to the maximum length or stretch cap.

Max Stretch

Limits motion-driven elongation of spray droplets.

- **Lowest value:** At 1, prevents elongation above the base stretch factor.
- **Highest value:** Allows very long motion-aligned droplets.

Tail

Controls how far the droplet tip or tail extends.

- **Lowest value:** Zero gives a rounder shape with no added tail length.
- **Highest value:** Produces a long, narrow trailing extension.

Oscillation

Controls the strength of the small details.

- **Lowest value:** Zero removes this variation.
- **Highest value:** Makes the variation more pronounced; large displacement values may overwhelm the base shape or path.

Oscillation Freq

Controls how frequently the named cyclic or random movement changes.

- **Lowest value:** Zero removes time progression of this component.
- **Highest value:** Produces more frequent oscillation or changes.

Wobble

Controls the strength of the small details.

- **Lowest value:** Zero removes this variation.
- **Highest value:** Makes the variation more pronounced; large displacement values may overwhelm the base shape or path.

Wobble Scale

Controls the size of the visible pattern without changing the size of the individual objects.

- **Lowest value:** Creates broader, smoother patterns.
- **Highest value:** Creates smaller, more closely spaced details.

Floor Height

Sets the signed height of the optional floor plane.

- **Lowest value:** Moves the contact or removal plane downward.
- **Highest value:** Moves the plane upward, intercepting particles earlier.

Time Scale

Controls the speed of the animated movement.

- **Lowest value:** Zero freezes the time-driven part; static offsets and other independent animation may remain.
- **Highest value:** Advances procedural motion much faster.

Time Offset

Shifts the animation or pattern to a different starting point.

- **Lowest value:** Uses an earlier or negative phase or a different noise slice.
- **Highest value:** Uses a later or positive phase or a different noise slice; does not inherently increase detail.

Show Emitter

Turns this option on or off.

- **Off:** Omits the source emitter from the combined output.
- **On:** Includes source emitter geometry with the spray output.

18. Tree Wind

Adds whole-tree wind and optional gusts, minor motion, secondary mode sway and leaf flutter. Direction is controlled by the Wind Axis gizmo or vector.

Setup and behavior notes

- Whole-tree sway, minor movement and mode or leaf motion are independent optional layers.
- Branch Start and Branch Full define an anchored-to-full response transition; keep them in a sensible order.

Practical workflow — Tree Wind and performance

Select one or more vegetation objects, add **Tree Wind**, then adjust the resulting Geometry Nodes modifier. The add-on can prepare simplified proxy geometry to make heavy-tree previews manageable. **Enable Viewport Optimization** provides a coarse convex-hull-style preview for very dense trees; where available, a fast-wind option simplifies the most expensive leaf movement. This is especially useful for tree assets approaching a million polygons. Simulation Nodes baking can create large caches: one test with five such trees over 1,000 frames reached roughly **10 GB**, so monitor available disk space before baking.

Begin with **Wind Strength**, **Wind Speed**, **Turbulence Size**, and the direction gizmo/axis. Bending controls set how far the trunk leans and where deformation begins. **Gust** and turbulence break up constant movement; greater **Wind Smoothness** (if exposed in this version) makes motion more fluid, while lower values can suit a storm. **Height Falloff** reduces influence around the base while allowing higher parts to bend; it is a normalized spatial influence control, not necessarily a height measured in metres.

Weight Paint gives deliberate control over moving regions. Create a Vertex Group on the plant, paint stronger weights on flexible areas and lower weights on the stable trunk, then assign it to the wind mask. Where dedicated vertex-group inputs are available, use separate leaf, branch and minor-wind masks. **Minor Wind** supplies subtle noise-based motion; **Branch Amount/Frequency** affect branch movement, **Branch Start/Full** define its progression by tree height, and **Leaf Flutter** creates leaf oscillation distinct from the general moving-noise pattern. Preview from the timeline start before judging cached motion.

Settings

Enable Viewport Optimization

Turns this option on or off.

- **Off:** Uses the regular Tree Wind output mode.
- **On:** Selects the simplified viewport-optimization mode; compare final rendering separately.

Wind Strength

Controls how strongly this movement affects the effect.

- **Lowest value:** Zero removes this movement. Negative values, if available, reverse its direction.
- **Highest value:** Makes the movement or deformation more noticeable.

Wind Speed

Controls how quickly this part of the effect changes over time.

- **Lowest value:** Zero stops change in this contribution; a static displacement or rotation may remain. Negative values reverse time where supported.
- **Highest value:** Produces rapid changes or rotation; may appear abrupt at the current frame rate.

Turbulence Size

Sets the spatial scale control used by the Tree Wind turbulence mode.

- **Lowest value:** Small or zero values reduce or collapse the coordinate-scale contribution. Signed negative values reverse it.
- **Highest value:** Changes the turbulence spatial pattern over the tree; it is not a leaf-size control.

Sagging Strength

Weights downward or vertical sag in Tree Wind.

- **Lowest value:** Zero removes this sag contribution; negative values reverse it.
- **Highest value:** Adds stronger sagging displacement.

Master Seed

Changes the repeatable random pattern.

- **Lowest value:** Selects one repeatable pattern, not zero randomness.
- **Highest value:** Selects a different repeatable pattern, not more detail or stronger wind.

Gust Strength

Adds force or intensity during gust events.

- **Lowest value:** Zero removes the added gust force; background wind can remain.

- **Highest value:** Produces stronger gust-driven bending or motion.

Gust Speed

Sets gust travel or phase speed. In Grass Wind the source describes downwind speed in metres per second.

- **Lowest value:** Zero stops gust travel or phase progression; other wind can still animate.
- **Highest value:** Moves gust fronts or gust phases rapidly.

Enable Minor Wind

Turns this option on or off.

- **Off:** Omits minor or ambient motion parts of the effect.
- **On:** Adds fine or ambient movement controlled by Details Scale.

Details Scale

Weights or scales the secondary detail-vector contribution in Tree Wind.

- **Lowest value:** Zero removes that detail-scale contribution.
- **Highest value:** Makes minor or detail movement more pronounced; Enable Minor Wind must select that mode.

Branch Amount

Sets secondary mode sway per distance from the trunk.

- **Lowest value:** Zero removes mode sway amplitude.
- **Highest value:** Produces greater mode movement, especially farther from the trunk.

Branch Frequency

Sets oscillation frequency of the named motion.

- **Lowest value:** Zero stops this oscillatory time contribution.
- **Highest value:** Produces rapid blade, leaf, or mode oscillation.

Branch Start

Sets the distance from the trunk below which secondary sway is suppressed, as a fraction of tree height.

- **Lowest value:** Allows secondary sway close to the trunk.
- **Highest value:** Anchors a larger inner region. Keep Branch Full above Branch Start.

Branch Full

Sets the trunk-distance threshold for full mode sway, as a fraction of tree height.

- **Lowest value:** Reaches full mode sway quickly after Branch Start.
- **Highest value:** Spreads the transition over a larger outer region.

Branch Size

Sets the grouping scale for nearby mode or frond parts, in metres according to the source.

- **Lowest value:** Nearby parts are divided into smaller independent-motion regions.
- **Highest value:** Larger groups move together, giving more coherent mode movement.

Branch Bounce

Weights vertical movement relative to horizontal mode sway.

- **Lowest value:** Zero removes the vertical bounce contribution.
- **Highest value:** Adds stronger up or down motion relative to horizontal sway.

Leaf Flutter

Sets angular amplitude of fast blade or leaf motion.

- **Lowest value:** Zero removes this flutter contribution.
- **Highest value:** Permits larger flutter angles; the main wind sway is separate.

Leaf Frequency

Sets oscillation frequency of the named motion.

- **Lowest value:** Zero stops this oscillatory time contribution.
- **Highest value:** Produces rapid blade, leaf, or mode oscillation.

Height Falloff

Shapes the Tree Wind height-dependent influence.

- **Lowest value:** Changes toward a weaker or flatter exponent contribution; negative values are artistic inversions and can exaggerate low-height regions.
- **Highest value:** Concentrates the height response more strongly according to the power mode; it is not a fixed height in metres.

Invert Falloff

Turns this option on or off.

- **Off:** Uses the normal Tree Wind falloff.
- **On:** Uses the inverted falloff mode, exchanging strongly or weakly affected regions.

Falloff From Center

Weights the radial or center-distance influence of Tree Wind.

- **Lowest value:** Zero removes this center-distance contribution.
- **Highest value:** Increases the effect of distance from the center on the wind response.

Vertex Group

Exact Weight Paint vertex-group name used by Tree Wind.

- **Unassigned or empty:** Leave this unassigned and this part of the effect may not appear.
- **Assigned or connected:** Choose the requested object, collection, or mask.

19. Spiderwebs

Creates webs between selected objects. Adjust strand density, sag, tears, and optional spiders.

Setup and behavior notes

- Assign Interaction Objects to supply the surfaces that the webs contact.

- Draft Mode and Strand Detail alter evaluation detail; Tear Count and Size alter openings, not the base density.

Practical workflow — Spiderwebs, Tears and Spiders

Click the Spiderwebs import tool to enter a two-surface drawing mode. Click points to outline the first controller polygon, then press **Enter**; draw and confirm a second polygon on another surface. The add-on makes helper meshes aligned to the chosen geometry and stores them in the **Spider Web Controllers** collection. These controllers are viewport wireframes, not rendered meshes; the strands span between their surfaces. Controller polygons can have four, five or more corners. Add further interaction objects if the web should connect to a prop such as a box.

If the web does not appear because its controllers are very small, try scaling down the web generator itself. Move the generator to refine suspended strands and reduce **Sag** when they droop too much. **Fill Volume** off limits web sampling to controller surfaces rather than interior points. **Draft Mode** shows lower-cost curve representations. **Primary Strands**, **Web Density**, **Min/Max Span**, **Strand Detail**, and **Wobble** control the structure; avoid unnecessarily high strand detail in a heavy scene.

Add collider meshes to the assigned collection to imitate contact with nearby furniture or a barrel. This is procedural visual avoidance, **not** a full cloth or string simulation; extreme movements can produce intersections. Use the collider margin and strength to soften contact and reposition the web generator if a strand appears awkward. **Tears** adds torn patches: adjust **Tear Count**, **Tear Size**, and **Tear Roughness** for larger, more irregular gaps. **Spiders** can be static or moving; **Spider Count/Scale**, movement speed, **Move Chance**, and **Min Path Length** control their distribution and travel. Where spider-web caching uses Simulation Nodes, ensure the relevant cache is enabled before baking.

Settings

Draft Mode

Turns this option on or off.

- **Off:** Uses the full web-detail mode.
- **On:** Uses a lighter draft or preview web mode.

Interaction Objects

Collection of meshes supplying web interaction or contact geometry.

- **Unassigned or empty:** Leave this unassigned and this part of the effect may not appear.
- **Assigned or connected:** Choose the requested object, collection, or mask.

Seed

Changes the repeatable random pattern.

- **Lowest value:** Selects one repeatable pattern, not zero randomness.
- **Highest value:** Selects a different repeatable pattern, not more detail or stronger wind.

Realize Output

Turns this option on or off.

- **Off:** Preserves instances where the web output uses them.
- **On:** Converts output instances into actual geometry.

Primary Strands

Sets the primary strand carrier or radial structure count.

- **Lowest value:** Uses fewer primary strands and a sparse support structure.

- **Highest value:** Uses many primary strands, increasing connection opportunities and output cost.

Web Density

Sets probability of retaining generated secondary web connections.

- **Lowest value:** At zero, suppresses the density-selected connections.
- **Highest value:** At one, retains all eligible connections; span and contact filters still apply.

Min Span

Sets the shortest or longest allowed web connections across the generated levels.

- **Lowest value:** A low minimum permits short connections; a low maximum rejects long ones.
- **Highest value:** A high minimum rejects short links; a high maximum allows long ones. Keep Min Span no greater than Max Span.

Max Span

Sets the shortest or longest allowed web connections across the generated levels.

- **Lowest value:** A low minimum permits short connections; a low maximum rejects long ones.
- **Highest value:** A high minimum rejects short links; a high maximum allows long ones. Keep Min Span no greater than Max Span.

Strand Detail

Controls point sampling along each web thread.

- **Lowest value:** Produces coarse threads with little shape resolution.
- **Highest value:** Produces smoother sag or wobble or collision curves at greater evaluation cost.

Sag

Adds signed sagging curvature to web strands.

- **Lowest value:** Negative values curve opposite the sag direction; zero removes sag.
- **Highest value:** Produces stronger positive sagging.

Thickness

Scales the web-thread thickness contribution.

- **Lowest value:** Zero collapses thread thickness.
- **Highest value:** Produces thicker, more visible threads; this is not the primary strand count.

Fill Volume

Turns this option on or off.

- **Off:** Uses the non-volume-fill connection mode.
- **On:** Enables the volume-fill connection gate. It still uses the selected interaction geometry and valid contact or span tests.

Material

Material assigned to web-thread geometry.

- **Unassigned or empty:** Leave this unassigned and this part of the effect may not appear.

- **Assigned or connected:** Choose the requested object, collection, or mask.

Wobble

Controls the strength of the small details.

- **Lowest value:** Zero removes this variation.
- **Highest value:** Makes the variation more pronounced; large displacement values may overwhelm the base shape or path.

Wobble Scale

Sets spatial frequency of the web-thread waviness.

- **Lowest value:** Produces broad, slowly varying bends.
- **Highest value:** Produces fine, tightly spaced waviness; Wobble still controls amplitude.

Use Colliders

Switches the named feature.

- **Off:** Bypasses the optional collider mode.
- **On:** Uses the assigned collider geometry for the generator-specific collision or fade behavior.

Collider Objects

Collection of meshes supplying web obstacles.

- **Unassigned or empty:** Leave this unassigned and this part of the effect may not appear.
- **Assigned or connected:** Choose the requested object, collection, or mask.

Collider Margin

Sets the gap and related avoidance reach around web colliders.

- **Lowest value:** Zero provides no extra gap and permits closer contact.
- **Highest value:** Keeps threads farther from obstacles and increases the affected region.

Collider Strength

Weights the web-thread displacement away from colliders.

- **Lowest value:** Zero removes the optional repulsion contribution.
- **Highest value:** Pushes strands farther or stronger away, subject to the solver mode.

Use Tears

Turns this option on or off.

- **Off:** Keeps uncut web output.
- **On:** Cuts the requested tear openings with their size and roughness.

Tear Count

Sets requested opening count when tears are enabled.

- **Lowest value:** Creates the fewest permitted tear openings.
- **Highest value:** Creates more tear candidates; overlapping openings can merge visually.

Tear Size

Sets radius of torn web openings.

- **Lowest value:** Creates small holes.
- **Highest value:** Creates large holes that can remove most of the web.

Tear Roughness

Controls irregularity at tear boundaries.

- **Lowest value:** Creates smoother, rounder opening edges.
- **Highest value:** Creates ragged, uneven opening edges.

Use Spiders

Turns this option on or off.

- **Off:** Omits spider output.
- **On:** Places spider instances on usable generated web routes.

Spider Count

Sets requested spider population on valid web paths.

- **Lowest value:** Uses a small spider population.
- **Highest value:** Requests many spiders, increasing instance and path-walking cost.

Spiders Move

Turns this option on or off.

- **Off:** Keeps spider path positions static.
- **On:** Enables spider walking using speed, move chance and valid path lengths.

Spider Speed

Sets fastest spider pace; the slowest pace is half of this value.

- **Lowest value:** Zero stops walking speed.
- **Highest value:** Makes spiders walk rapidly; shorter paths may be traversed quickly.

Spider Scale

Scales spider instances.

- **Lowest value:** Zero collapses or hides the spider model.
- **Highest value:** Makes spiders larger without automatically widening strands or changing path clearance.

Move Chance

Sets the percentage chance of a spider entering a movement decision.

- **Lowest value:** At zero, no eligible decision initiates movement.
- **Highest value:** At 100%, every eligible decision initiates movement.

Min Path Length

Rejects spider trips below this path length.

- **Lowest value:** Zero permits short trips.
- **Highest value:** Requires longer paths and can leave no usable walking route.

Spider Rotation Offset

Sets extra per-axis texture mapping or spider-model rotation, as named. Rotation units are angular.

- **Lowest value:** Zero adds no rotation; negative values rotate in the opposite direction.
- **Highest value:** Larger angles rotate the mapped pattern or model farther. Full turns repeat orientation.

20. 3D Puddles Pro

Builds actual puddle surfaces from detected basin data. The add-on creates a hidden full-basin cache; water-level and shoreline controls operate on that cached basin structure. Centimetre and square-centimetre units are explicit in this interface.

Setup and behavior notes

- Fill To Rim selects rim-based fill. With it off, the add-on uses Puddle Height and forces Rim Clearance to zero.
- A hidden cache is created from the source terrain. Recreate it when the basin-defining terrain changes.
- Enable Wet Mask stores lrfx_puddle_wet_mask and lrfx_puddle_distance; a compatible terrain shader is still required.
- Preview Mode affects viewport evaluation; final renders use full quality.
- Preview Flood Iterations and Adaptive Preview Flood currently have no visible effect.

Practical workflow — 3D Puddles Pro

Select a target mesh and launch **3D Puddles Pro**. When prompted, choose which existing materials should receive the surrounding **wetness** effect, then confirm. The add-on creates a separate, hidden **Basin Object**/reference cache in a collection; the actual water remains generated geometry. The cache is used to identify basin heights and form rounded filled puddles efficiently. **Do not manually edit or delete the hidden basin object.** If the ground geometry changes later, remove and recreate the puddle effect so the cache reflects the new surface.

Choose a fill mode in **Puddle Formation**. With **Fill To Rim** off, **Puddle Height** produces a relative water level across uneven basins. With it on, **Rim Clearance** governs the gap below each basin rim. **Depth Threshold** filters the shallower depressions; larger values generally require deeper basins. **Minimum Area** suppresses tiny puddles; increasing it reduces scattered micro-puddles. **Enable Open Spawning** permits edge-crossing puddles but is experimental; **Disable Height Limit** bypasses a normal height restriction, and **Surface Offset** moves the water surface up or down.

The companion wetness node group changes the ground material near the puddles, and its smoothness depends partly on target mesh resolution. Adjust the wet-distance and **Wet Falloff** controls if transitions appear too abrupt. Puddle appearance can come from the supplied **Dirt**, **Vegetation**, **Cold**, and **Clean** materials, a custom material, and optional ripple motion. Once satisfied, apply/convert the generated puddles to geometry if you need a lighter static result.

Settings

Basin Object

Hidden full-basin cache object created for this terrain by the 3D Puddles Pro operator. It is not an arbitrary decorative mesh.

- **Unassigned or empty:** Leave this unassigned and this part of the effect may not appear.
- **Assigned or connected:** Choose the requested object, collection, or mask.

Depth Threshold

Rejects basins whose full spill depth is below this threshold, measured in centimetres from the basin floor.

- **Lowest value:** Zero permits shallow detected basins.
- **Highest value:** Rejects increasingly deeper basins too; this may leave no puddles.

Puddle Height

Sets the live water height in centimetres above the cached full-basin floor. Used when Fill To Rim is off.

- **Lowest value:** Zero places the level at the floor and can leave no water area.
- **Highest value:** Fills more of the basin until its natural rim limit, unless Disable Height Limit allows overflow.

Rim Clearance

Keeps full-basin water a specified number of centimetres below the escape rim.

- **Lowest value:** Zero permits filling to the natural spill rim.
- **Highest value:** Lowers the water below the rim and can remove shallow puddles. The add-on forces it to zero when Fill To Rim is off.

Minimum Area

Rejects filled ponds smaller than this area in square centimetres.

- **Lowest value:** Zero permits very small water patches.
- **Highest value:** Rejects more small ponds and can remove all output.

Enable Open Spawning

Turns this option on or off.

- **Off:** Rejects shoreline regions requiring closure at an open mesh border.
- **On:** Closes eligible shoreline endpoints at open boundaries to allow puddle planes there.

Disable Height Limit

Turns this option on or off.

- **Off:** Stops filling at the local natural spill or rim limit.
- **On:** Allows overflowing water to merge and continue toward terrain boundaries.

Fill To Rim

Turns this option on or off.

- **Off:** Uses live Puddle Height measured from the cached basin floor.
- **On:** Fills each detected basin toward its lowest escape rim, adjusted by Rim Clearance.

Surface Offset

Adds a small signed display offset to the water surface in centimetres. It does not move the source terrain.

- **Lowest value:** Negative values lower the visible surface. Zero leaves the actual water level.
- **Highest value:** Positive values lift the visible surface; excessive offset can float the water above its shoreline.

Wet Distance

Sets the fully wet distance from generated puddle surfaces in centimetres.

- **Lowest value:** Zero confines the fully wet core to the water boundary.
- **Highest value:** Makes a broad fully wet surrounding zone.

Wet Falloff

Sets the additional distance for the wet mask to fade to dry in centimetres.

- **Lowest value:** Zero gives an abrupt wet or dry transition.
- **Highest value:** Produces a wider gradual transition beyond Wet Distance.

21. 3D Tire Marks

Displaces source terrain using repeated tire-image sampling along a Path curve. Track spacing, individual tire width, tread phase, impression depth and raised or crumbled rims are separate controls.

Setup and behavior notes

- Mesh Resolution increases mesh density; Resolution and Precision set spatial sampling steps, so larger spacing is coarser.
- Footprints use centimetre-labeled inputs, while Tire Marks uses its own scene-scale dimension controls. Do not transfer values unchanged between the two interfaces.

Practical workflow — 3D Tire Marks

Select the deformable ground, add **3D Tire Marks**, and draw the curve along the desired wheel route. This is actual Geometry Nodes displacement, not a projected shader. For a non-displacing alternative, see Tire Marks — Path Data and Tire Marks — Material. Use **Track Width** for the spacing between left/right tire traces and **Tire Width** for the individual trace. Change **Type** or assign a custom tread image for different tire patterns. **Tread Depth** controls tread relief; **Shoulder** shapes the outer track edge, while **Rim Height/Width** and **Crumble** create displaced, irregular ground around the groove. Blur or reduce relief detail if the tread becomes unnaturally sharp.

For overlapping footprint and tire effects, avoid strong displacement from both in the same place. Reduce the strength/depth of one of them or turn off adaptive mesh resolution if the shared ground topology becomes unstable. After finalizing the track shape, apply the modifier if a static mesh is preferable.

Settings

Path

Curve object defining the tire or footprint route. Assign a separate curve, not the terrain host.

- **Unassigned or empty:** Leave this unassigned and this part of the effect may not appear.
- **Assigned or connected:** Choose the requested object, collection, or mask.

Type

Selects a discrete mode. Available choices: `1`, `2`, `3`, `4`.

- **1:** Chooses this option.
- **2:** Chooses this option.
- **3:** Chooses this option.
- **4:** Chooses this option.

Tire Image

Image sampled instead of the built-in tire tread when Use Custom Image? is on.

- **Unassigned or empty:** Leave this unassigned and this part of the effect may not appear.
- **Assigned or connected:** Choose the requested object, collection, or mask.

Invert Image

Turns this option on or off.

- **Off:** Uses the original image-value polarity.
- **On:** Inverts the image sampling polarity, exchanging light or dark relief contributions.

Use Alpha

Turns this option on or off.

- **Off:** Uses the grayscale or color-value image sampling mode.
- **On:** Uses image alpha in the footprint or tread mask. An image with solid alpha can produce a solid silhouette.

Mesh Resolution

Caps the adaptive Subdivide Mesh level calculated from source edge statistics and Resolution spacing. It allows refinement rather than unconditionally forcing that many levels.

- **Lowest value:** At zero, permits no extra subdivision. Fine image relief may not resolve on the source mesh.
- **Highest value:** Allows greater adaptive refinement up to the calculated requirement, with higher potential memory cost. Resolution spacing still controls the requested refinement.

Track Width

Sets the overall lateral track-layout width.

- **Lowest value:** Brings the track layout toward the guide center; zero can overlap the rows.
- **Highest value:** Moves wheel rows farther apart. Tire Width still controls an individual tread width.

Tire Width

Sets the width of an individual tire impression.

- **Lowest value:** Creates narrow tire marks.
- **Highest value:** Creates wider marks, which can overlap neighboring rows.

Tile Length

Sets the length of one repeated tire-image segment along the path.

- **Lowest value:** Repeats the tread more frequently.
- **Highest value:** Makes tread repetitions farther apart and stretches the image longitudinally.

Wheel Offset

Offsets the longitudinal tread phase between wheel tracks.

- **Lowest value:** At zero, uses matching base phase.
- **Highest value:** At one, applies the full normalized phase offset; this changes alignment rather than physical track width.

Mirror Left

Turns this option on or off.

- **Off:** Leaves the left tire-image orientation unchanged.
- **On:** Mirrors the left tread-image coordinate mode.

Depth

Sets the base signed terrain depression, separate from texture relief. Footprint Base Depth uses centimetres.

- **Lowest value:** Zero removes the base depression; negative values reverse the signed displacement.
- **Highest value:** Produces a deeper base impression in the depression mode.

Tread Depth

Weights the tire-image relief within the base impression.

- **Lowest value:** Zero suppresses tread relief.
- **Highest value:** At one, applies the full texture-relief contribution.

Shoulder

Sets the width of the transition at tire shoulders through distance remapping.

- **Lowest value:** Produces a narrow or sharp shoulder transition.
- **Highest value:** Produces a broader or softer shoulder transition.

Rim Height

Sets signed raised or lowered displacement around an impression. Footprint Rim Height uses centimetres.

- **Lowest value:** Zero removes the rim height; negative values lower it.
- **Highest value:** Raises a taller rim around the depression.

Rim Width

Sets the width of the rim region around an impression. Footprint Rim Width uses centimetres.

- **Lowest value:** Produces a narrow rim; zero removes its usable width where accepted.
- **Highest value:** Produces a broad rim region.

Crumble

Sets the amplitude of noise roughening along impression edges. Footprint Crumble uses centimetres.

- **Lowest value:** Zero leaves cleaner impression edges.
- **Highest value:** Adds stronger irregular displacement around the rim.

Crumble Scale

Sets the spatial frequency of the rim-roughening noise.

- **Lowest value:** Low values create broad roughness; zero makes the spatial noise constant.
- **Highest value:** Creates fine, tightly spaced rim irregularity.

End Taper

Sets the distance over which tire impressions fade near the path ends.

- **Lowest value:** Zero minimizes end taper, producing abrupt ends.

- **Highest value:** Creates long gradually fading starts or ends and may weaken most of a short path.

Resolution

Sets the spatial mesh sampling step in the tracks generator. Footprint Resolution uses centimetres.

- **Lowest value:** Small positive spacing gives more samples and finer impressions, at greater cost.
- **Highest value:** Large spacing produces a coarse surface and can miss small impressions. This control works in the opposite direction to subdivision level.

Precision

Sets the curve-resampling or proximity sampling step used for locating impressions. Footprint Precision uses centimetres.

- **Lowest value:** Small spacing samples curved paths more densely and improves placement continuity.
- **Highest value:** Coarse spacing can miss turns or produce uneven or cut-off impression areas.

Flatten Z

Turns this option on or off.

- **Off:** Preserves the path's original height contribution.
- **On:** Flattens the path height in the guide-coordinate mode before locating the impression.

Vertex Group Name

Exact Weight Paint and float-attribute name used by Only In Vertex Group?.

- **Unassigned or empty:** Leave this unassigned and this part of the effect may not appear.
- **Assigned or connected:** Choose the requested object, collection, or mask.

22. 3D Footprints

Displaces terrain with repeated footprint images along a Path curve. Foot placement, base depression, internal tread relief and raised rims are independent; length controls labeled cm are converted internally.

Setup and behavior notes

- Mirror Left is a float weight in this version; Tire Marks uses a Boolean switch with the same label.
- Base Depth is the flat silhouette depression; Relief Depth is the extra tread inside it.

Practical workflow — 3D Footprints

Choose **3D Footprints**, select the ground mesh, and draw the walking path. For a material-only alternative, see Footprints — Path Data and Footprints — Material. The effect locally refines geometry near the drawn curve so the whole ground need not be subdivided uniformly, but adequate mesh quality is still important. **Apply Effect** converts the finished deformation into ordinary mesh geometry. The effect can also be edited directly in Blender's modifier panel.

Use **Depth** or the base-depth controls for how deeply a shoe sinks; **Print Fill** determines how broadly the footprint presses into ground; **Depth Random** makes successive steps less identical; and **Wall Sharpness** controls the edge profile (extremely low values may damage the effect). **Texture Contrast** and relief settings control sole-pattern detail. **Rim Height/Width** and **Crumble** build raised loose material at the edge; **Blur Geometry** softens the overall deformation, while relief-specific smoothing softens the tread/sole imprint. Use a custom image to stamp your own pattern.

Assign **Vertex Group Name** when only part of one mesh should deform—for example, a combined ground/metal-plate/ground object where only the soil should receive footprints. Avoid stacking aggressive 3D Footprints and 3D Tire Marks on the same area; lower deformation depth or disable adaptive resolution if the target mesh breaks.

Settings

Path

Curve object defining the tire or footprint route. Assign a separate curve, not the terrain host.

- **Unassigned or empty:** Leave this unassigned and this part of the effect may not appear.
- **Assigned or connected:** Choose the requested object, collection, or mask.

Type

Selects a discrete mode. Available choices: `Boot`, `Barefoot`, `Animal`.

- **Boot:** Uses the built-in boot footprint.
- **Barefoot:** Uses the built-in barefoot print.
- **Animal:** Uses the built-in animal print.

Custom Footprint Image

Image sampled instead of the built-in footprint when Custom Image? is on.

- **Unassigned or empty:** Leave this unassigned and this part of the effect may not appear.
- **Assigned or connected:** Choose the requested object, collection, or mask.

Invert Image

Turns this option on or off.

- **Off:** Uses the original image-value polarity.
- **On:** Inverts the image sampling polarity, exchanging light or dark relief contributions.

Use Alpha

Turns this option on or off.

- **Off:** Uses the grayscale or color-value image sampling mode.
- **On:** Uses image alpha in the footprint or tread mask. An image with solid alpha can produce a solid silhouette.

Mesh Resolution

Caps the adaptive Subdivide Mesh level calculated from source edge statistics and Resolution spacing. It allows refinement rather than unconditionally forcing that many levels.

- **Lowest value:** At zero, permits no extra subdivision. Fine image relief may not resolve on the source mesh.
- **Highest value:** Allows greater adaptive refinement up to the calculated requirement, with higher potential memory cost. Resolution spacing still controls the requested refinement.

Toe Out

Adds signed outward toe rotation to the footprint placement.

- **Lowest value:** Negative angles rotate in the opposite sense. Zero removes toe-out.
- **Highest value:** Positive angles rotate the toes outward; large angles turn the footprint substantially.

Alternate

Weights the alternating left or right placement contribution.

- **Lowest value:** At zero, removes that alternating contribution.
- **Highest value:** At one, applies full alternating placement.

Mirror Left

Weights horizontal mirroring of the left footprint image. This is a float control in Footprints, not a Boolean.

- **Lowest value:** At zero, leaves the left image unmirrored.
- **Highest value:** At one, fully mirrors it; intermediate values blend the coordinate reflection.

Rotation Jitter

Sets stable random variation of footprint rotation, size, step spacing or depth, as named.

- **Lowest value:** Zero makes the named property uniform.
- **Highest value:** Produces wider per-step variation. It does not necessarily increase the mean size, spacing or depth.

Scale Jitter

Sets stable random variation of footprint rotation, size, step spacing or depth, as named.

- **Lowest value:** Zero makes the named property uniform.
- **Highest value:** Produces wider per-step variation. It does not necessarily increase the mean size, spacing or depth.

Step Jitter

Sets stable random variation of footprint rotation, size, step spacing or depth, as named.

- **Lowest value:** Zero makes the named property uniform.
- **Highest value:** Produces wider per-step variation. It does not necessarily increase the mean size, spacing or depth.

Seed

Changes the repeatable random pattern.

- **Lowest value:** Selects one repeatable pattern, not zero randomness.
- **Highest value:** Selects a different repeatable pattern, not more detail or stronger wind.

Print Fill

Raises the footprint grayscale silhouette response toward a more solid base.

- **Lowest value:** At one, follows the texture grays more closely.
- **Highest value:** Produces a more solid, flatter sole depression while texture relief remains separately controlled.

Texture Contrast

Multiplies or shapes the footprint texture response used by the relief.

- **Lowest value:** Low values soften or suppress image contrast.
- **Highest value:** Makes relief transitions more pronounced; values can saturate or clamp.

Depth Random

Sets stable random variation of footprint rotation, size, step spacing or depth, as named.

- **Lowest value:** Zero makes the named property uniform.
- **Highest value:** Produces wider per-step variation. It does not necessarily increase the mean size, spacing or depth.

Wall Sharpness

Shapes the falloff at the footprint walls.

- **Lowest value:** Produces softer, more gradually sloping sides.
- **Highest value:** Produces sharper transitions at the silhouette boundary.

Crumble Scale

Sets the spatial frequency of the rim-roughening noise.

- **Lowest value:** Low values create broad roughness; zero makes the spatial noise constant.
- **Highest value:** Creates fine, tightly spaced rim irregularity.

Flatten Z

Turns this option on or off.

- **Off:** Preserves the path's original height contribution.
- **On:** Flattens the path height in the guide-coordinate mode before locating the impression.

Vertex Group Name

Exact Weight Paint and float-attribute name used by Only In Vertex Group?.

- **Unassigned or empty:** Leave this unassigned and this part of the effect may not appear.
- **Assigned or connected:** Choose the requested object, collection, or mask.

Blur Geometry

Sets blur iterations for the displacement or internal relief field, as named.

- **Lowest value:** Zero preserves the unblurred detail.
- **Highest value:** Smooths more strongly and may wash out fine tread; many iterations increase evaluation cost.

23. Edge Dirt Shader

Layers procedural dirt over an incoming shader using nested edge-finding, directional or Z masks, grunge colors, roughness and bump.

Practical workflow — Edge Dirt

Apply **Edge Dirt** to the material whose edges need additional grime. This is a simpler edge-based alternative to the more layered **Pro Dirt** and **Pro Rust** setups. Start with **Mask Strength** to establish how much dirt is visible, then set **Spread** to decide how far the effect reaches from the detected edges. The material blends **Color A**, **Color B**, and **Color C**, allowing different tones within the dirt pattern. **C Color Amount** adjusts the contribution of the third color.

Use **Texture Scale** to change the size of the procedural grunge pattern and **Texture Contrast** to make its breakup softer or more selective. **Mask Saturation** adjusts the mask processing; it is not the same as overall material opacity. Finish with **Roughness**, **Dirt Roughness Scale**, and **Bump Strength** to give the dirt its surface character. Texture-coordinate changes, where exposed by the shader, can be used to change the mapping or use an appropriate UV map.

Settings

Enable

Turns this option on or off.

- **Off:** Uses the incoming shader without the enabled edge-dirt contributions.
- **On:** Applies the edge-dirt shading or mask parts of the effect.

Mask Strength

Multiplies the edge-dirt mask contribution.

- **Lowest value:** Zero removes this mask contribution.
- **Highest value:** Strengthens the mask and can saturate it.

Color A

Sets the color of the effect. For A, B, or C colors, the procedural mixing mask decides where each color appears.

- **Lowest value:** Black or darker colors darken the affected area. RGB zero does not disable the whole effect.
- **Highest value:** White or lighter colors brighten the affected area. Hue changes color, not particle count or geometry depth.

Color B

Sets the color of the effect. For A, B, or C colors, the procedural mixing mask decides where each color appears.

- **Lowest value:** Black or darker colors darken the affected area. RGB zero does not disable the whole effect.
- **Highest value:** White or lighter colors brighten the affected area. Hue changes color, not particle count or geometry depth.

Spread

Sets the reach of edge-dirt detection through the nested edge-finder mode.

- **Lowest value:** Restricts influence close to the detected features.
- **Highest value:** Extends the edge-dirt influence across a wider region; the underlying detection still depends on geometry and shading.

Texture Scale

Controls the size of the procedural edge-dirt grunge pattern by scaling its texture coordinates.

- **Lowest value:** Low or zero values give broad or constant spatial variation.
- **Highest value:** Creates finer, more frequent variation. This setting controls texture coordinates rather than random particle count.

Roughness

Controls how matte or glossy the affected area looks.

- **Lowest value:** Low values create sharper reflections; zero is the smooth limit. Values below zero are outside the useful physical range and are clamped by shading.
- **Highest value:** High values broaden reflections. One is the fully rough physical limit; accepting a larger input does not establish additional physical roughness.

Dirt Roughness Scale

Sets coordinate scale of the grunge used by dirt roughness.

- **Lowest value:** Makes large-scale or spatially constant roughness variation.
- **Highest value:** Produces fine-grained roughness variation.

Bump Strength

Controls how strong the surface bump appears.

- **Lowest value:** Zero leaves the normal unchanged by this bump contribution.
- **Highest value:** Creates stronger apparent relief. Bump changes shading normals; it does not by itself displace mesh vertices.

Color C

Sets the color of the effect. For A, B, or C colors, the procedural mixing mask decides where each color appears.

- **Lowest value:** Black or darker colors darken the affected area. RGB zero does not disable the whole effect.
- **Highest value:** White or lighter colors brighten the affected area. Hue changes color, not particle count or geometry depth.

★Texture Contrast

Remaps contrast of the edge-dirt grunge parts of the effect.

- **Lowest value:** Creates softer, less selective grunge.
- **Highest value:** Creates stronger grunge contrast or selection, subject to its internal thresholds.

★Mask Saturation

Adjusts the saturation control inside the grunge-mask parts of the effect.

- **Lowest value:** Suppresses the saturation contribution.
- **Highest value:** Applies the full saturation adjustment. This is a texture-mask processing control, not material opacity.

★C Color Amount

Sets the lower output value of the field used to blend Color C.

- **Lowest value:** Zero leaves Color C controlled entirely by the original field.
- **Highest value:** Makes this part of the effect stronger; very high values may overpower the material.

24. Dust Shader

Adds a dust material over the incoming shader using surface normal Z, procedural noise and a shaded blend mask. It produces shading, not airborne dust particles.

Setup and behavior notes

- Height thresholds normal Z rather than measuring world-space altitude.
- Amount multiplies the blend-mask value. Zero selects the original shader.

Practical workflow — regular Dust shader

Use **Dust** for a lightweight dusty coating rather than the more detailed top-and-crevice layering of **Pro Dust**. Apply it to the target material, choose **Base Color**, and set **Amount** to control how strongly the dust blends with the original surface. **Scale** and **Detail** change the grain of the procedural mask.

The **Height** control responds to surface orientation, not the object's height above the ground. Use it to make dust favor upward-facing faces or extend more gradually onto sloping faces. For dust that gathers distinctly inside corners, use **Pro Dust** instead; for three-dimensional fibers and specks, see **Dust on Object**.

Settings

Base Color

Sets the color of the effect. For A, B, or C colors, the procedural mixing mask decides where each color appears.

- **Lowest value:** Black or darker colors darken the affected area. RGB zero does not disable the whole effect.
- **Highest value:** White or lighter colors brighten the affected area. Hue changes color, not particle count or geometry depth.

Amount

Scales the brightness or value of the dust blend mask before Mix Shader.

- **Lowest value:** At zero, the mask becomes black and the original Shader is selected.
- **Highest value:** At one, preserves the mask's full value; surface orientation and noise still control coverage.

Scale

Changes the scale of the texture pattern without resizing the object.

- **Lowest value:** Low positive values create larger marks; zero makes spatial input constant. Negative values reflect coordinates.
- **Highest value:** High positive values create smaller, more tightly repeated marks. Texture resolution remains unchanged.

Height

Sets the lower threshold of the remapped surface-normal Z field. Despite its name, it is an orientation threshold, not world-space height.

- **Lowest value:** A lower threshold admits more normal directions into the dust mask.
- **Highest value:** A higher threshold restricts coverage toward upward normals; extreme values can collapse the remapping.

Detail

Adds higher-frequency octaves to the named procedural noise.

- **Lowest value:** Zero uses the base noise layer only.
- **Highest value:** Adds fine noise layers and potentially greater shader or geometry evaluation cost.

25. Rust Shader

Blends a procedural two-color rust material over an incoming BSDF, with separate selection contrast, texture-coordinate scale and bump.

Practical workflow — regular Rust shader

Choose the regular **Rust** shader for a relatively quick two-color corrosion effect when you do not need the separate **Edge Rust** and **Wall/Ground Rust** modules of **Pro Rust**. Start with **Intensity**, select **A Color** and **B Color**, and tune **Contrast** until the rust appears in the desired areas. **B Color Amount** changes the selection of the second rust tone, while **Scale** controls the size of the pattern. Adjust **Roughness** and **Bump Strength** to make the coating look appropriately weathered. This effect changes the material; it does not physically corrode the mesh.

Settings

Intensity

Weights the added effect against the incoming material shader.

- **Lowest value:** At zero, selects the original-material side of the effect blend.
- **Highest value:** At one, allows the full added effect wherever its procedural mask is active. The original shader can remain outside that mask.

Roughness

Controls how matte or glossy the affected area looks.

- **Lowest value:** Low values create sharper reflections; zero is the smooth limit. Values below zero are outside the useful physical range and are clamped by shading.
- **Highest value:** High values broaden reflections. One is the fully rough physical limit; accepting a larger input does not establish additional physical roughness.

Contrast

Sets the lower threshold used to remap the procedural or image mask.

- **Lowest value:** Low thresholds admit more of the mask field.
- **Highest value:** High thresholds retain only stronger field values, often creating smaller, more selective patches. It changes mask selection rather than image resolution.

Scale

Changes the scale of the texture pattern without resizing the object.

- **Lowest value:** Low positive values create larger marks; zero makes spatial input constant. Negative values reflect coordinates.
- **Highest value:** High positive values create smaller, more tightly repeated marks. Texture resolution remains unchanged.

Bump Strength

Controls how strong the surface bump appears.

- **Lowest value:** Zero leaves the normal unchanged by this bump contribution.
- **Highest value:** Creates stronger apparent relief. Bump changes shading normals; it does not by itself displace mesh vertices.

A Color

Sets the color of the effect. For A, B, or C colors, the procedural mixing mask decides where each color appears.

- **Lowest value:** Black or darker colors darken the affected area. RGB zero does not disable the whole effect.
- **Highest value:** White or lighter colors brighten the affected area. Hue changes color, not particle count or geometry depth.

B Color

Sets the color of the effect. For A, B, or C colors, the procedural mixing mask decides where each color appears.

- **Lowest value:** Black or darker colors darken the affected area. RGB zero does not disable the whole effect.

- **Highest value:** White or lighter colors brighten the affected area. Hue changes color, not particle count or geometry depth.

B Color Amount

Changes the remapping threshold used to mix the rust B color.

- **Lowest value:** A lower threshold admits a wider range of the second-color field.
- **Highest value:** A higher threshold makes the second-color response more selective. This input is a mask threshold rather than a literal percentage of B color.

26. Scratches Shader

Blends image or procedural scratch shading over an incoming BSDF. Scale maps texture coordinates; bump alters shading normals rather than cutting mesh grooves.

Practical workflow — Scratches

Apply **Scratches** to the relevant material. Begin with **Intensity** so the scratches remain in proportion to the underlying texture. Then choose **Base Color** and **Roughness**, and use **Contrast** to make the lines more or less defined. **Scale** adjusts the size or repetition of the scratch texture. **Bump Strength** provides apparent depth in the shading without cutting grooves into the actual mesh.

Settings

Intensity

Weights the added effect against the incoming material shader.

- **Lowest value:** At zero, selects the original-material side of the effect blend.
- **Highest value:** At one, allows the full added effect wherever its procedural mask is active. The original shader can remain outside that mask.

Base Color

Sets the color of the effect. For A, B, or C colors, the procedural mixing mask decides where each color appears.

- **Lowest value:** Black or darker colors darken the affected area. RGB zero does not disable the whole effect.
- **Highest value:** White or lighter colors brighten the affected area. Hue changes color, not particle count or geometry depth.

Roughness

Controls how matte or glossy the affected area looks.

- **Lowest value:** Low values create sharper reflections; zero is the smooth limit. Values below zero are outside the useful physical range and are clamped by shading.
- **Highest value:** High values broaden reflections. One is the fully rough physical limit; accepting a larger input does not establish additional physical roughness.

Contrast

Sets the lower threshold used to remap the procedural or image mask.

- **Lowest value:** Low thresholds admit more of the mask field.

- **Highest value:** High thresholds retain only stronger field values, often creating smaller, more selective patches. It changes mask selection rather than image resolution.

Scale

Changes the scale of the texture pattern without resizing the object.

- **Lowest value:** Low positive values create larger marks; zero makes spatial input constant. Negative values reflect coordinates.
- **Highest value:** High positive values create smaller, more tightly repeated marks. Texture resolution remains unchanged.

Bump Strength

Controls how strong the surface bump appears.

- **Lowest value:** Zero leaves the normal unchanged by this bump contribution.
- **Highest value:** Creates stronger apparent relief. Bump changes shading normals; it does not by itself displace mesh vertices.

27. Dirt Particles Shader

Uses image or noise masks and bump to shade fine dirt-like marks over the incoming BSDF. Despite the name, this group does not spawn Geometry Nodes particles.

Practical workflow — Dirt Particles shader

Use this shader to add small, fine dirt marks directly to a material. Despite its name, it is a **shader effect**, not the geometry-based **Dust on Object** generator. Select **Base Color**, balance **Intensity** and **Contrast** for visibility, and tune **Scale** for the size of the dirt pattern. **Roughness** controls how the affected areas reflect light. Use **Dust on Object** if you need individually modeled fibers or dust particles instead.

Settings

Intensity

Weights the added effect against the incoming material shader.

- **Lowest value:** At zero, selects the original-material side of the effect blend.
- **Highest value:** At one, allows the full added effect wherever its procedural mask is active. The original shader can remain outside that mask.

Base Color

Sets the color of the effect. For A, B, or C colors, the procedural mixing mask decides where each color appears.

- **Lowest value:** Black or darker colors darken the affected area. RGB zero does not disable the whole effect.
- **Highest value:** White or lighter colors brighten the affected area. Hue changes color, not particle count or geometry depth.

Roughness

Controls how matte or glossy the affected area looks.

- **Lowest value:** Low values create sharper reflections; zero is the smooth limit. Values below zero are outside the useful physical range and are clamped by shading.

- **Highest value:** High values broaden reflections. One is the fully rough physical limit; accepting a larger input does not establish additional physical roughness.

Contrast

Sets the lower threshold used to remap the procedural or image mask.

- **Lowest value:** Low thresholds admit more of the mask field.
- **Highest value:** High thresholds retain only stronger field values, often creating smaller, more selective patches. It changes mask selection rather than image resolution.

Scale

Changes the scale of the texture pattern without resizing the object.

- **Lowest value:** Low positive values create larger marks; zero makes spatial input constant. Negative values reflect coordinates.
- **Highest value:** High positive values create smaller, more tightly repeated marks. Texture resolution remains unchanged.

28. Fingerprints Shader

Applies fingerprint-like texture shading over the incoming BSDF, with separate blend intensity, roughness, threshold contrast and coordinate scale.

Practical workflow — Fingerprints

Apply **Fingerprints** to the material where hand contact should be visible. Adjust **Intensity** first, then balance **Contrast** so the prints are readable without overpowering the original surface. Use **Base Color** and **Roughness** to match the surface, and **Scale** to size the fingerprint pattern. These marks are material shading, not decals or physical surface geometry.

Settings

Intensity

Weights the added effect against the incoming material shader.

- **Lowest value:** At zero, selects the original-material side of the effect blend.
- **Highest value:** At one, allows the full added effect wherever its procedural mask is active. The original shader can remain outside that mask.

Base Color

Sets the color of the effect. For A, B, or C colors, the procedural mixing mask decides where each color appears.

- **Lowest value:** Black or darker colors darken the affected area. RGB zero does not disable the whole effect.
- **Highest value:** White or lighter colors brighten the affected area. Hue changes color, not particle count or geometry depth.

Roughness

Controls how matte or glossy the affected area looks.

- **Lowest value:** Low values create sharper reflections; zero is the smooth limit. Values below zero are outside the useful physical range and are clamped by shading.

- **Highest value:** High values broaden reflections. One is the fully rough physical limit; accepting a larger input does not establish additional physical roughness.

Contrast

Sets the lower threshold used to remap the procedural or image mask.

- **Lowest value:** Low thresholds admit more of the mask field.
- **Highest value:** High thresholds retain only stronger field values, often creating smaller, more selective patches. It changes mask selection rather than image resolution.

Scale

Changes the scale of the texture pattern without resizing the object.

- **Lowest value:** Low positive values create larger marks; zero makes spatial input constant. Negative values reflect coordinates.
- **Highest value:** High positive values create smaller, more tightly repeated marks. Texture resolution remains unchanged.

29. Edge Wear

Edge Wear combines geometry-derived edge data with a material shader. Keep both parts on the receiving object: the **Edges Data** stage defines the affected edges, and the **Shader** stage controls their visual appearance.

Edge Wear — Edges Data (Geometry Nodes)

Prepares an object for the Edge Wear material. Add this effect to the same object before adjusting the Edge Wear shader.

Setup and behavior notes

- Add Edges Data to the object, then apply the matching Edge Wear material to see the worn areas.

Practical workflow — preparing Edge Wear geometry

Edge Wear uses geometry-derived edge data together with its companion shader. Add the **Edges Data** effect to the target object, then apply the **Edge Wear — Shader** to the material. **Angle Threshold** determines which changes in face angle count as wear-producing edges. **Edge Width** (the label retained in this version) controls the edge-region width, while **On Z Surfaces Amount** decides how much additional wear appears on upward-facing flat areas. Set that last value to zero if you do not want the extra top-surface contribution.

Settings

Angle Threshold

Sets the mesh edge-angle threshold used for detecting sharp changes.

- **Lowest value:** A small angle admits shallow changes and selects more edges.
- **Highest value:** A large angle selects only strong changes and can exclude nearly all edges.

Edge Windth

Sets edge-mask sample radius or distance. The original interface spelling is preserved.

- **Lowest value:** Creates a narrow edge sampling region.
- **Highest value:** Samples a wider region and broadens the edge influence.

On Z Surfaces Amount

Weights the top-facing contribution used by edge sampling.

- **Lowest value:** Zero removes the extra top-facing multiplier contribution.
- **Highest value:** Increases top-facing influence; it does not change the sharp-edge angle threshold.

Edge Wear — Shader (Material)

Uses Edges Data attributes to blend a built-in or custom worn material over the incoming shader. Detailed overlay, noise breakup, color, roughness and bump are exposed separately.

Setup and behavior notes

- Keep Edges Data on the same object as the Edge Wear material.
- Custom Shader IN takes a shader, not image color.

Practical workflow — Edge Wear shader

Apply the Edge Wear shader to a material on an object that already has the companion **Edges Data** effect. The geometry stage identifies eligible edges using **Angle Threshold**, sets their reach with **Edge Windth**, and can also add wear on top-facing surfaces. The shader then uses this information to blend worn material around those areas.

Set **Edge Color** and **Edge Roughness**, or connect a material shader to **Custom Shader IN** where available. **Invert Mask** exchanges worn and unworn regions. Adjust **Bump Strength** and **Filter Width** to refine the apparent surface relief. In **Detail Mask Overlay**, use **Noise Offset** and the noise-scale controls to introduce irregularity instead of perfectly consistent edge wear. The **Add Noise Breakup?** option and **Noise Breakup Scale** belong to the additional breakup layer; if an older asset library shows no change when enabled, check that the correct node-group version is installed and connected before depending on that layer.

Settings

Master Seed

Changes the repeatable random pattern.

- **Lowest value:** Selects one repeatable pattern, not zero randomness.
- **Highest value:** Selects a different repeatable pattern, not more detail or stronger wind.

Edge Color

Sets the color of the effect. For A, B, or C colors, the procedural mixing mask decides where each color appears.

- **Lowest value:** Black or darker colors darken the affected area. RGB zero does not disable the whole effect.
- **Highest value:** White or lighter colors brighten the affected area. Hue changes color, not particle count or geometry depth.

Edge Roughness

Controls how matte or glossy the affected area looks.

- **Lowest value:** Low values create sharper reflections; zero is the smooth limit. Values below zero are outside the useful physical range and are clamped by shading.
- **Highest value:** High values broaden reflections. One is the fully rough physical limit; accepting a larger input does not establish additional physical roughness.

Noise Offset

Shifts the procedural mask or texture coordinates or remapping in the named mode.

- **Lowest value:** Negative or low values select an earlier or shifted pattern or threshold.
- **Highest value:** Positive or high values select another shifted pattern; this does not intrinsically add geometry detail.

#1 Noise Scale

Controls the size of the visible texture pattern.

- **Lowest value:** Low positive values create broad features; zero creates constant spatial input.
- **Highest value:** Creates fine, tightly spaced texture features. Higher coordinate scale makes smaller marks rather than larger objects.

#2 Noise Scale

Controls the size of the visible texture pattern.

- **Lowest value:** Low positive values create broad features; zero creates constant spatial input.
- **Highest value:** Creates fine, tightly spaced texture features. Higher coordinate scale makes smaller marks rather than larger objects.

Add Noise Breakup?

Enables the additional procedural breakup layer where that branch is connected in the installed shader group.

- **Off:** Uses the normal detail-mask pattern without this extra breakup contribution.
- **On:** Adds another source of irregularity. If nothing changes, check whether your installed node-group version has this branch connected.

Noise Breakup Scale

Scales the optional additional breakup pattern, when Add Noise Breakup? is enabled and the branch is connected.

- **Lowest value:** Creates broader breakup features.
- **Highest value:** Creates finer, more closely spaced breakup features.

Invert Mask

Turns this option on or off.

- **Off:** Applies the normal edge selection.
- **On:** Exchanges selected and unselected edge-mask regions.

Bump Strength

Controls how strong the surface bump appears.

- **Lowest value:** Zero leaves the normal unchanged by this bump contribution.
- **Highest value:** Creates stronger apparent relief. Bump changes shading normals; it does not by itself displace mesh vertices.

Filter Width

Sets the bump filter footprint in pixels.

- **Lowest value:** Small positive widths preserve fine or subpixel relief but can reveal sharper transitions.
- **Highest value:** Larger widths smooth stepwise height transitions and can give a bevel-like shading edge; they do not bevel geometry.

30. Ground Effect

Ground Effect uses a geometry-based ground reference and a matching dirt material. The **Ground Mask** stage determines where the dirt should appear; the **Shader** stage controls the visible stain.

Ground Effect — Ground Mask (Geometry Nodes)

Creates the ground mask used by the Ground Effect material. Choose the ground object and adjust how far the effect reaches above it.

Setup and behavior notes

- Choose the ground object that the effect should follow.
- Use the matching Ground Effect material to show the mask.

Practical workflow — selecting the ground reference

The **Ground Mask** is the geometry-dependent part of Ground Effect. After starting the effect, select the object that represents the ground and confirm the choice with **Enter** when the placement tool requests it. The add-on uses the spatial relationship between this object and the target mesh to generate a mask. Reassign the ground object if the scene layout changes; **Dirt Height** sets how high the contact stain is allowed to extend. Use **Ground Effect — Shader** on the same receiving object to see the dirt material. The ground-reference object and the object receiving the shader can be different objects.

Settings

Ground Effect — Shader (Material)

Layers ground dirt over Material using the ground attribute and procedural breakup. It includes built-in dirt shading or a custom dirt shader, cutout controls and dirt or splash mapping.

Setup and behavior notes

- Add Ground Mask to the object before using this material.

Practical workflow — ground-contact dirt material

Apply **Ground Effect** to the wall or object that should show staining near another surface, and choose the object that serves as **Ground** using the companion **Ground Mask** effect. Select the affected material and allow the ground mask to drive the shader. The mask is based on the distance between the target and the ground object, so it is useful for walls, foundations, and objects touching the floor.

Start with **Dirt Height** in the ground-mask setup to control the vertical reach. In the material settings, use **Ignore Tops** to avoid staining upward-facing surfaces and **Fade with height** to fade the effect as it rises. **Mask Contrast** changes the sharpness of that boundary. For a less uniform contact band, adjust **Noise Mask Offset**, the two **Noise Scale** values, and **Master Seed**. If you prefer your own material for the ground dirt, enable **Custom Dirt Shader?** and connect a shader to **Custom Dirt Shader**; the built-in dirt shading remains available when this override is disabled.

Settings

Custom Dirt Shader

Supplies the custom effect shader used when the associated Custom Shader toggle is on.

- **Unassigned or empty:** Leave this empty and the affected material area may not appear.
- **Assigned or connected:** Choose a shader for the material surface.

Custom Dirt Shader?

Turns this option on or off.

- **Off:** Uses the internal dirt shader.
- **On:** Uses the connected Custom Dirt Shader as the dirt-material mode.

Ignore Tops

Turns this option on or off.

- **Off:** Retains top-facing parts in the ground-shader mask.
- **On:** Suppresses upward-facing surfaces in that mask.

Fade with height

Weights a Generated-Z fade in the ground shader. This is a float influence rather than a Boolean switch.

- **Lowest value:** Zero omits the added upward fade contribution.
- **Highest value:** One applies the full Generated-Z fade; intermediate values blend it.

Mask Contrast

Changes the remapping response of the ground dirt mask.

- **Lowest value:** Produces a softer mask response.
- **Highest value:** Makes the ground mask more selective or contrasty, subject to its internal remapping.

Noise Mask Offset

Shifts the procedural mask or texture coordinates or remapping in the named mode.

- **Lowest value:** Negative or low values select an earlier or shifted pattern or threshold.
- **Highest value:** Positive or high values select another shifted pattern; this does not intrinsically add geometry detail.

#1 Noise Scale

Controls the size of the visible texture pattern.

- **Lowest value:** Low positive values create broad features; zero creates constant spatial input.
- **Highest value:** Creates fine, tightly spaced texture features. Higher coordinate scale makes smaller marks rather than larger objects.

#2 Noise Scale

Controls the size of the visible texture pattern.

- **Lowest value:** Low positive values create broad features; zero creates constant spatial input.
- **Highest value:** Creates fine, tightly spaced texture features. Higher coordinate scale makes smaller marks rather than larger objects.

Master Seed

Changes the repeatable random pattern.

- **Lowest value:** Selects one repeatable pattern, not zero randomness.
- **Highest value:** Selects a different repeatable pattern, not more detail or stronger wind.

Dirt Color A

Sets the color of the effect. For A, B, or C colors, the procedural mixing mask decides where each color appears.

- **Lowest value:** Black or darker colors darken the affected area. RGB zero does not disable the whole effect.
- **Highest value:** White or lighter colors brighten the affected area. Hue changes color, not particle count or geometry

depth.

Dirt Color B

Sets the color of the effect. For A, B, or C colors, the procedural mixing mask decides where each color appears.

- **Lowest value:** Black or darker colors darken the affected area. RGB zero does not disable the whole effect.
- **Highest value:** White or lighter colors brighten the affected area. Hue changes color, not particle count or geometry depth.

Dirt Level

Controls how high the shaded dirt region climbs from the ground-mask contribution.

- **Lowest value:** Keeps the dirt region low.
- **Highest value:** Extends the dirt farther upward, interacting with the ground data and masks.

Contact Dirt

Adds extra dirt where the ground or contact mask is active.

- **Lowest value:** Zero removes extra contact concentration.
- **Highest value:** Makes contact regions dirtier, subject to mask clamping.

Dirt Seed

Offsets the fourth-dimension noise coordinate in dirt grain, level warp and splash textures.

- **Lowest value:** Selects one repeatable texture pattern.
- **Highest value:** Selects a different repeatable pattern, not a higher detail level.

Dirt Roughness

Controls how matte or glossy the affected area looks.

- **Lowest value:** Low values create sharper reflections; zero is the smooth limit. Values below zero are outside the useful physical range and are clamped by shading.
- **Highest value:** High values broaden reflections. One is the fully rough physical limit; accepting a larger input does not establish additional physical roughness.

Dirt Bump

Controls how strong the surface bump appears.

- **Lowest value:** Zero leaves the normal unchanged by this bump contribution.
- **Highest value:** Creates stronger apparent relief. Bump changes shading normals; it does not by itself displace mesh vertices.

Dirt Grain Scale

Controls the size of the visible texture pattern.

- **Lowest value:** Low positive values create broad features; zero creates constant spatial input.
- **Highest value:** Creates fine, tightly spaced texture features. Higher coordinate scale makes smaller marks rather than larger objects.

Dirt Location

Offsets the dirt texture coordinates in XYZ without moving the object.

- **Lowest value:** Negative components move the pattern in the negative coordinate direction.
- **Highest value:** Positive components move it in the positive direction; they do not increase detail.

Dirt Rotation

Sets extra per-axis texture mapping or spider-model rotation, as named. Rotation units are angular.

- **Lowest value:** Zero adds no rotation; negative values rotate in the opposite direction.
- **Highest value:** Larger angles rotate the mapped pattern or model farther. Full turns repeat orientation.

Dirt Scale

Scales dirt texture coordinates per axis, not the object mesh.

- **Lowest value:** Small positive components stretch texture features along that axis; zero collapses spatial variation.
- **Highest value:** Large components create smaller or repeated texture features; negative components reflect coordinates.

Splash Scale

Controls the size of the visible texture pattern.

- **Lowest value:** Low positive values create broad features; zero creates constant spatial input.
- **Highest value:** Creates fine, tightly spaced texture features. Higher coordinate scale makes smaller marks rather than larger objects.

Splash Detail

Adds octaves to the procedural dirt-splash pattern.

- **Lowest value:** Uses broad base splash noise.
- **Highest value:** Adds finer detail at greater shader evaluation cost.

Splash Distortion

Warps the coordinates of the corresponding procedural noise.

- **Lowest value:** Zero leaves the noise undistorted.
- **Highest value:** Creates more twisted or irregular features. Signed negative distortion changes the warp direction where accepted.

31. Leaks Shader

Adds directional moist and additive leak shading using nested ambient-occlusion tracing.

Practical workflow — Leaks

Leaks is intended primarily for **Cycles** and usually behaves most predictably on architectural meshes—walls, ledges, façades, and roofs. It can also add interest to statues and other detailed models, but complex shapes may require more manual adjustment. Select the receiving object, add Leaks, and choose the materials to be affected. The generator looks for suitable upward-facing surfaces and protrusions, from which it creates downward moisture streaks. Even small ledges may produce visible leaking; surface angle matters, so check the shape of the source geometry when the streaks appear in unexpected places.

Set **Moist Distance** to determine how far the moisture influence reaches and **Moist Scale** to control its pattern. Adjust the moisture color, then refine the **Drips** settings—particularly the streak length and drip color. A custom shader may be used for the effect instead of the built-in appearance. Disable **Enable on Top** when you do not want leaks originating on top faces; use **Flip Direction** when the effect needs to flow in the opposite direction. Review the outcome in rendered Cycles

view rather than relying solely on a material-preview approximation.

Settings

32. Quick Puddles Shader

Blends procedural or painted puddles into an existing material and supports ripple bump, displacement passthrough and animated boundary ripples. It is separate from the basin-based 3D Puddles Pro generator.

Setup and behavior notes

- Actual shader displacement requires the displacement output and an appropriate render or material setup.

Practical workflow — scattered and painted puddles

Quick Puddles is a material-based effect, unlike **3D Puddles Pro**, which creates separate puddle geometry. Select an asphalt, pavement, or ground object; add the shader and select the materials to receive it. The operator may add a **Surface Flatness** Geometry Nodes modifier to help decide where puddles can appear. **Keep this modifier**, as its generated data can be needed for the flatness-aware distribution.

Choose **Spawn Everywhere** to distribute puddles over the selected material without the flatness restriction, or **Spawn Only on Flat** to limit puddles to suitably level areas. In the second mode, adjust **Sensitivity** to keep pools away from steep or irregular parts of the mesh. **Wetness** is the primary overall coverage control; **Puddles Contrast** refines the transition and the surrounding wetness. Use **Seed**, **Scale**, **Detail**, and **Distortion** to vary puddle shapes. Texture mapping can use **Object** coordinates or the object's active **UV** map, depending on the mode in your installed version.

For the water appearance, start with **Water Inside Strength** and **Depth Water Factor**, then adjust roughness, darkness, and the reflective/transmissive properties of the water material. The edge control is now named **Water Edge Sharpness**: a value near **0** produces a softer boundary, while a value near **1** produces a sharp transition. A soft transition often blends more naturally into asphalt. **Ripple Height**, **Ripple Scale**, **Ripple Bump**, and **Slow down time Ripples** control the animated surface ripples. The separate **Edge Ripple** settings determine the ripples generated around the water boundary, including strength, speed, frequency, and how far the wave area reaches inward and outward. Edge ripples are easier to see with a more defined puddle edge.

Painting puddles. To place puddles precisely, assign the required UV map and use **Create Mask + Paint** in the painted-mask workflow. Paint the desired wet region, then enable **Distort Painted Edges** and adjust its scale and strength to break up unnaturally perfect boundaries. **Draw Puddles** starts directly in the painting workflow; the material controls otherwise remain substantially the same. Painted and procedural puddles can be refined using the same water and ripple settings.

Settings

Your Displacement

Passes the original displacement vector into the existing-material side of the puddle displacement blend.

- **Lowest value:** Zero removes this original displacement contribution. Negative components reverse the axis displacement.
- **Highest value:** Larger components add larger displacement on the applicable mode; the output must be connected and supported by the render or material setup.

Color

Sets the color of the effect. For A, B, or C colors, the procedural mixing mask decides where each color appears.

- **Lowest value:** Black or darker colors darken the affected area. RGB zero does not disable the whole effect.
- **Highest value:** White or lighter colors brighten the affected area. Hue changes color, not particle count or geometry

depth.

Height

Supplies a color or scalar height-like source to puddle height or inside remapping. It is a texture input rather than one global depth in centimetres.

- **Lowest value:** Black gives the lowest value for this texture.
- **Highest value:** White gives the highest value, with grays in between.

Normal

Use a grayscale height texture to control bumps. A standard normal-map image will not give the intended result here.

- **Lowest value:** Black gives the lowest value for this texture.
- **Highest value:** White gives the highest value, with grays in between.

Wetness

Strengthens the procedural wet or puddle response through its mask or remapping parts of the effect.

- **Lowest value:** Zero suppresses this wetness contribution.
- **Highest value:** Raises the wet or puddle response and can saturate coverage. Other masks still determine placement.

Puddles Contrast

Shapes the contrast or threshold response of the puddle noise.

- **Lowest value:** Leaves softer, less sharply separated transitions.
- **Highest value:** Produces stronger separation between puddle and dry regions; coverage may change.

Water Inside Strength

Scales the maximum interior-water response in the puddle mask remapping.

- **Lowest value:** Zero removes this interior strength.
- **Highest value:** Strengthens the puddle interior response and can saturate the downstream blend.

Seed

Changes the repeatable random pattern.

- **Lowest value:** Selects one repeatable pattern, not zero randomness.
- **Highest value:** Selects a different repeatable pattern, not more detail or stronger wind.

Scale

Changes the scale of the texture pattern without resizing the object.

- **Lowest value:** Low positive values create larger marks; zero makes spatial input constant. Negative values reflect coordinates.
- **Highest value:** High positive values create smaller, more tightly repeated marks. Texture resolution remains unchanged.

Detail

Adds higher-frequency octaves to the named procedural noise.

- **Lowest value:** Zero uses the base noise layer only.
- **Highest value:** Adds fine noise layers and potentially greater shader or geometry evaluation cost.

Distortion

Warps the coordinates of the corresponding procedural noise.

- **Lowest value:** Zero leaves the noise undistorted.
- **Highest value:** Creates more twisted or irregular features. Signed negative distortion changes the warp direction where accepted.

Water Edge Sharpness

Controls how clearly the water boundary is defined. This setting was formerly called Water Edge Softness. It affects the shader transition rather than changing the geometric radius of the puddle.

- **Lowest value:** At 0, the boundary is softer, with a gradual transition into surrounding wetness.
- **Highest value:** At 1, the water edge is sharp and clearly defined; other wetness and depth controls still affect the final appearance.

Depth Water Factor

Sets the second value in the water-depth or edge-factor mix.

- **Lowest value:** Reduces or reverses this contribution for signed negative values.
- **Highest value:** Increases the contribution where its mix mode is selected; this is not a guaranteed physical water depth.

Water darkness

Sets the color of the effect. For A, B, or C colors, the procedural mixing mask decides where each color appears.

- **Lowest value:** Black or darker colors darken the affected area. RGB zero does not disable the whole effect.
- **Highest value:** White or lighter colors brighten the affected area. Hue changes color, not particle count or geometry depth.

Ripple Height

Weights the procedural ripple-height field passed to the ripple module.

- **Lowest value:** Zero removes this height contribution.
- **Highest value:** Creates stronger ripple field relief. Mesh displacement requires the displacement output and an appropriate render setup.

Ripple Scale

Sets spatial frequency or scale of the procedural ripple pattern.

- **Lowest value:** Low values make broad ripple features.
- **Highest value:** Creates more tightly spaced ripples in the coordinate field.

Ripple Bump

Controls how strong the surface bump appears.

- **Lowest value:** Zero leaves the normal unchanged by this bump contribution.
- **Highest value:** Creates stronger apparent relief. Bump changes shading normals; it does not by itself displace mesh

vertices.

Slow down time Ripples

Divides the time-driven ripple phase by this control.

- **Lowest value:** Small positive values make the ripple animation faster. Zero is a singular divisor and should be avoided.
- **Highest value:** Large positive values slow ripple animation; negative values reverse phase progression.

Edge Ripple Inner Width

Sets the width of the ripple zone inside the puddle boundary.

- **Lowest value:** Confines edge ripples to a narrow inner strip.
- **Highest value:** Extends ripples farther into the water surface.

Edge Ripple Outer Width

Sets permitted ripple reach outside the puddle boundary.

- **Lowest value:** At zero, confines ripples to the water-side zone.
- **Highest value:** Allows ripples farther onto the outside surface; keep small if dry surroundings should remain unaffected.

Edge Ripple Frequency

Sets ripple-band density across the edge-distance field.

- **Lowest value:** Zero removes spatial band repetition.
- **Highest value:** Creates many tightly spaced ripple bands.

Edge Ripple Speed

Sets signed phase speed of the animated edge ripples.

- **Lowest value:** Negative values reverse wave-front travel; zero freezes this phase.
- **Highest value:** Positive values move wave fronts inward, with higher values moving faster.

Edge Ripple Strength

Controls how strong the surface bump appears.

- **Lowest value:** Zero leaves the normal unchanged by this bump contribution.
- **Highest value:** Creates stronger apparent relief. Bump changes shading normals; it does not by itself displace mesh vertices.

33. Atmosphere

Adds controllable atmospheric haze to a scene. Use the two fog layers for broad depth and low-lying mist.

How to use

Place the atmosphere asset in the scene. Adjust Master Density before balancing the two fog layers.

- The Atmosphere and Low Atmosphere panels control different height layers. Balance them with Master Density.

Atmosphere placement and visibility. The asset estimates scene dimensions and creates an atmosphere volume automatically. Choose **Only Emissive**, **Only Volume**, or **Mixed** (the terminology in the Type selector may be ordered differently). The upper atmosphere is more distant and lighter; the **Low Atmosphere** produces stronger low-lying haze. Balance their densities and height falloffs for atmospheric perspective. The operator can extend the viewport's view distance and the active camera's clip end to about **10,000 m**. When switching to a different camera, check its **Clip End** manually or the distant fog may appear missing or cut off.

Rendering performance. In Blender 5.x, volume scattering can be expensive. Where the volume shader exposes **Bias**, enabling it may noticeably reduce render time; compare the final image before treating it as a universal speed improvement.

Settings

Type

Chooses how the atmosphere is rendered.

- **Mixed:** Combines the volume and emissive appearance.
- **Only Volume:** Shows volumetric fog.
- **Only Emissive:** Shows the emissive layer.

Master Density

Controls the overall amount of fog.

- **Lowest value:** Reduces or removes this part of the effect.
- **Highest value:** Makes it more visible or affects more elements.

Color

Sets the color of the visible effect.

- **Lowest value:** A darker color produces a darker result; hue also changes its appearance.
- **Highest value:** A lighter color produces a lighter result; choose any hue that fits the scene.

Noise Scale

Sets the size of cloudy variations in the fog.

- **Lowest value:** Shows larger, broader pattern features.
- **Highest value:** Shows smaller, more closely repeated features.

Noise Strength

Controls how uneven or cloudy the fog looks.

- **Lowest value:** Reduces or removes this part of the effect.
- **Highest value:** Makes it more visible or affects more elements.

34. Moss Shader

The regular **Moss Shader** uses a geometry-derived **Surface Mask** together with the **Lazy Moss Material**. These are parts of the same effect; it is distinct from **Pro Moss** and from the Geometry Nodes moss scatter generator.

Moss Shader — Surface Mask (Geometry Nodes)

Prepares where the Lazy Moss material appears on an object, with controls for top faces, edges, and downward streaks.

How to use

The regular **Moss** material is different from **Pro Moss**: it uses a companion geometry-derived surface mask as well as shader processing. Apply **Moss Material — Surface Mask** to the receiving mesh and the **Lazy Moss Material** to its material. The mask determines where top growth, downward moss streaks, and edge-related effects occur; the material provides the color and surface detail.

Start by adjusting the downward growth distribution. **Growth Bias**, where exposed in the effect's main settings, changes the direction or tendency of moss growth, including mossy streaks along the object. **Randomize Growth** makes these streaks uneven, **Mossy Leaks Length** changes how far they extend, and **Split Angle** limits the face-angle transitions used for the mask. **Reduce Moss On Vertical** removes growth from steep surfaces when that appearance is not wanted.

For edge effects, enable **Add Near Edges** and adjust **Moss Edge Width**. On meshes with projecting ledges, the near-edge component may *reduce* moss beside exposed edges rather than add moss directly on those edges. It is easier to inspect on a simple wall or cube with a projecting top than on an intricate statue. **Moss On Top Amount** separately influences the top-facing growth component. When comparing masks, disable other moss modules temporarily so you can see one distribution method at a time.

Settings

Reduce Moss On Vertical

Removes moss from steep and downward-facing parts of the object.

- **Lowest value:** Keeps more moss on walls and steep surfaces.
- **Highest value:** Removes more moss from steep and downward-facing surfaces.

Randomize Growth

Adds uneven variation to the moss coverage.

- **Lowest value:** Keeps moss coverage more even.
- **Highest value:** Adds more uneven patches and growth variation.

Split Angle

Decides when differently angled faces are treated as separate moss areas.

- **Lowest value:** Uses a smaller or weaker setting.
- **Highest value:** Uses a larger or stronger setting.

Mossy Leaks Length

Controls how far moss streaks run down from upper areas.

- **Lowest value:** Keeps downward moss streaks short.
- **Highest value:** Lets the streaks extend farther down.

Add Near Edges

Allows moss to grow around eligible edges.

- **Off:** Disables add near edges.
- **On:** Enables add near edges.

Moss Edge Width

Sets how far moss spreads from those edges.

- **Lowest value:** Keeps this part of the effect smaller or closer to its source.

- **Highest value:** Makes it larger, longer, or farther reaching.

Moss On Top Amount

Controls how much moss appears on upward-facing surfaces.

- **Lowest value:** Shows less moss on upper surfaces.
- **Highest value:** Adds more moss to upper surfaces.

Moss Shader — Material

Adds moss color, surface detail, and optional displacement over an existing material.

How to use

Apply **Lazy Moss Material** together with the **Moss Material — Surface Mask** modifier on the same target. Connect the underlying material to the **Shader** input in the Shader Editor. Unlike **Pro Moss**, the regular Moss material combines the geometry-driven mask with moss shading and can produce characteristic top-down streaks.

Moss Mask Scale changes the size of the *distribution mask* (the larger regions where moss appears). This is distinct from the **Texture Scale** setting, which controls the actual moss surface texture. Use **Moss Color Overlay** to tint the result, then adjust **Roughness** and **Bump Strength**. The mask can also use **Moss in Ambient Occlusion**: enable it to place moss in recessed or sheltered areas, adjust **AO Distance**, use **Invert** to reverse the selection, and set **AO Texture Scale** for the breakup pattern. With **Only Local** enabled, nearby separate objects do not contribute to the same occlusion mask; turning it off allows surrounding geometry to influence it.

Moss Displacement adds physical surface detail when supported by the current rendering and modifier setup. Adaptive Subdivision may be required and can become expensive on complex meshes. Save the scene before switching to this mode, and use a reasonably optimized target. For a lighter look or a quick preview, keep to the material's normal and bump details.

Settings

Moss Color Overlay

Tints the visible moss layer.

- **Lowest value:** A darker color produces a darker result; hue also changes its appearance.
- **Highest value:** A lighter color produces a lighter result; choose any hue that fits the scene.

Texture Scale

Scale of the moss textures (color and height). Higher values give smaller details.

- **Lowest value:** Shows larger, broader pattern features.
- **Highest value:** Shows smaller, more closely repeated features.

Roughness

Surface roughness of the moss. 0 is a perfect mirror, 1 is fully rough.

- **Lowest value:** The surface looks smoother and more reflective.
- **Highest value:** The surface looks more matte and diffuse.

Bump Strength

Strength of the bump effect created from the moss height.

- **Lowest value:** Reduces or removes this part of the effect.

- **Highest value:** Makes it more visible or affects more elements.

Seed

Random seed that shifts the moss pattern. Change it to get a different layout.

- **Lowest value:** Selects one repeatable variation.
- **Highest value:** Selects another repeatable variation; a larger number does not mean better quality.

Invert

Inverts the moss pattern, so covered and empty areas swap places.

- **Off:** Disables invert.
- **On:** Enables invert.

Moss Displacement Height

Height of the moss displacement in scene units. Set to 0 to disable moss displacement.

- **Lowest value:** Keeps this part of the effect smaller or closer to its source.
- **Highest value:** Makes it larger, longer, or farther reaching.

Only Local

Limits sheltered-area detection to this object.

- **Off:** Disables only local.
- **On:** Enables only local.

AO Distance

Sets how far the sheltered-area effect reaches.

- **Lowest value:** Keeps this part of the effect smaller or closer to its source.
- **Highest value:** Makes it larger, longer, or farther reaching.

AO Texture Scale

Controls the size of the breakup pattern in sheltered areas.

- **Lowest value:** Shows larger, broader pattern features.
- **Highest value:** Shows smaller, more closely repeated features.

35. Snow Material

Adds a snow appearance to a material, with controls for coverage, melting, wetness, and small dirt particles.

How to use

Open the **Shader Editor** when setting up the Snow material. The effect blends a snow shader over the original material; a conventional **Principled BSDF** setup with diffuse/base-color information and a normal or height map is recommended for a more convincing blend, although it is not strictly mandatory. To use your own underlying shader, open the Snow node group and connect that shader to its **Material** input.

Begin with the main mask controls: **Snow Coverage** determines the overall covered area, **Details** adds fine variation, **Noisiness** makes the snow boundary irregular, and **Transition Softness** controls whether the border is sharply defined or gradual. Set **Snow Color**, **Surface Tint**, and **Translucency Amount** for the snow's appearance. The primary melting adjustments are **Melt**, **Melt Amount**, and **Melt Variation**; **Melt From Bottom** and **Melt Softness** offer more subtle

refinements. Inspect the result under scene lighting rather than judging coverage solely from the mask preview.

Dirt Particles adds small dark or colored specks within the snow. Start with the particle noise pattern and **Bump Texture Scale**, then refine the dirt density, size, and color to avoid an overly uniform surface. Displacement or **Adaptive Subdivision** can increase detail but also cost substantial memory on dense geometry. Enable these only after the base material looks right and the target mesh is reasonably optimized.

Settings

Snow Color

Sets the snow color of the visible effect.

- **Lowest value:** A darker color produces a darker result; hue also changes its appearance.
- **Highest value:** A lighter color produces a lighter result; choose any hue that fits the scene.

Surface Tint

Sets the surface tint of the visible effect.

- **Lowest value:** A darker color produces a darker result; hue also changes its appearance.
- **Highest value:** A lighter color produces a lighter result; choose any hue that fits the scene.

Translucency Amount

Controls how strongly translucency amount appears.

- **Lowest value:** Reduces or removes this part of the effect.
- **Highest value:** Makes it more visible or affects more elements.

Translucency Color

Sets the translucency color of the visible effect.

- **Lowest value:** A darker color produces a darker result; hue also changes its appearance.
- **Highest value:** A lighter color produces a lighter result; choose any hue that fits the scene.

Melt Amount

Controls how strongly melt amount appears.

- **Lowest value:** Reduces or removes this part of the effect.
- **Highest value:** Makes it more visible or affects more elements.

Melt From Bottom

Adjusts melt from bottom for this effect.

- **Lowest value:** Uses a smaller or weaker setting.
- **Highest value:** Uses a larger or stronger setting.

Melt Softness

Softens the change in melt softness.

- **Lowest value:** The transition looks sharper.
- **Highest value:** The transition becomes broader and softer.

Melt Variation

Adjusts melt variation for this effect.

- **Lowest value:** Makes the result more even and predictable.
- **Highest value:** Adds more variation between elements or across the surface.

Wet Roughness

Sets how glossy or matte the wet area looks.

- **Lowest value:** The surface looks smoother and more reflective.
- **Highest value:** The surface looks more matte and diffuse.

Dirt Particle Density

Controls how strongly dirt particle density appears.

- **Lowest value:** Reduces or removes this part of the effect.
- **Highest value:** Makes it more visible or affects more elements.

Dirt Particle Size

Changes the size of dirt particle size.

- **Lowest value:** Keeps this part of the effect smaller or closer to its source.
- **Highest value:** Makes it larger, longer, or farther reaching.

Dirt Color

Sets the dirt color of the visible effect.

- **Lowest value:** A darker color produces a darker result; hue also changes its appearance.
- **Highest value:** A lighter color produces a lighter result; choose any hue that fits the scene.

Bump Texture Scale

Changes the size of the bump texture pattern.

- **Lowest value:** Shows larger, broader pattern features.
- **Highest value:** Shows smaller, more closely repeated features.

36. Pro Rust

Adds layered rust to edges, walls, and upward-facing surfaces.

How to use

Pro Rust creates layered corrosion in two distribution modules: **Wall/Ground Rust** and **Edge Rust**. Apply it to the chosen materials, preferably one or more materials using a conventional shader setup, and first choose **Rust Color A** and **Rust Color B**. Color **B** corresponds to the more concentrated **Brown Rust** layer and is usually best kept darker. Color **A** corresponds to the outward-spreading **Orange Rust** layer. **Rust Shader Specular**, **Bump Strength**, and **Rust Mask Bump Strength** control reflection, fine rust texture, and the apparent separation between the rust layer and the original material.

Wall and Ground Rust. Use **Global Contrast** for the overall distribution, then independently set **Wall Rust Scale**, **Floor Rust Scale**, **Wall Contrast**, and **Ground Contrast**. These masks use different surface mapping and are intended for vertical and upward-facing areas. You can choose texture coordinates or a UV map; if you change the coordinates, use **Convert to Box** where the add-on provides that action to rebuild a correct triplanar mapping. The relevant projection

setup must remain attached to the object.

Edge Rust. Use **Rust Amount** for the darker brown rust, and **Orange Rust Amount** for the lighter orange contribution. **Orange Rust Distance** controls how far orange rust extends away from the edge. **Texture Contrast** breaks up the edge-rust distribution, making the pattern less regular. An effective setup is to establish a narrow dark corrosion core, then expand the lighter orange layer to suggest corrosion spreading from the edges.

Settings

Bump Strength

Controls the strength of bump.

- **Lowest value:** Reduces or removes this part of the effect.
- **Highest value:** Makes it more visible or affects more elements.

Filter Width

Softens fine bumps and helps avoid harsh shading edges.

- **Lowest value:** Keeps small bumps sharp.
- **Highest value:** Softens small bumps and may remove fine detail.

Rust Color A

Chooses the first rust color.

- **Lowest value:** A darker color produces a darker result; hue also changes its appearance.
- **Highest value:** A lighter color produces a lighter result; choose any hue that fits the scene.

Rust Color B

Sets the rust color b of the visible effect.

- **Lowest value:** A darker color produces a darker result; hue also changes its appearance.
- **Highest value:** A lighter color produces a lighter result; choose any hue that fits the scene.

Rust Mask Bump Strength

Controls how strongly rust mask bump strength appears.

- **Lowest value:** Reduces or removes this part of the effect.
- **Highest value:** Makes it more visible or affects more elements.

Rust Shader Specular

Adjusts rust shader specular for this effect.

- **Lowest value:** Uses a smaller or weaker setting.
- **Highest value:** Uses a larger or stronger setting.

Inside Rust Bump Scale

Changes the size of inside rust bump scale.

- **Lowest value:** Keeps this part of the effect smaller or closer to its source.
- **Highest value:** Makes it larger, longer, or farther reaching.

★ Texture Contrast

Controls how sharply texture contrast is separated from surrounding areas.

- **Lowest value:** The pattern has softer transitions.
- **Highest value:** The affected and unaffected areas separate more sharply.

Rust Amount

Controls how strongly rust amount appears.

- **Lowest value:** Reduces or removes this part of the effect.
- **Highest value:** Makes it more visible or affects more elements.

Rust Texture Scale

Changes the size of the rust texture pattern.

- **Lowest value:** Shows larger, broader pattern features.
- **Highest value:** Shows smaller, more closely repeated features.

Orange Rust Amount

Controls how strongly orange rust amount appears.

- **Lowest value:** Reduces or removes this part of the effect.
- **Highest value:** Makes it more visible or affects more elements.

Orange Rust Distance

Sets the extent of orange rust distance.

- **Lowest value:** Keeps this part of the effect smaller or closer to its source.
- **Highest value:** Makes it larger, longer, or farther reaching.

Global Contrast

Controls how sharply global contrast is separated from surrounding areas.

- **Lowest value:** The pattern has softer transitions.
- **Highest value:** The affected and unaffected areas separate more sharply.

Wall Rust Scale

Changes the size of the wall rust pattern.

- **Lowest value:** Keeps this part of the effect smaller or closer to its source.
- **Highest value:** Makes it larger, longer, or farther reaching.

Floor Rust Scale

Changes the size of the floor rust pattern.

- **Lowest value:** Keeps this part of the effect smaller or closer to its source.
- **Highest value:** Makes it larger, longer, or farther reaching.

Wall Contrast

Controls how sharply wall contrast is separated from surrounding areas.

- **Lowest value:** The pattern has softer transitions.
- **Highest value:** The affected and unaffected areas separate more sharply.

Ground Contrast

Controls how sharply ground contrast is separated from surrounding areas.

- **Lowest value:** The pattern has softer transitions.
- **Highest value:** The affected and unaffected areas separate more sharply.

Location

Moves the placement of the surface pattern.

- **Lowest value:** Moves the pattern toward negative axes.
- **Highest value:** Moves the pattern toward positive axes.

Wall Texture Coordinates

Choose the mode used for wall texture coordinates.

Choose the option that best matches the scene.

Ground Texture Coordinates

Choose the mode used for ground texture coordinates.

Choose the option that best matches the scene.

37. Pro Dirt

Adds dirt to edges and to wall and ground areas, with separate colors and surface detail.

How to use

Pro Dirt distributes two overlapping dirt tones through three areas: **Wall Dirt**, **Ground Dirt**, and **Edge Dirt**. Add it to the desired materials, then reduce the initial coverage if needed before exploring the individual masks. **Dirt Color A** describes the deeper or more concentrated dirt region; **Dirt Color B** is used for a broader surrounding region. Adjust material **Roughness** and **Specular** to fit the surface. **Bump Strength** adds fine relief within the dirty patches, while **Dirt Mask Bump Strength** changes how pronounced the whole dirt layer appears against the underlying material.

Wall and ground masks. The setup is semi-procedural: it maps texture detail onto the object and separates vertical from upward-facing areas. The **Ground Dirt** distribution uses triplanar projection and needs the accompanying **Triplanar Projection** modifier, which the operator creates automatically. Do not remove that modifier while using the effect. **Wall Dirt** commonly uses Object coordinates; the mapping may be changed to a suitable UV map when the shader supports it. Use **Global Contrast** first, then **Wall Dirt Scale** and **Floor Dirt Scale** to set the two texture sizes. A **Location** offset moves the texture pattern. Higher **Wall Contrast** or **Ground Contrast** makes the corresponding mask more selective; at its strongest setting, a layer may almost disappear. This is useful for keeping only wall dirt or only ground dirt.

Edge Dirt. Its detection mask follows eligible edges, while texture coordinates drive the separate breakup texture rather than the underlying geometry selection. The darker concentrated layer is controlled by **A Dirt Amount** and **A Dirt Width**; **B Dirt Amount** and **B Dirt Distance** set the strength and outward reach of the second layer. **Texture Contrast** makes the breakup pattern more selective and irregular. Build the effect one layer at a time: adjust wall and ground distribution, then add edge buildup rather than raising all layers at once.

Settings

Bump Strength

Controls the strength of bump.

- **Lowest value:** Reduces or removes this part of the effect.
- **Highest value:** Makes it more visible or affects more elements.

Filter Width

Softens fine bumps and helps avoid harsh shading edges.

- **Lowest value:** Keeps small bumps sharp.
- **Highest value:** Softens small bumps and may remove fine detail.

Roughness

Specifies microfacet roughness of the surface for specular reflection and transmission (0.0 is a perfect mirror reflection, 1.0 is completely rough).

- **Lowest value:** The surface looks smoother and more reflective.
- **Highest value:** The surface looks more matte and diffuse.

Texture Coordinates

Coordinates of the edge dirt texture. With UV, pick the UV map in the add-on panel.

Choose the option that best matches the scene.

★ Texture Contrast

Controls how sharply texture contrast is separated from surrounding areas.

- **Lowest value:** The pattern has softer transitions.
- **Highest value:** The affected and unaffected areas separate more sharply.

Global Contrast

Contrast of the edge dirt mask.

- **Lowest value:** The pattern has softer transitions.
- **Highest value:** The affected and unaffected areas separate more sharply.

Wall Texture Coordinates

Choose the mode used for wall texture coordinates.

Choose the option that best matches the scene.

Ground Texture Coordinates

Choose the mode used for ground texture coordinates.

Choose the option that best matches the scene.

Wall Contrast

Controls how sharply wall contrast is separated from surrounding areas.

- **Lowest value:** The pattern has softer transitions.
- **Highest value:** The affected and unaffected areas separate more sharply.

Ground Contrast

Controls how sharply ground contrast is separated from surrounding areas.

- **Lowest value:** The pattern has softer transitions.
- **Highest value:** The affected and unaffected areas separate more sharply.

Location

Moves the placement of the surface pattern.

- **Lowest value:** Moves the pattern toward negative axes.
- **Highest value:** Moves the pattern toward positive axes.

38. Pro Moss

Grows a material layer of moss over tops, crevices, and edges.

How to use

Pro Moss is a shader-driven moss layer with separate controls for top-facing surfaces, crevices, exposed edges, and downward-facing growth. Add it to the receiving materials and begin with **Amount** for overall strength and **Coverage** for the area occupied. Lower coverage produces scattered moss patches; higher coverage can eventually cover most of the object. **Texture Scale** adjusts the moss texture itself, while **Edge Softness** blends the boundaries. The **Clump By Height** control (as labeled in this reference) changes the distribution according to height-related detail.

Top and underside growth. Amount On Top adds moss to upward-facing surfaces; **Grow Falloff** (when present in the effect panel) controls how quickly the mask falls off as those faces turn into slopes or vertical walls. **Grow From Below** instead emphasizes downward-facing geometry. Temporarily turn off other placement layers to judge these masks independently.

Crevice and edge moss. Crevice Moss increases buildup in recessed corners. Tune **Crevice Reach** for the overall distance, **Crevice Spread** for the width of the surrounding influence, then **Crevice Focus** to sharpen or soften that buildup. **Own Corners Only** limits the geometry used to calculate these recesses to the current object; with it disabled, other nearby objects can also affect the mask. **Edge Moss** adds a distinct worn or overgrown edge contribution, controlled by **Edge Width**. The global breakup texture—**Breakup Amount**, **Breakup Balance**, and **Breakup Softness**—applies across the top, edge, and crevice distributions, unlike the independently broken-up regions in **Pro Dust**.

Material appearance. Tint and **Soft Light** alter the moss tones, while **Rim Color** and **Rim Amount** can introduce another color near concave detail to add depth. Finish with **Roughness**, **Specular**, **Sheen**, **Normal Strength**, and **Bump**. Keep the moss modulation subtle enough that the underlying model remains readable.

Settings

Material Displacement

Passes through displacement from the material underneath this effect.

- **Lowest value:** Zero components remove this directional contribution; negative components point along negative axes.
- **Highest value:** Positive or larger components point along positive axes or increase the response.

Material Height

Supplies a height map for the underlying material.

- **Lowest value:** Keeps this part of the effect smaller or closer to its source.
- **Highest value:** Makes it larger, longer, or farther reaching.

Amount On Top

Surfaces tilted less than this get full moss.

- **Lowest value:** Reduces or removes this part of the effect.
- **Highest value:** Makes it more visible or affects more elements.

Grow From Below

Invert: tops <-> undersides and upper <-> lower part.

- **Off:** Disables grow from below.
- **On:** Enables grow from below.

Texture Scale

Texture repeats per meter.

- **Lowest value:** Shows larger, broader pattern features.
- **Highest value:** Shows smaller, more closely repeated features.

Crevice Moss

Extra moss in corners and cavities.

- **Lowest value:** Uses a smaller or weaker setting.
- **Highest value:** Uses a larger or stronger setting.

Crevice Focus

Adjusts crevice focus for this effect.

- **Lowest value:** Uses a smaller or weaker setting.
- **Highest value:** Uses a larger or stronger setting.

Crevice Reach

Sets the extent of crevice reach.

- **Lowest value:** Keeps this part of the effect smaller or closer to its source.
- **Highest value:** Makes it larger, longer, or farther reaching.

Crevice Spread

Controls how far the effect spreads away from crevices.

- **Lowest value:** Keeps this part of the effect smaller or closer to its source.
- **Highest value:** Makes it larger, longer, or farther reaching.

Own Corners Only

Limits crevice effects to corners of the same object.

- **Off:** Nearby objects can influence sheltered-area placement.
- **On:** Only crevices on this object affect the placement.

Edge Moss

Extra moss on convex edges.

- **Lowest value:** Uses a smaller or weaker setting.
- **Highest value:** Uses a larger or stronger setting.

Edge Width

Sets the extent of edge width.

- **Lowest value:** Keeps this part of the effect smaller or closer to its source.
- **Highest value:** Makes it larger, longer, or farther reaching.

Breakup Amount

Controls how strongly breakup amount appears.

- **Lowest value:** Makes the result more even and predictable.
- **Highest value:** Adds more variation between elements or across the surface.

Breakup Balance

Controls how much of the dusty or mossy pattern survives.

- **Lowest value:** Keeps more of the effect visible.
- **Highest value:** Leaves more gaps and clean patches.

Breakup Softness

Softens the change in breakup softness.

- **Lowest value:** The transition looks sharper.
- **Highest value:** The transition becomes broader and softer.

Amount

Overall moss opacity.

- **Lowest value:** Reduces or removes this part of the effect.
- **Highest value:** Makes it more visible or affects more elements.

Coverage

Controls how much of the target is covered.

- **Lowest value:** Leaves more of the target exposed.
- **Highest value:** Covers more of the target.

Edge Softness

Softens the change in edge softness.

- **Lowest value:** The transition looks sharper.
- **Highest value:** The transition becomes broader and softer.

Clump By Height

Moss borders follow the moss height (tufts).

- **Lowest value:** Keeps this part of the effect smaller or closer to its source.
- **Highest value:** Makes it larger, longer, or farther reaching.

Tint

Sets the tint of the visible effect.

- **Lowest value:** A darker color produces a darker result; hue also changes its appearance.
- **Highest value:** A lighter color produces a lighter result; choose any hue that fits the scene.

Soft Light

Sets the soft light of the visible effect.

- **Lowest value:** A darker color produces a darker result; hue also changes its appearance.
- **Highest value:** A lighter color produces a lighter result; choose any hue that fits the scene.

Rim Color

Sets the rim color of the visible effect.

- **Lowest value:** A darker color produces a darker result; hue also changes its appearance.
- **Highest value:** A lighter color produces a lighter result; choose any hue that fits the scene.

Rim Amount

Controls how strongly rim amount appears.

- **Lowest value:** Reduces or removes this part of the effect.
- **Highest value:** Makes it more visible or affects more elements.

Roughness

Sets how glossy or matte the surface looks.

- **Lowest value:** The surface looks smoother and more reflective.
- **Highest value:** The surface looks more matte and diffuse.

Specular

Adjusts specular for this effect.

- **Lowest value:** Uses a smaller or weaker setting.
- **Highest value:** Uses a larger or stronger setting.

Sheen

Adjusts sheen for this effect.

- **Lowest value:** Uses a smaller or weaker setting.
- **Highest value:** Uses a larger or stronger setting.

Normal Strength

Controls the strength of normal.

- **Lowest value:** Reduces or removes this part of the effect.
- **Highest value:** Makes it more visible or affects more elements.

Bump

Adjusts bump for this effect.

- **Lowest value:** Uses a smaller or weaker setting.
- **Highest value:** Uses a larger or stronger setting.

Thickness

Moss relief (displacement).

- **Lowest value:** Uses a smaller or weaker setting.
- **Highest value:** Uses a larger or stronger setting.

39. Pro Dust

Adds a soft dust layer to upper surfaces and crevices.

How to use

Pro Dust combines two major distribution modes: dust on upper-facing surfaces and accumulated dust in crevices. Apply it to the desired material or materials. For a clear preview, temporarily give the base material a dark color; compare presets before refining the default setup. **Texture Scale** controls the grain of the dust, while **Amount** is the main intensity multiplier. **Color**, **Color Variation**, **Roughness**, **Fluff**, **Thickness**, and **Grain Bump** determine the dusty material appearance. **Fluff** tends to lighten or soften dust in some corners, while **Thickness** adds apparent body and **Grain Bump** strengthens fine relief.

Dust On Top. Begin with **Top Strength** and **Top Falloff**. A narrow falloff tends to concentrate dust on faces pointing upward, while a broader falloff includes slopes. **Invert** reverses the placement. Use **Top Breakup** and **Top Breakup Balance** to fragment the top mask; **Top Breakup Softness** reduces overly sharp transitions. **Top Color Noise** and **Top Noise Scale** introduce additional color variation specifically within the top layer. Temporarily reduce crevice accumulation when fine-tuning top dust.

Crevice Dust. This uses corner and contact shading to place dirt in recessed areas. Set **Crevice Reach** first, then **Crevice Spread**, and finally **Crevice Focus** to control how concentrated or soft the deposits appear. For example, reduce a broad reach to a few centimeters to emphasize small grooves on a detailed model. In this setup, setting **Crevice Reach** to zero may revert to the built-in reach behavior rather than simply switching the layer off; use **Crevice Dust** to reduce its strength. **Own Corners Only** excludes other scene objects from influencing the mask, while disabling it permits adjacent objects to affect dust accumulation. Unlike Dust On Top's dedicated breakup controls, the general **Breakup Amount**, **Breakup Balance**, and **Breakup Softness** principally refine the near-edge or crevice contribution.

Settings

Material Displacement

Passes through displacement from the material underneath this effect.

- **Lowest value:** Zero components remove this directional contribution; negative components point along negative axes.
- **Highest value:** Positive or larger components point along positive axes or increase the response.

Material Height

Supplies a height map for the underlying material.

- **Lowest value:** Keeps this part of the effect smaller or closer to its source.
- **Highest value:** Makes it larger, longer, or farther reaching.

Texture Scale

Texture repeats per meter.

- **Lowest value:** Shows larger, broader pattern features.
- **Highest value:** Shows smaller, more closely repeated features.

Amount

Overall dust opacity.

- **Lowest value:** Reduces or removes this part of the effect.
- **Highest value:** Makes it more visible or affects more elements.

Color

Sets the color of the visible effect.

- **Lowest value:** A darker color produces a darker result; hue also changes its appearance.
- **Highest value:** A lighter color produces a lighter result; choose any hue that fits the scene.

Color Variation

Grain shading inside the dust.

- **Lowest value:** Makes the result more even and predictable.
- **Highest value:** Adds more variation between elements or across the surface.

Roughness

Sets how glossy or matte the surface looks.

- **Lowest value:** The surface looks smoother and more reflective.
- **Highest value:** The surface looks more matte and diffuse.

Fluff

Soft fuzzy sheen at grazing angles.

- **Lowest value:** Uses a smaller or weaker setting.
- **Highest value:** Uses a larger or stronger setting.

Thickness

Dust layer height (displacement).

- **Lowest value:** Uses a smaller or weaker setting.
- **Highest value:** Uses a larger or stronger setting.

Grain Bump

Adjusts grain bump for this effect.

- **Lowest value:** Uses a smaller or weaker setting.
- **Highest value:** Uses a larger or stronger setting.

Dust On Top

Force dust on every up-facing face, broken up by the Dust Breakup image.

- **Off:** Disables dust on top.
- **On:** Enables dust on top.

Invert

Controls whether invert is used.

- **Off:** Disables invert.
- **On:** Enables invert.

Top Breakup Balance

Lower = more top dust survives.

- **Lowest value:** Keeps more of the surface effect.
- **Highest value:** Leaves more clean gaps in the pattern.

Top Strength

Controls the strength of top.

- **Lowest value:** Reduces or removes this part of the effect.
- **Highest value:** Makes it more visible or affects more elements.

Top Falloff

How far down the sides the top dust fades (.

- **Lowest value:** The transition looks sharper.
- **Highest value:** The transition becomes broader and softer.

Top Breakup

How much the Dust Breakup image cuts the top dust.

- **Lowest value:** Makes the result more even and predictable.
- **Highest value:** Adds more variation between elements or across the surface.

Top Breakup Softness

Softens the change in top breakup softness.

- **Lowest value:** The transition looks sharper.
- **Highest value:** The transition becomes broader and softer.

Top Color Noise

Lighter or darker or tinted patches in the top dust.

- **Lowest value:** Uses a smaller or weaker setting.
- **Highest value:** Uses a larger or stronger setting.

Top Noise Scale

Changes the size of the top noise pattern.

- **Lowest value:** Shows larger, broader pattern features.
- **Highest value:** Shows smaller, more closely repeated features.

Crevice Dust

Extra dust in corners, even on walls.

- **Lowest value:** Uses a smaller or weaker setting.
- **Highest value:** Uses a larger or stronger setting.

Crevice Focus

Adjusts crevice focus for this effect.

- **Lowest value:** Uses a smaller or weaker setting.
- **Highest value:** Uses a larger or stronger setting.

Crevice Reach

Sets the extent of crevice reach.

- **Lowest value:** Keeps this part of the effect smaller or closer to its source.
- **Highest value:** Makes it larger, longer, or farther reaching.

Crevice Spread

Controls how far the effect spreads away from crevices.

- **Lowest value:** Keeps this part of the effect smaller or closer to its source.
- **Highest value:** Makes it larger, longer, or farther reaching.

Height Influence

Low areas of Material Height collect dust.

- **Lowest value:** Keeps this part of the effect smaller or closer to its source.
- **Highest value:** Makes it larger, longer, or farther reaching.

Own Corners Only

Limits crevice effects to corners of the same object.

- **Off:** Nearby objects can influence sheltered-area placement.
- **On:** Only crevices on this object affect the placement.

Breakup Amount

Clean or dusty patches.

- **Lowest value:** Makes the result more even and predictable.
- **Highest value:** Adds more variation between elements or across the surface.

Breakup Balance

Controls how much of the dusty or mossy pattern survives.

- **Lowest value:** Keeps more of the effect visible.
- **Highest value:** Leaves more gaps and clean patches.

Breakup Softness

Softens the change in breakup softness.

- **Lowest value:** The transition looks sharper.
- **Highest value:** The transition becomes broader and softer.

40. Oil Stains

Oil Stains consists of placement **Controllers** and the corresponding **Material**. Controllers and drawn curves determine the location and shape of the stains; the shader supplies the visual effect.

Oil Stains — Controllers (Geometry Nodes)

Defines the shapes and paths that the Oil Stains material uses.

How to use

Assign controller objects or curves, then place them where stains should appear. Use the companion Oil Stains material on the receiving surface.

- The controllers and curves shape the mask; the companion material supplies the oil appearance.

Creating stains and controllers. Switch to **Cycles** for the intended volumetric/oil-film appearance. Select the receiving mesh, position the 3D Cursor where the oil should appear, and add **Oil Stains**. Controller objects are collected in **Oil Controllers** and place individual stain patches. To paint more deliberate shapes, enable **Spawn On Curves**, use the drawing action, and draw a curve; its width, width variation and precision control the spread of the mark. The generated curve collection keeps these drawing guides organized. These controls produce the stain *location*; the companion Oil Stains material produces its *look*.

Settings

Controllers

Collection containing the objects that shape the oil marks.

- **Unassigned:** This source is not used; effects that require it may not appear.
- **Assigned:** The effect uses the selected source.

Subdivisions

Adjusts subdivisions for this effect.

- **Lowest value:** Uses less detail and is faster to preview.
- **Highest value:** Produces a smoother or more accurate result but takes longer to evaluate.

Spawn On Curves

Places oil marks on the assigned curves.

- **Off:** Disables spawn on curves.
- **On:** Enables spawn on curves.

Curves

Collection containing drawn curves used to place oil streaks.

- **Unassigned:** This source is not used; effects that require it may not appear.
- **Assigned:** The effect uses the selected source.

Curve Width

Sets the width of the oil marks around assigned curves.

- **Lowest value:** Keeps this part of the effect smaller or closer to its source.
- **Highest value:** Makes it larger, longer, or farther reaching.

Width Variation

Sets the extent of width variation.

- **Lowest value:** Keeps this part of the effect smaller or closer to its source.
- **Highest value:** Makes it larger, longer, or farther reaching.

Curve Precision

Controls how closely the shape follows the assigned curves.

- **Lowest value:** Uses less detail and is faster to preview.
- **Highest value:** Produces a smoother or more accurate result but takes longer to evaluate.

Oil Stains — Material

Shades oil marks with wet darkening, a reflective film, drips, and colored swirls.

How to use

Apply the material to the surface and supply the Oil Stains controllers if the marks do not appear.

Shader connections and appearance. The material expects an appropriate **Principled BSDF** PBR setup with Base Color, Roughness and Normal inputs. The add-on can reconnect multiple Principled BSDF nodes combined through Mix nodes. Unusual procedural materials may not reconnect automatically: if warned, locate the inserted node group and manually connect its Base Color, Roughness and Normal inputs and outputs to the correct material paths.

Adjust the main **Oil Color**, **Oil Roughness** and **Stain Strength** first. **Outline Breakup** interrupts an otherwise uniform boundary; **Mottling**, **Tide Ring**, and **Drips** introduce film variation, transitional rings, and scattered detail. **Wet Darkening**, **Wet Roughness**, **Wet Flatten**, and **Wet Spread** shape the damp zone surrounding the stain. **Film Strength** and related film controls determine the rainbow-like iridescence. Save successful combinations as presets for repeated use.

Settings

Base Color

Sets the base color of the visible effect.

- **Lowest value:** A darker color produces a darker result; hue also changes its appearance.
- **Highest value:** A lighter color produces a lighter result; choose any hue that fits the scene.

Base Roughness

Controls how matte or glossy base roughness appears.

- **Lowest value:** The surface looks smoother and more reflective.
- **Highest value:** The surface looks more matte and diffuse.

Oil Color

Chooses the color of the oil marks.

- **Lowest value:** A darker color produces a darker result; hue also changes its appearance.
- **Highest value:** A lighter color produces a lighter result; choose any hue that fits the scene.

Stain Strength

Controls the strength of stain.

- **Lowest value:** Reduces or removes this part of the effect.

- **Highest value:** Makes it more visible or affects more elements.

Oil Roughness

Sets how glossy or matte the oil looks.

- **Lowest value:** The surface looks smoother and more reflective.
- **Highest value:** The surface looks more matte and diffuse.

Wet Darkening

Controls how much the stained area darkens when it looks wet.

- **Lowest value:** Keeps the wet patch closer to the original surface color.
- **Highest value:** Makes the wet patch darker.

Wet Roughness

Sets how glossy or matte the wet area looks.

- **Lowest value:** The surface looks smoother and more reflective.
- **Highest value:** The surface looks more matte and diffuse.

Wet Flatten

Controls how much the wet area looks smoother and flatter.

- **Lowest value:** Keeps more of the underlying surface detail.
- **Highest value:** Makes the wet area look flatter and smoother.

Base IOR

Controls how strongly the original surface reflects light at different viewing angles.

- **Lowest value:** Uses a smaller or weaker setting.
- **Highest value:** Uses a larger or stronger setting.

Film Strength

Controls the strength of film.

- **Lowest value:** Reduces or removes this part of the effect.
- **Highest value:** Makes it more visible or affects more elements.

Film IOR

Changes the reflections of the thin oil film.

- **Lowest value:** Uses a smaller or weaker setting.
- **Highest value:** Uses a larger or stronger setting.

Outline Breakup

Adjusts outline breakup for this effect.

- **Lowest value:** Makes the result more even and predictable.
- **Highest value:** Adds more variation between elements or across the surface.

Mottling

Adjusts mottling for this effect.

- **Lowest value:** Uses a smaller or weaker setting.
- **Highest value:** Uses a larger or stronger setting.

Tide Ring

Adjusts tide ring for this effect.

- **Lowest value:** Uses a smaller or weaker setting.
- **Highest value:** Uses a larger or stronger setting.

Drips

Adjusts drips for this effect.

- **Lowest value:** Uses a smaller or weaker setting.
- **Highest value:** Uses a larger or stronger setting.

Freshness

Adjusts freshness for this effect.

- **Lowest value:** Uses a smaller or weaker setting.
- **Highest value:** Uses a larger or stronger setting.

Wet Spread

Adjusts wet spread for this effect.

- **Lowest value:** Keeps this part of the effect smaller or closer to its source.
- **Highest value:** Makes it larger, longer, or farther reaching.

Film Coverage

Controls how strongly film coverage appears.

- **Lowest value:** Reduces or removes this part of the effect.
- **Highest value:** Makes it more visible or affects more elements.

Film Swirl

Adjusts film swirl for this effect.

- **Lowest value:** Uses a smaller or weaker setting.
- **Highest value:** Uses a larger or stronger setting.

Film Min nm

Controls the low end of the thin-film color variation.

- **Lowest value:** Shifts one end of the film colors toward shorter wavelengths.
- **Highest value:** Shifts it toward longer wavelengths.

Film Max nm

Controls the high end of the thin-film color variation.

- **Lowest value:** Limits the upper end of the color range.
- **Highest value:** Allows a wider range of film colors.

41. Tire Marks

Tire Marks combines a Geometry Nodes **Path Data** component with its companion shader **Material**. These projected tire marks are distinct from the separate **3D Tire Marks** displacement generator.

Tire Marks — Path Data (Geometry Nodes)

Creates the path data used by the matching tire marks material.

How to use

Assign a curve to Path and place it along the intended track. Keep this effect on the same receiving object as the matching shader.

- The path effect and tire mark material belong on the same receiving surface.

Drawing a natural tire path. Select the receiving ground, add Tire Marks Shader, and draw a smooth route. The Geometry Nodes path group supplies coordinates; its companion Tire Marks material generates the color and breakup. Use **Track Width** for distance between tire lanes, **Tire Width** for a single tire imprint, and **Precision/Subdivisions** for projection quality. Avoid implausibly sharp turns. If corners become visible, select the guide curve, **Subdivide** the problem area a few times, and run **Smooth**; simply adding many vertices without smoothing is not necessary.

Settings

Path

Selects the curve that guides the marks.

- **Unassigned:** This source is not used; effects that require it may not appear.
- **Assigned:** The effect uses the selected source.

Track Width

Sets the extent of track width.

- **Lowest value:** Keeps this part of the effect smaller or closer to its source.
- **Highest value:** Makes it larger, longer, or farther reaching.

Tire Width

Sets the extent of tire width.

- **Lowest value:** Keeps this part of the effect smaller or closer to its source.
- **Highest value:** Makes it larger, longer, or farther reaching.

Subdivisions

Adjusts subdivisions for this effect.

- **Lowest value:** Uses less detail and is faster to preview.
- **Highest value:** Produces a smoother or more accurate result but takes longer to evaluate.

Precision

Adjusts precision for this effect.

- **Lowest value:** Uses less detail and is faster to preview.
- **Highest value:** Produces a smoother or more accurate result but takes longer to evaluate.

Flatten Z

Keeps the generated path at a consistent height.

- **Off:** Disables flatten z.
- **On:** Enables flatten z.

Tire Marks — Material

Draws tire impressions in the material along a prepared path.

How to use

Use the companion path effect first. Then set tire type, mark color, and strength in the material.

Material refinement. Choose the tread **Tire Type**, then adjust **Marks Color**, roughness, and strength. **Tile Length** controls repetition along the path; **Breakup/Breakup Scale** make the marks patchy so they do not look perfectly continuous. If a sharply bent path produces angular texture artifacts, smooth the guide curve first, then adjust the repetition and breakup. Keep sufficient subdivisions on the receiving ground for stable projection.

Settings

Tire Type

Chooses one of the available tire patterns.

- **Lowest value:** Uses a smaller or weaker setting.
- **Highest value:** Uses a larger or stronger setting.

Marks Color

Sets the marks color of the visible effect.

- **Lowest value:** A darker color produces a darker result; hue also changes its appearance.
- **Highest value:** A lighter color produces a lighter result; choose any hue that fits the scene.

Mark Roughness

Controls how matte or glossy mark roughness appears.

- **Lowest value:** The surface looks smoother and more reflective.
- **Highest value:** The surface looks more matte and diffuse.

Strength

Controls how strongly strength appears.

- **Lowest value:** Reduces or removes this part of the effect.
- **Highest value:** Makes it more visible or affects more elements.

Tile Length

Sets the extent of tile length.

- **Lowest value:** Keeps this part of the effect smaller or closer to its source.
- **Highest value:** Makes it larger, longer, or farther reaching.

Wheel Offset

Adjusts wheel offset for this effect.

- **Lowest value:** Uses a smaller or weaker setting.
- **Highest value:** Uses a larger or stronger setting.

Mirror Left

Adjusts mirror left for this effect.

- **Lowest value:** Uses a smaller or weaker setting.
- **Highest value:** Uses a larger or stronger setting.

Breakup

Adjusts breakup for this effect.

- **Lowest value:** Makes the result more even and predictable.
- **Highest value:** Adds more variation between elements or across the surface.

Breakup Scale

Changes the size of the breakup pattern.

- **Lowest value:** Keeps this part of the effect smaller or closer to its source.
- **Highest value:** Makes it larger, longer, or farther reaching.

42. Footprints

Footprints combines Geometry Nodes **Path Data** with a footprint **Material**. This projected material effect is distinct from the separate **3D Footprints** geometry generator.

Footprints — Path Data (Geometry Nodes)

Prepares a drawn path for material-based footprints.

How to use

Assign a curve to Path and place it across the receiving surface. Pair it with the matching footprint material.

- The path effect and footprint material belong on the same receiving surface.

Creating shader footprints. Select the target ground and draw the path using the Footprints shader tool. The Geometry Nodes modifier produces path/placement coordinates that drive the material rather than displacing the surface into deep footprints. **Path** selects the guide curve, while **Precision** controls surface projection. Redraw or edit the curve if the walking route changes. The path-data generator and its companion shader must remain on the same receiving object.

Settings

Path

Selects the curve that guides the marks.

- **Unassigned:** This source is not used; effects that require it may not appear.
- **Assigned:** The effect uses the selected source.

Subdivisions

Adjusts subdivisions for this effect.

- **Lowest value:** Uses less detail and is faster to preview.
- **Highest value:** Produces a smoother or more accurate result but takes longer to evaluate.

Precision

Adjusts precision for this effect.

- **Lowest value:** Uses less detail and is faster to preview.
- **Highest value:** Produces a smoother or more accurate result but takes longer to evaluate.

Flatten Z

Keeps the generated path at a consistent height.

- **Off:** Disables flatten z.
- **On:** Enables flatten z.

Footprints — Material

Places boot, barefoot, or animal footprints along the prepared path.

How to use

Apply the companion path effect first. Adjust step spacing and print size to fit the surface.

Footprint texture sizing. The shader supports shoes, bare feet and animal tracks, with separate color, roughness and strength. Each footprint texture fits inside an area generated along the drawn path. If **Step Length** is reduced too far, prints can become cropped because the texture exceeds its available area. Reduce **Print Length** and **Print Width** (or use any shared print-scale control exposed in the interface), then adjust stride spacing and jitter to restore a natural walk. Material footprints depend on sufficient target mesh density for reliable projection; add subdivisions if details look incorrect.

Settings

Print Color

Sets the print color of the visible effect.

- **Lowest value:** A darker color produces a darker result; hue also changes its appearance.
- **Highest value:** A lighter color produces a lighter result; choose any hue that fits the scene.

Print Strength

Controls the strength of print.

- **Lowest value:** Reduces or removes this part of the effect.
- **Highest value:** Makes it more visible or affects more elements.

Print Roughness

Controls how matte or glossy print roughness appears.

- **Lowest value:** The surface looks smoother and more reflective.
- **Highest value:** The surface looks more matte and diffuse.

Step Length

Sets the extent of step length.

- **Lowest value:** Keeps this part of the effect smaller or closer to its source.
- **Highest value:** Makes it larger, longer, or farther reaching.

Stride Width

Sets the extent of stride width.

- **Lowest value:** Keeps this part of the effect smaller or closer to its source.

- **Highest value:** Makes it larger, longer, or farther reaching.

Print Length

Sets the extent of print length.

- **Lowest value:** Keeps this part of the effect smaller or closer to its source.
- **Highest value:** Makes it larger, longer, or farther reaching.

Print Width

Sets the extent of print width.

- **Lowest value:** Keeps this part of the effect smaller or closer to its source.
- **Highest value:** Makes it larger, longer, or farther reaching.

Toe Out

Adjusts toe out for this effect.

- **Lowest value:** Uses a smaller or weaker setting.
- **Highest value:** Uses a larger or stronger setting.

Alternate

Adjusts alternate for this effect.

- **Lowest value:** Uses a smaller or weaker setting.
- **Highest value:** Uses a larger or stronger setting.

Mirror Left

Adjusts mirror left for this effect.

- **Lowest value:** Uses a smaller or weaker setting.
- **Highest value:** Uses a larger or stronger setting.

Rotation Jitter

Adjusts rotation jitter for this effect.

- **Lowest value:** Makes the result more even and predictable.
- **Highest value:** Adds more variation between elements or across the surface.

Scale Jitter

Changes the size of scale jitter.

- **Lowest value:** Keeps this part of the effect smaller or closer to its source.
- **Highest value:** Makes it larger, longer, or farther reaching.

Step Jitter

Adjusts step jitter for this effect.

- **Lowest value:** Makes the result more even and predictable.
- **Highest value:** Adds more variation between elements or across the surface.

Side Jitter

Adjusts side jitter for this effect.

- **Lowest value:** Makes the result more even and predictable.
- **Highest value:** Adds more variation between elements or across the surface.

End Trim

Adjusts end trim for this effect.

- **Lowest value:** Uses a smaller or weaker setting.
- **Highest value:** Uses a larger or stronger setting.

Seed

Chooses a different repeatable pattern without changing the overall amount.

- **Lowest value:** Selects one repeatable variation.
- **Highest value:** Selects another repeatable variation; a larger number does not mean better quality.

43. Ground Leaves

Scatters leaves on the ground and optionally lifts them into the air with gusts.

How to use

Assign Ground Collection and visible leaf objects. Start with density and size, then enable wind and collision as needed.

- Choose Ground Collection before increasing Density. Use Remove from Vertical Surfaces to keep leaves off steep faces.
- Wind pickup affects only eligible settled leaves. Raise Affected Leaves if gusts seem too subtle.

Settings

Max Leaf Count

Limits the number of active leaves.

- **Lowest value:** Limits the number of active leaves and may stop new emission.
- **Highest value:** Allows more leaves at once but can slow down the scene.

Gravity

Sets the direction and strength of gravity acting on leaves.

- **Lowest value:** Stronger downward gravity makes leaves fall faster.
- **Highest value:** An upward direction can lift leaves or slow their fall.

Wind Visualizer

Shows a viewport arrow for wind direction and strength.

- **Off:** Disables wind visualizer.
- **On:** Enables wind visualizer.

Ground Collection

Chooses the surfaces on which leaves can settle.

- **Unassigned:** This source is not used; effects that require it may not appear.
- **Assigned:** The effect uses the selected source.

Density

Controls how thickly this layer appears.

- **Lowest value:** Reduces or removes this part of the effect.
- **Highest value:** Makes it more visible or affects more elements.

Leaf Size

Changes the size of leaf size.

- **Lowest value:** Keeps this part of the effect smaller or closer to its source.
- **Highest value:** Makes it larger, longer, or farther reaching.

Random Size

Changes the size of random size.

- **Lowest value:** Keeps this part of the effect smaller or closer to its source.
- **Highest value:** Makes it larger, longer, or farther reaching.

Ground Offset

Adjusts ground offset for this effect.

- **Lowest value:** Uses a smaller or weaker setting.
- **Highest value:** Uses a larger or stronger setting.

Remove from Vertical Surfaces

Exclude surface points whose world slope exceeds Vertical Angle.

- **Off:** Allows leaves on steep surfaces.
- **On:** Removes leaves from surfaces steeper than Vertical Angle.

Vertical Angle (degrees)

Maximum allowed slope:.

- **Lowest value:** Uses a smaller or weaker setting.
- **Highest value:** Uses a larger or stronger setting.

Enable Vertex Paint

Controls whether vertex paint is used.

- **Off:** Disables vertex paint.
- **On:** Enables vertex paint.

Paint Vertex Group

Adjusts paint vertex group for this effect.

- **Unassigned:** This source is not used; effects that require it may not appear.
- **Assigned:** The effect uses the selected source.

Enable Noise Mask?

Controls whether noise mask? is used.

- **Off:** Disables noise mask?.
- **On:** Enables noise mask?.

Dissolve Mask

Adjusts dissolve mask for this effect.

- **Lowest value:** Uses a smaller or weaker setting.
- **Highest value:** Uses a larger or stronger setting.

Detail

The number of noise octaves. Higher values give more detailed noise but increase render time.

- **Lowest value:** Uses a smaller or weaker setting.
- **Highest value:** Uses a larger or stronger setting.

Roughness

Blend factor between an octave and its previous one. A value of zero corresponds to zero detail.

- **Lowest value:** The surface looks smoother and more reflective.
- **Highest value:** The surface looks more matte and diffuse.

Noise Mask Seed

Chooses a different repeatable pattern without changing the overall amount.

- **Lowest value:** Selects one repeatable variation.
- **Highest value:** Selects another repeatable variation; a larger number does not mean better quality.

★Wind

Uses the separate Angle, Lift, and Intensity wind controls.

- **Off:** Uses the main Wind direction.
- **On:** Uses the separate Angle, Lift, and Intensity controls.

Angle

Wind direction on the ground plane (.

- **Lowest value:** Uses a smaller or weaker setting.
- **Highest value:** Uses a larger or stronger setting.

Lift

Vertical wind speed: positive lifts the leaves, negative presses them down.

- **Lowest value:** Uses a smaller or weaker setting.
- **Highest value:** Uses a larger or stronger setting.

Intensity

Horizontal wind speed.

- **Lowest value:** Reduces or removes this part of the effect.
- **Highest value:** Makes it more visible or affects more elements.

Enable Leaf Wind

Allows wind to affect leaf flight.

- **Off:** Disables leaf wind.
- **On:** Enables leaf wind.

Collider

Assign the collider used by this effect.

- **Unassigned:** This source is not used; effects that require it may not appear.
- **Assigned:** The effect uses the selected source.

Remove Initial Inside Collider

Delete particles that start inside the Collider when the simulation initializes. This is evaluated only for the initial state.

- **Off:** Disables remove initial inside collider.
- **On:** Enables remove initial inside collider.

Bounce

Adjusts bounce for this effect.

- **Lowest value:** Uses a smaller or weaker setting.
- **Highest value:** Uses a larger or stronger setting.

Friction

Adjusts friction for this effect.

- **Lowest value:** Uses a smaller or weaker setting.
- **Highest value:** Uses a larger or stronger setting.

Leaf Radius Factor

Sets the extent of leaf radius factor.

- **Lowest value:** Keeps this part of the effect smaller or closer to its source.
- **Highest value:** Makes it larger, longer, or farther reaching.

Leaf Thickness Ratio

Adjusts leaf thickness ratio for this effect.

- **Lowest value:** Uses a smaller or weaker setting.
- **Highest value:** Uses a larger or stronger setting.

★Leaf Contact Margin

Adjusts leaf contact margin for this effect.

- **Lowest value:** Uses a smaller or weaker setting.
- **Highest value:** Uses a larger or stronger setting.

★ Soft Contact Distance

Sets the extent of soft contact distance.

- **Lowest value:** Keeps this part of the effect smaller or closer to its source.
- **Highest value:** Makes it larger, longer, or farther reaching.

Contact Rotation Response

Adjusts contact rotation response for this effect.

- **Lowest value:** Uses a smaller or weaker setting.
- **Highest value:** Uses a larger or stronger setting.

Colliders Affect Ground Particles

Controls whether colliders affect ground particles is used.

- **Off:** Disables colliders affect ground particles.
- **On:** Enables colliders affect ground particles.

Ground Collider Push Speed

Controls how quickly ground collider push speed changes.

- **Lowest value:** Movement or variation changes slowly; zero may stop this part of the animation.
- **Highest value:** Movement or variation changes faster.

Ground Collider Rotation Turbulence

Adjusts ground collider rotation turbulence for this effect.

- **Lowest value:** Uses a smaller or weaker setting.
- **Highest value:** Uses a larger or stronger setting.

Ground Collider Lift Falloff

Softens the change in ground collider lift falloff.

- **Lowest value:** The transition looks sharper.
- **Highest value:** The transition becomes broader and softer.

Instance Object

Assign the instance object used by this effect.

- **Unassigned:** This source is not used; effects that require it may not appear.
- **Assigned:** The effect uses the selected source.

Instance Collection

Assign the instance collection used by this effect.

- **Unassigned:** This source is not used; effects that require it may not appear.
- **Assigned:** The effect uses the selected source.

Size Scale

Controls whether size scale is used.

- **Off:** Disables size scale.
- **On:** Enables size scale.

Offset Rotation

Sets the direction or orientation used for offset rotation.

- **Lowest value:** Zero components remove this directional contribution; negative components point along negative axes.
- **Highest value:** Positive or larger components point along positive axes or increase the response.

★ Affected Leaves %

Sets the share of grounded leaves that can be picked up by wind.

- **Lowest value:** Reduces or removes this part of the effect.
- **Highest value:** Makes it more visible or affects more elements.

★Flying Leaves %

Sets the share of lifted leaves that stay airborne longer.

- **Lowest value:** Reduces or removes this part of the effect.
- **Highest value:** Makes it more visible or affects more elements.

Flying Height Min

Minimum soft target height above the last ground anchor for continued-flight leaves.

- **Lowest value:** Keeps this part of the effect smaller or closer to its source.
- **Highest value:** Makes it larger, longer, or farther reaching.

Flying Height Max

Maximum soft target height above the last ground anchor. Each continued-flight leaf keeps a stable target inside this range.

- **Lowest value:** Keeps this part of the effect smaller or closer to its source.
- **Highest value:** Makes it larger, longer, or farther reaching.

Flying Lift Min

Minimum strength of the damped continued-flight height response.

- **Lowest value:** Moves this end of the variation range down.
- **Highest value:** Moves this end of the variation range up.

Flying Lift Max

Maximum strength of the damped continued-flight height response. Each leaf keeps a stable lift value.

- **Lowest value:** Moves this end of the variation range down.
- **Highest value:** Moves this end of the variation range up.

Flight Turbulence Strength

Controls the strength of flight turbulence.

- **Lowest value:** Reduces or removes this part of the effect.
- **Highest value:** Makes it more visible or affects more elements.

Flight Turbulence Frequency

Controls how quickly flight turbulence frequency changes.

- **Lowest value:** Movement or variation changes slowly; zero may stop this part of the animation.
- **Highest value:** Movement or variation changes faster.

Flight Rotation Turbulence

Adjusts flight rotation turbulence for this effect.

- **Lowest value:** Uses a smaller or weaker setting.
- **Highest value:** Uses a larger or stronger setting.

Flight Turbulence Fade In

Adjusts flight turbulence fade in for this effect.

- **Lowest value:** The effect reaches full strength quickly.
- **Highest value:** The effect takes longer to reach full strength.

Gust Strength

Variation around the existing Wind vector. Zero gives constant wind.

- **Lowest value:** Reduces or removes this part of the effect.
- **Highest value:** Makes it more visible or affects more elements.

Gust Size

Spatial size of coherent gust patches.

- **Lowest value:** Keeps this part of the effect smaller or closer to its source.
- **Highest value:** Makes it larger, longer, or farther reaching.

Gust Duration

Controls how long each gust persists.

- **Lowest value:** Gusts change quickly.
- **Highest value:** Gusts linger longer.

Direction Variation

Smooth sideways meandering. Does not create wind when Wind is zero.

- **Lowest value:** Makes the result more even and predictable.
- **Highest value:** Adds more variation between elements or across the surface.

Lift Coefficient

Sets how easily wind lifts leaves from the ground.

- **Lowest value:** Leaves are harder for ground wind to lift.
- **Highest value:** Leaves lift more readily when wind passes over them.

Ground Grip

Sets how strongly settled leaves resist being lifted by wind.

- **Lowest value:** Leaves lift off the ground more easily.
- **Highest value:** Leaves stay grounded until stronger wind reaches them.

Leaf Variation

Persistent differences in aerodynamic response, grip and recovery time.

- **Lowest value:** Makes the result more even and predictable.
- **Highest value:** Adds more variation between elements or across the surface.

Response Time

Sets how quickly leaves respond to changes in wind.

- **Lowest value:** Leaves respond to gusts quickly.
- **Highest value:** Leaves respond gradually to changing wind.

Rest Time

Minimum settled time before pickup. A pressure drop is also required to re-arm after a launch.

- **Lowest value:** The transition or pause is shorter.
- **Highest value:** The transition or pause lasts longer.

Lift Duration Min

Minimum duration of the finite surface-entrainment force pulse. Each leaf gets a stable value inside the Min or Max range.

- **Lowest value:** Moves this end of the variation range down.
- **Highest value:** Moves this end of the variation range up.

Lift Duration Max

Maximum duration of the finite surface-entrainment force pulse. Min or Max are automatically ordered internally.

- **Lowest value:** Moves this end of the variation range down.
- **Highest value:** Moves this end of the variation range up.

Lift Height Min

Minimum height scale where the additional pickup lift fades. This is not an exact trajectory ceiling.

- **Lowest value:** Keeps this part of the effect smaller or closer to its source.
- **Highest value:** Makes it larger, longer, or farther reaching.

Lift Height Max

Maximum height scale where the additional pickup lift fades. Each leaf keeps a stable value.

- **Lowest value:** Keeps this part of the effect smaller or closer to its source.
- **Highest value:** Makes it larger, longer, or farther reaching.

Ground Carry Min

Minimum additional near-ground relaxation toward tangential wind, in 1 or s.

- **Lowest value:** Moves this end of the variation range down.
- **Highest value:** Moves this end of the variation range up.

Ground Carry Max

Maximum additional near-ground relaxation toward tangential wind, in 1 or s.

- **Lowest value:** Moves this end of the variation range down.
- **Highest value:** Moves this end of the variation range up.

Tumble Min

Adjusts tumble min for this effect.

- **Lowest value:** Moves this end of the variation range down.
- **Highest value:** Moves this end of the variation range up.

Tumble Max

Maximum pickup torque scale. Each leaf gets a stable value in this range.

- **Lowest value:** Moves this end of the variation range down.
- **Highest value:** Moves this end of the variation range up.

Rotation Strength

Controls how strongly leaves turn into the wind. In flight it follows apparent airflow (Wind - particle Velocity). Zero disables the added balance torque.

- **Lowest value:** Reduces or removes this part of the effect.
- **Highest value:** Makes it more visible or affects more elements.

Rotation Speed

Limits how quickly leaves turn toward the wind. Near the target the requested rate automatically falls to zero.

- **Lowest value:** Movement or variation changes slowly; zero may stop this part of the animation.
- **Highest value:** Movement or variation changes faster.

Fade Delay Min

Minimum simulation seconds of full Ground Wind after pickup. Sampled once for each pickup.

- **Lowest value:** The transition or pause is shorter.
- **Highest value:** The transition or pause lasts longer.

Fade Delay Max

Maximum simulation seconds before fading starts. Min or Max are automatically ordered.

- **Lowest value:** The transition or pause is shorter.
- **Highest value:** The transition or pause lasts longer.

Fade Duration Min

Minimum simulation seconds for the smooth transition from full Ground Wind influence to zero.

- **Lowest value:** The transition or pause is shorter.
- **Highest value:** The transition or pause lasts longer.

Fade Duration Max

Maximum fade duration. A stable random value is sampled once for each pickup.

- **Lowest value:** The transition or pause is shorter.
- **Highest value:** The transition or pause lasts longer.

Fade Falloff

Shape of the smooth fade.

- **Lowest value:** The transition looks sharper.
- **Highest value:** The transition becomes broader and softer.

Settle Time

Adjusts settle time for this effect.

- **Lowest value:** Uses a smaller or weaker setting.
- **Highest value:** Uses a larger or stronger setting.

Landing Friction

Adjusts landing friction for this effect.

- **Lowest value:** Uses a smaller or weaker setting.
- **Highest value:** Uses a larger or stronger setting.

Ground Clearance

Adjusts ground clearance for this effect.

- **Lowest value:** Uses a smaller or weaker setting.
- **Highest value:** Uses a larger or stronger setting.

Landing Yaw Randomness

Adjusts landing yaw randomness for this effect.

- **Lowest value:** Makes the result more even and predictable.
- **Highest value:** Adds more variation between elements or across the surface.

Maximum Ground Slope To Land

Adjusts maximum ground slope to land for this effect.

- **Lowest value:** Uses a smaller or weaker setting.
- **Highest value:** Uses a larger or stronger setting.

Wind

Sets the direction or orientation used for wind.

- **Lowest value:** Zero components remove this directional contribution; negative components point along negative axes.
- **Highest value:** Positive or larger components point along positive axes or increase the response.

Time Scale

Changes the size of the time pattern.

- **Lowest value:** Movement or variation changes slowly; zero may stop this part of the animation.
- **Highest value:** Movement or variation changes faster.

Substep Iterations

Controls how accurately fast leaf movement and collisions are calculated.

- **Lowest value:** Uses less detail and is faster to preview.
- **Highest value:** Produces a smoother or more accurate result but takes longer to evaluate.

Air Drag

Adjusts air drag for this effect.

- **Lowest value:** Uses a smaller or weaker setting.
- **Highest value:** Uses a larger or stronger setting.

Edge Drag

Adjusts edge drag for this effect.

- **Lowest value:** Uses a smaller or weaker setting.
- **Highest value:** Uses a larger or stronger setting.

Drag Randomness

Adjusts drag randomness for this effect.

- **Lowest value:** Makes the result more even and predictable.
- **Highest value:** Adds more variation between elements or across the surface.

Leaf Lift

Adjusts leaf lift for this effect.

- **Lowest value:** Uses a smaller or weaker setting.
- **Highest value:** Uses a larger or stronger setting.

Flutter Strength

Controls the strength of flutter.

- **Lowest value:** Reduces or removes this part of the effect.
- **Highest value:** Makes it more visible or affects more elements.

Flutter Frequency

Controls how quickly flutter frequency changes.

- **Lowest value:** Movement or variation changes slowly; zero may stop this part of the animation.
- **Highest value:** Movement or variation changes faster.

Spin Speed

Controls how quickly spin speed changes.

- **Lowest value:** Movement or variation changes slowly; zero may stop this part of the animation.
- **Highest value:** Movement or variation changes faster.

Air Alignment

Adjusts air alignment for this effect.

- **Lowest value:** Uses a smaller or weaker setting.
- **Highest value:** Uses a larger or stronger setting.

Angular Damping

Adjusts angular damping for this effect.

- **Lowest value:** Uses a smaller or weaker setting.
- **Highest value:** Uses a larger or stronger setting.

Maximum Leaf Speed

Controls how quickly maximum leaf speed changes.

- **Lowest value:** Movement or variation changes slowly; zero may stop this part of the animation.
- **Highest value:** Movement or variation changes faster.

44. Quick Snow

Builds a snow cover over selected objects with optional drifts, mounds, collision clearing, and a compressed path.

How to use

Assign a target object or collection. Set Snow Height and Snow Coverage before adding secondary detail.

- The viewport optimization controls affect preview speed; the final render can use full detail.
- Enable Snow Path and choose a curve before adjusting Path Width and Path Depth.

Quick Snow setup and surface control. Select the desired target and add **Quick Snow**; the volume/SDF-based mesh operation can take time on dense scenes. A compatible snow material is assigned automatically, and the generated object can use saved presets. **Snow Height** and **Snow Coverage** control bulk and distribution. Where the current operator exposes **Remove from Vertical Surfaces**, use its angle threshold to clear steep geometry—such as the sides of a sphere—while retaining snow on horizontal faces. **Mesh Quality** changes resolution; very low values may produce holes. Disabling the **Micro Hills Layer** can sometimes avoid those artifacts when quality must remain low, but may not fix every case.

Interacting with objects and terrain. Put collision objects into a collection, enable the snow collision controls, and assign it. **Collision Distance**, **Displacement Width**, and **Displacement Height** create a raised or displaced border around contacts. Enable **Snow Path**, use the draw/pencil control, and sketch a curve to carve a path through the snow. **Windy Snow Details** creates streaks and height variation; **Snow Mounds** adds broader drifts, each of which can be disabled for a plain cover. **Snow Edges** adds irregularity to the snow boundary; adjust its transition and distortion for a softer or rougher margin. Dense targets can overproduce microscopic bumps; reduce the relevant **Micro Hills Layer** density/step controls or disable the detail layer. Use viewport-optimization switches while editing, then inspect full quality for the final render.

Settings

Target Collection

Objects to cover with snow.

- **Unassigned:** This source is not used; effects that require it may not appear.
- **Assigned:** The effect uses the selected source.

Target Object

Assign the target object used by this effect.

- **Unassigned:** This source is not used; effects that require it may not appear.
- **Assigned:** The effect uses the selected source.

Snow Height

Sets how thick the snow cover is.

- **Lowest value:** Creates a thinner snow cover.
- **Highest value:** Creates a thicker snow cover.

Snow Coverage

Sets how much of the target is covered by snow.

- **Lowest value:** Leaves more of the target exposed.
- **Highest value:** Covers more of the target with snow.

Mesh Quality

Controls snow smoothness and detail.

- **Lowest value:** Uses less detail and is faster to preview.
- **Highest value:** Produces a smoother or more accurate result but takes longer to evaluate.

Seed

Chooses a different repeatable pattern without changing the overall amount.

- **Lowest value:** Selects one repeatable variation.
- **Highest value:** Selects another repeatable variation; a larger number does not mean better quality.

Smooth Mesh - For Better Shadows?

Smooths the snow surface to improve the appearance of its shadows.

- **Off:** Keeps the original snow surface.
- **On:** Smooths the snow for softer shadows.

Reduce Point Density

Uses fewer snow details in the viewport for a faster preview.

- **Off:** Keeps the usual viewport detail.
- **On:** Reduces preview detail for faster interaction.

Viewport Density Factor

Multiplier for viewport snow point density. Lower values are faster but can make coarse or sparse snow.

- **Lowest value:** Reduces or removes this part of the effect.
- **Highest value:** Makes it more visible or affects more elements.

Viewport Voxel Scale

Controls how coarse the temporary viewport preview looks.

- **Lowest value:** Preserves finer preview detail but may run slower.

- **Highest value:** Uses a coarser, faster preview; check the final render separately.

Reduce Smooth Iterations

Override expensive smoothing iterations with a smaller viewport value.

- **Off:** Disables reduce smooth iterations.
- **On:** Enables reduce smooth iterations.

Viewport Smooth Iterations

smoothing iterations used only in the viewport when optimization is enabled.

- **Lowest value:** Uses less detail and is faster to preview.
- **Highest value:** Produces a smoother or more accurate result but takes longer to evaluate.

Disable Collision in Viewport

Skip Object Collision calculations only in the viewport. Final render still uses collision if the main collision switch is enabled.

- **Off:** Disables disable collision in viewport.
- **On:** Enables disable collision in viewport.

Disable Wind Layer in Viewport

Skip the primary windy snow layer only in the viewport.

- **Off:** Disables disable wind layer in viewport.
- **On:** Enables disable wind layer in viewport.

Disable Micro Details in Viewport

Skip scatter-based micro details only while Viewport Optimization is active in the 3D Viewport.

- **Off:** Disables disable micro details in viewport.
- **On:** Enables disable micro details in viewport.

Collision Objects

Collection containing objects that displace and block snow.

- **Unassigned:** This source is not used; effects that require it may not appear.
- **Assigned:** The effect uses the selected source.

Collision Distance

Actual no-snow clearance from collider surfaces. radius is feathered to zero at this edge instead of creating a vertical wall.

- **Lowest value:** Keeps this part of the effect smaller or closer to its source.
- **Highest value:** Makes it larger, longer, or farther reaching.

Displacement Width

Total width of the displaced-snow hump. Maximum displacement is always at exactly half this width.

- **Lowest value:** Keeps this part of the effect smaller or closer to its source.
- **Highest value:** Makes it larger, longer, or farther reaching.

Displacement Height

Maximum displaced-snow height at the middle of Displacement Width. Height is zero at both inner and outer edges.

- **Lowest value:** Keeps this part of the effect smaller or closer to its source.
- **Highest value:** Makes it larger, longer, or farther reaching.

Variation Amount

Maximum vertical amplitude of the primary wind variation layer.

- **Lowest value:** Makes the result more even and predictable.
- **Highest value:** Adds more variation between elements or across the surface.

Drift Size

Size of the primary broad wind drifts.

- **Lowest value:** Keeps this part of the effect smaller or closer to its source.
- **Highest value:** Makes it larger, longer, or farther reaching.

Wind Stretch

Elongates the snow drifts in the wind direction.

- **Lowest value:** Makes snow drifts rounder.
- **Highest value:** Stretches snow drifts along the wind direction.

Wind Direction

Direction of the elongated wind drifts around the local Z axis.

- **Lowest value:** Uses a smaller or weaker setting.
- **Highest value:** Uses a larger or stronger setting.

Fine Detail

Amount of secondary small detail mixed into the primary wind layer.

- **Lowest value:** Uses a smaller or weaker setting.
- **Highest value:** Uses a larger or stronger setting.

Distortion

Amount of distortion.

- **Lowest value:** Uses a smaller or weaker setting.
- **Highest value:** Uses a larger or stronger setting.

Wind Seed

Stable seed offset shared by the wind layer and the secondary snow mound layer.

- **Lowest value:** Selects one repeatable variation.
- **Highest value:** Selects another repeatable variation; a larger number does not mean better quality.

Enable Snow Mounds Layer

Second layer: photo-like snow mounds based on Noise Texture. Adds soft raised piles and shallow depressions on top of the primary wind layer.

- **Off:** Disables snow mounds layer.
- **On:** Enables snow mounds layer.

Snow Mounds Amount

Height of the secondary photo-like snow mound layer.

- **Lowest value:** Reduces or removes this part of the effect.
- **Highest value:** Makes it more visible or affects more elements.

Snow Mounds Size

Size of the secondary photo-like snow mounds.

- **Lowest value:** Keeps this part of the effect smaller or closer to its source.
- **Highest value:** Makes it larger, longer, or farther reaching.

Snow Mounds Contrast

Separates the secondary layer into more distinct snow mounds.

- **Lowest value:** The pattern has softer transitions.
- **Highest value:** The affected and unaffected areas separate more sharply.

Snow Mounds Depression

Carves soft depressions between the secondary snow mounds.

- **Lowest value:** Uses a smaller or weaker setting.
- **Highest value:** Uses a larger or stronger setting.

Snow Mounds Fine Detail

Additional small breakup on the secondary snow mound layer.

- **Lowest value:** Uses a smaller or weaker setting.
- **Highest value:** Uses a larger or stronger setting.

Micro Details Density

Number of micro-detail scatter points per square meter of snow surface.

- **Lowest value:** Reduces or removes this part of the effect.
- **Highest value:** Makes it more visible or affects more elements.

Micro Detail Radius

Approximate influence radius around every scattered micro-detail point.

- **Lowest value:** Keeps this part of the effect smaller or closer to its source.
- **Highest value:** Makes it larger, longer, or farther reaching.

Micro Detail Depth

Base depth or height of the micro details before the per-point random multiplier.

- **Lowest value:** Uses a smaller or weaker setting.
- **Highest value:** Uses a larger or stronger setting.

Micro Detail Randomness

Randomizes radius and depth independently for every scattered point.

- **Lowest value:** Makes the result more even and predictable.
- **Highest value:** Adds more variation between elements or across the surface.

Micro Hills Probability

Probability that a micro detail becomes a tiny raised hill instead of a dent. Set to 0 for holes only.

- **Lowest value:** Reduces or removes this part of the effect.
- **Highest value:** Makes it more visible or affects more elements.

Micro Detail Seed

Seed for point locations and per-point radius or depth variation.

- **Lowest value:** Selects one repeatable variation.
- **Highest value:** Selects another repeatable variation; a larger number does not mean better quality.

Edge Smoothness

Strength of horizontal edge relaxation. The upper snow surface keeps its height.

- **Lowest value:** Uses a smaller or weaker setting.
- **Highest value:** Uses a larger or stronger setting.

Edge Smooth Iterations

Number of local relaxation passes used to remove spikes and stair-stepping.

- **Lowest value:** Uses less detail and is faster to preview.
- **Highest value:** Produces a smoother or more accurate result but takes longer to evaluate.

Edge Transition

Normal-Z threshold controlling how far edge treatment blends onto rounded snow lips.

- **Lowest value:** Uses a smaller or weaker setting.
- **Highest value:** Uses a larger or stronger setting.

Edge Pullback

Small inward erosion of the rim that suppresses unnatural protruding lips.

- **Lowest value:** Uses a smaller or weaker setting.
- **Highest value:** Uses a larger or stronger setting.

Edge Distortion

Natural irregular horizontal offset of the rim.

- **Lowest value:** Uses a smaller or weaker setting.
- **Highest value:** Uses a larger or stronger setting.

Edge Distortion Size

Size of edge irregularities at the 10 m reference domain. It scales with the target domain.

- **Lowest value:** Keeps this part of the effect smaller or closer to its source.
- **Highest value:** Makes it larger, longer, or farther reaching.

Edge Droop

Very small downward settling on the upper snow lip.

- **Lowest value:** Uses a smaller or weaker setting.
- **Highest value:** Uses a larger or stronger setting.

Snow Path Curve

Chooses the curve used to press a path into the snow.

- **Unassigned:** This source is not used; effects that require it may not appear.
- **Assigned:** The effect uses the selected source.

Path Width

Total width of the compressed snow path.

- **Lowest value:** Keeps this part of the effect smaller or closer to its source.
- **Highest value:** Makes it larger, longer, or farther reaching.

Path Depth

Maximum compression depth at the center of the path. Depth is automatically capped against the target surface.

- **Lowest value:** Uses a smaller or weaker setting.
- **Highest value:** Uses a larger or stronger setting.

Path Edge Width

Half-width of the displaced snow ridge around each path edge.

- **Lowest value:** Keeps this part of the effect smaller or closer to its source.
- **Highest value:** Makes it larger, longer, or farther reaching.

Path Edge Height

Maximum raised snow height along both path edges.

- **Lowest value:** Keeps this part of the effect smaller or closer to its source.
- **Highest value:** Makes it larger, longer, or farther reaching.

Path Edge Distortion

Natural lateral irregularity applied mainly near the path edges.

- **Lowest value:** Uses a smaller or weaker setting.
- **Highest value:** Uses a larger or stronger setting.

Path Distortion Size

World or local size of the edge distortion pattern.

- **Lowest value:** Keeps this part of the effect smaller or closer to its source.
- **Highest value:** Makes it larger, longer, or farther reaching.

Path Seed

Seed for the path edge distortion pattern.

- **Lowest value:** Selects one repeatable variation.
- **Highest value:** Selects another repeatable variation; a larger number does not mean better quality.

45. Quick Thin Snow

Adds a thin snow layer with the same main controls as Quick Snow, tuned for shallow coverage.

How to use

Assign a target and start with Height and Coverage. Use Quality and surface smoothing when a shallow layer looks uneven.

- Thin coverage benefits from a finer Quality setting and gentle smoothing.
- Viewport optimization changes only the preview; check the full render before final delivery.

Quick Thin Snow workflow. Choose **Quick Thin Snow** to create a much thinner cover than the full Quick Snow generator typically produces. The effect assigns snow shading automatically and supports presets. Its major controls parallel Quick Snow: target, **Height**, **Coverage**, preview quality, collision objects, wind deformation, snow mounds, edge transitions and a drawn Snow Path. Because this is still a volume/SDF-based geometric snow effect, expect heavier evaluations than a simple surface shader. Use viewport optimizations while arranging the scene. For more extensive examples of collisions, edge breakup and detailed snow patterns, see Quick Snow.

Settings

Target

Objects to cover with snow.

- **Unassigned:** This source is not used; effects that require it may not appear.
- **Assigned:** The effect uses the selected source.

Target Object

Assign the target object used by this effect.

- **Unassigned:** This source is not used; effects that require it may not appear.
- **Assigned:** The effect uses the selected source.

Height

Sets how thick the thin snow layer is.

- **Lowest value:** Creates a very thin layer.
- **Highest value:** Creates a thicker thin-snow layer.

Coverage

Controls how much of the target is covered.

- **Lowest value:** Leaves more of the target exposed.
- **Highest value:** Covers more of the target.

Seed

Chooses a different repeatable pattern without changing the overall amount.

- **Lowest value:** Selects one repeatable variation.
- **Highest value:** Selects another repeatable variation; a larger number does not mean better quality.

Quality

Controls the detail of the thin snow surface.

- **Lowest value:** Uses less detail and is faster to preview.
- **Highest value:** Produces a smoother or more accurate result but takes longer to evaluate.

Enable Surface Smooth?

Controls whether surface smooth? is used.

- **Off:** Disables surface smooth?.
- **On:** Enables surface smooth?.

Reduce Point Density

Uses fewer snow details in the viewport for a faster preview.

- **Off:** Keeps the usual viewport detail.
- **On:** Reduces preview detail for faster interaction.

Viewport Density Factor

Multiplier for viewport snow point density. Lower values are faster but can make coarse or sparse snow.

- **Lowest value:** Reduces or removes this part of the effect.
- **Highest value:** Makes it more visible or affects more elements.

Viewport Voxel Scale

Controls how coarse the temporary viewport preview looks.

- **Lowest value:** Preserves finer preview detail but may run slower.
- **Highest value:** Uses a coarser, faster preview; check the final render separately.

Reduce Smooth Iterations

Override expensive smoothing iterations with a smaller viewport value.

- **Off:** Disables reduce smooth iterations.
- **On:** Enables reduce smooth iterations.

Viewport Smooth Iterations

smoothing iterations used only in the viewport when optimization is enabled.

- **Lowest value:** Uses less detail and is faster to preview.
- **Highest value:** Produces a smoother or more accurate result but takes longer to evaluate.

Disable Collision in Viewport

Skip Object Collision calculations only in the viewport. Final render still uses collision if the main collision switch is enabled.

- **Off:** Disables disable collision in viewport.
- **On:** Enables disable collision in viewport.

Disable Wind Layer in Viewport

Skip the primary windy snow layer only in the viewport.

- **Off:** Disables disable wind layer in viewport.
- **On:** Enables disable wind layer in viewport.

Disable Mounds Layer in Viewport

Skip the secondary Noise Texture snow-mounds layer only in the viewport.

- **Off:** Disables disable mounds layer in viewport.
- **On:** Enables disable mounds layer in viewport.

Collision Objects

Collection containing objects that displace and block snow.

- **Unassigned:** This source is not used; effects that require it may not appear.
- **Assigned:** The effect uses the selected source.

Collision Distance

Actual no-snow clearance from collider surfaces. radius is feathered to zero at this edge instead of creating a vertical wall.

- **Lowest value:** Keeps this part of the effect smaller or closer to its source.
- **Highest value:** Makes it larger, longer, or farther reaching.

Displacement Width

Total width of the displaced-snow hump. Maximum displacement is always at exactly half this width.

- **Lowest value:** Keeps this part of the effect smaller or closer to its source.
- **Highest value:** Makes it larger, longer, or farther reaching.

Displacement Height

Maximum displaced-snow height at the middle of Displacement Width. Height is zero at both inner and outer edges.

- **Lowest value:** Keeps this part of the effect smaller or closer to its source.
- **Highest value:** Makes it larger, longer, or farther reaching.

Variation Amount

Maximum vertical amplitude of the primary wind variation layer.

- **Lowest value:** Makes the result more even and predictable.
- **Highest value:** Adds more variation between elements or across the surface.

Drift Size

Size of the primary broad wind drifts.

- **Lowest value:** Keeps this part of the effect smaller or closer to its source.
- **Highest value:** Makes it larger, longer, or farther reaching.

Wind Stretch

Elongates the snow drifts in the wind direction.

- **Lowest value:** Makes snow drifts rounder.
- **Highest value:** Stretches snow drifts along the wind direction.

Wind Direction

Direction of the elongated wind drifts around the local Z axis.

- **Lowest value:** Uses a smaller or weaker setting.
- **Highest value:** Uses a larger or stronger setting.

Fine Detail

Amount of secondary small detail mixed into the primary wind layer.

- **Lowest value:** Uses a smaller or weaker setting.
- **Highest value:** Uses a larger or stronger setting.

Wind Seed

Stable seed offset shared by the wind layer and the secondary snow mound layer.

- **Lowest value:** Selects one repeatable variation.
- **Highest value:** Selects another repeatable variation; a larger number does not mean better quality.

Enable Snow Mounds Layer

Second layer: photo-like snow mounds based on Noise Texture. Adds soft raised piles and shallow depressions on top of the primary wind layer.

- **Off:** Disables snow mounds layer.
- **On:** Enables snow mounds layer.

Snow Mounds Amount

Height of the secondary photo-like snow mound layer.

- **Lowest value:** Reduces or removes this part of the effect.
- **Highest value:** Makes it more visible or affects more elements.

Snow Mounds Size

Size of the secondary photo-like snow mounds.

- **Lowest value:** Keeps this part of the effect smaller or closer to its source.
- **Highest value:** Makes it larger, longer, or farther reaching.

Snow Mounds Contrast

Separates the secondary layer into more distinct snow mounds.

- **Lowest value:** The pattern has softer transitions.
- **Highest value:** The affected and unaffected areas separate more sharply.

Snow Mounds Depression

Carves soft depressions between the secondary snow mounds.

- **Lowest value:** Uses a smaller or weaker setting.

- **Highest value:** Uses a larger or stronger setting.

Snow Mounds Fine Detail

Additional small breakup on the secondary snow mound layer.

- **Lowest value:** Uses a smaller or weaker setting.
- **Highest value:** Uses a larger or stronger setting.

Edge Smoothness

Strength of horizontal edge relaxation. The upper snow surface keeps its height.

- **Lowest value:** Uses a smaller or weaker setting.
- **Highest value:** Uses a larger or stronger setting.

Edge Smooth Iterations

Number of local relaxation passes used to remove spikes and stair-stepping.

- **Lowest value:** Uses less detail and is faster to preview.
- **Highest value:** Produces a smoother or more accurate result but takes longer to evaluate.

Edge Transition

Normal-Z threshold controlling how far edge treatment blends onto rounded snow lips.

- **Lowest value:** Uses a smaller or weaker setting.
- **Highest value:** Uses a larger or stronger setting.

Edge Pullback

Small inward erosion of the rim that suppresses unnatural protruding lips.

- **Lowest value:** Uses a smaller or weaker setting.
- **Highest value:** Uses a larger or stronger setting.

Edge Distortion

Natural irregular horizontal offset of the rim.

- **Lowest value:** Uses a smaller or weaker setting.
- **Highest value:** Uses a larger or stronger setting.

Edge Distortion Size

Size of edge irregularities at the 10 m reference domain. It scales with the target domain.

- **Lowest value:** Keeps this part of the effect smaller or closer to its source.
- **Highest value:** Makes it larger, longer, or farther reaching.

Edge Droop

Very small downward settling on the upper snow lip.

- **Lowest value:** Uses a smaller or weaker setting.
- **Highest value:** Uses a larger or stronger setting.

Snow Path Curve

Chooses the curve used to press a path into the snow.

- **Unassigned:** This source is not used; effects that require it may not appear.
- **Assigned:** The effect uses the selected source.

Path Width

Total width of the compressed snow path.

- **Lowest value:** Keeps this part of the effect smaller or closer to its source.
- **Highest value:** Makes it larger, longer, or farther reaching.

Path Depth

Maximum compression depth at the center of the path. Depth is automatically capped against the target surface.

- **Lowest value:** Uses a smaller or weaker setting.
- **Highest value:** Uses a larger or stronger setting.

Path Edge Width

Half-width of the displaced snow ridge around each path edge.

- **Lowest value:** Keeps this part of the effect smaller or closer to its source.
- **Highest value:** Makes it larger, longer, or farther reaching.

Path Edge Height

Maximum raised snow height along both path edges.

- **Lowest value:** Keeps this part of the effect smaller or closer to its source.
- **Highest value:** Makes it larger, longer, or farther reaching.

Path Edge Distortion

Natural lateral irregularity applied mainly near the path edges.

- **Lowest value:** Uses a smaller or weaker setting.
- **Highest value:** Uses a larger or stronger setting.

Path Distortion Size

World or local size of the edge distortion pattern.

- **Lowest value:** Keeps this part of the effect smaller or closer to its source.
- **Highest value:** Makes it larger, longer, or farther reaching.

Path Seed

Seed for the path edge distortion pattern.

- **Lowest value:** Selects one repeatable variation.
- **Highest value:** Selects another repeatable variation; a larger number does not mean better quality.